Affine Unique Games on noisy Johnson graphs with near-one completeness

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Editorial note. Fixed native arXiv v1 author TeX, licensed CC BY 4.0. Counted result thm:main_johnson_close_t
Both Johnson analyses are retained for complete source-local support and precise parameter comparison; each record counts only its named main theorem. All deferred iteration, rounding and Johnson SoS proofs are included. Grassmann/HDX theorem statements appear only as original contextual statements; their sketch/omitted-detail generalizations are not counted. Source files and bibliography are retained unchanged; no proof is supplied by the collector.

Allowed standard prerequisites (editorial).

- Sum-of-squares pseudodistributions and standard conditioning/information inequalities. Author complete preliminaries, cited BBKSS, RT12 and related sources; exact invoked facts retained
- Polynomial approximations and global hypercontractivity background. Author preliminary statements and full source-local Johnson SoS certificate retained

1 Introduction

1.1 Unique-Games

The Unique Games Conjecture (UGC in short) is a central open problems in Complexity Theory [Kho02]. The primary reason for the interest in UGC is that, if true, it implies a large number of hardness of approximation results that are often times tight [KR08, KKMO07, Aus07, KNS10, Rag08] (see [Kho10, Tre12]). For example, one of the most striking consequences of UGC is that it implies that a class of semi-definite programs achieves the best possible approximation ratio (among all efficient algorithms) for constraint satisfaction problems [Rag08].

Research towards a proof of UGC, on the other hand, has stalled for the most part up until recently. Unlike problems such as 3-SAT, a randomly sampled instance of Unique-Games (UG) is easy to solve, raising the question of how do hard instances of UG even look like. To discuss this, we first give a formal definition of the Unique-Games problem and the statement of the Unique-Games Conjecture.

Definition 1.1. *A instance of Unique-Games Ψ consists of a graph $G = (V, E)$, a finite alphabet Σ and a collection of constraints, $\Phi = \{\Phi_e\}_{e \in E}$, one for each edge in G. For all $e \in E$, the constraint Φ_e takes the form $\Phi_e = \{(\sigma, \phi_e(\sigma)) \mid \sigma \in \Sigma\}$, where $\phi_e \colon \Sigma \to \Sigma$ is a 1-to-1 map.*

The goal in the Unique-Games problem is to find an assignment $A\colon V \to \Sigma$ that satisfies the maximum number of constraints possible, that is, satisfies that $(A(u), A(v)) \in \Phi_e$ for the largest number of edges $e = (u, v) \in E$ as possible. We define the value of the instance Ψ by:

$$\mathsf{val}(\Psi) = \max_{A\colon V\to\Sigma} \frac{\#\{e \mid A \text{ satisfies } e\}}{|E|}.$$

With this in mind, the Unique-Games Conjecture is the following statement:

Conjecture 1.2. *For all $\varepsilon, \delta > 0$ there is $k \in \mathbb{N}$ such that given a Unique-Games instance Ψ with alphabet size at most k, it is NP-hard to distinguish between:*
YES case: $\mathsf{val}(\Psi) \geq 1 - \varepsilon$.
NO case: $\mathsf{val}(\Psi) \leq \delta$.

It turns out that the topology of the underlying graph G plays a crucial role in the complexity of the UG instance defined over it [AKK+08, MM10, AIMS10]. In particular, it turns out that UG over expander graphs is easy:

Definition 1.3. *Given a regular graph $G = (V, E)$ and a set of vertices $S \subseteq V$, the edge expansion of S is defined by:*

$$\Phi(S) = \Pr_{u\in S, v\in\Gamma(u)} [v \notin S].$$

A graph G is a called a (γ, ξ)-small set-expander if for every $S \subseteq V$ of size at most $\xi|V|$ it holds that $\Phi(S) \geq \gamma$. Informally, we say that G is a small set expander if it is a (γ, ξ)-small set expander for ξ that is a small constant, and we say G is an expander if $\xi = 1/2$. The results of [AKK+08, MM10, AIMS10, BBK+21] assert that UG instances with completeness close to 1 over small-set expanders are easy. Thus, to have any chance of proving UGC, one must use graphs which are outside the scope of expanders and small set expanders.

Indeed, recent progress towards UGC [KMS17, DKK+18a, DKK+18b, KMS18] has utilized graphs which are not small-set expanders. In these works it is proved that 2-to-1-Games are NP-hard (which is a very similar problem to UG, except that each one of the maps ϕ_e defining the constraints is a 2-to-1 map). Among others, this implies that for all $\varepsilon > 0$, given a UG instance Ψ over sufficiently large alphabet, it is NP-hard to distinguish between the case that $\mathsf{val}(\Psi) \geq 1/2$ and the case that $\mathsf{val}(\Psi) \leq \varepsilon$. To prove these results, these works use graphs that are not small set expanders in two different ways:

1. Smooth Parallel Repetition: A key step in the reduction of [KMS17, DKK+18a, DKK+18b, KMS18] is an application of the Parallel Repetition Theorem [Raz98] to get a hardness result for a sufficiently smooth outer PCP construction. Roughly speaking, this step in the process may be associated with the Johnson graph with a large intersection parameter. That is, with the graph $J(n, \ell, t)$ in which the vertices are $\binom{[n]}{\ell}$, and two vertices A and B are adjacent if $|A \cap B| = \ell - t$, and we think of t as much smaller than ℓ (say, $t = \sqrt{\ell}$).

2. Composition with the Grassmann encoding: The Grassmann encoding is an encoding of linear functions based on the Grassmann graph $\mathsf{Grass}(n, \ell)$ over $\mathbb{F}_2$. The Grassmann graph over $\mathbb{F}_2$ is the graph whose vertices are all ℓ-dimensional subspaces of $\mathbb{F}_2^n$, denoted by $\begin{bmatrix} n \\ \ell \end{bmatrix}$, and two vertices L and L' are adjacent if $\mathsf{dim}(L \cap L') = \ell - 1$.

Both of the graphs above, namely the Johnson graph with large intersection sizes, as well as the Grassmann graph, are not small set expanders. However very importantly, the class of small sets in the Grassmann graph with bad expansion has a succinct and intuitive characterization, and the proof of the 2-to-1 Games Theorem heavily relies on this characterization.

Though the term is not formally defined, we refer to graphs such as the Grassmann graph above as globally hypercontractive graphs. By that, we mean that there is a collection of "obviously-non-expanding local sets", such that any small set that doesn't expand well must have a large intersection with one of the sets from the collection (see Section 5.1 for a semi-formal definition). Aside from the Grassmann graph, this class of graphs includes Johnson graphs with small intersection sizes [KMMS18], certain Cayley graphs over the symmetric group [FKLM20], p-biased cubes for $p = o(1)$, other product domains [KLLM21] as well as high dimensional expanders [GLL22, BHKL22b].

A natural question to consider then refers to the complexity of UG over graphs that are globally hypercontractive, which is the main question of study in this paper. The motivation for this is two-fold:

1. First, in light of the reduction of [KMS17, DKK+18a, DKK+18b, KMS18] and the above discussion, it is an interesting question whether there may be a reduction showing the NP-hardness of UG that only uses globally hypercontractive graphs. Gaining a better understanding for these questions will shed further light on the source of hardness in this reduction, as well as explain the distinct roles of the "smooth parallel repetition" and "composition with the Grassmann encoding" steps above.

2. Second, the UGC is related to another well-known computational problem known as the Small-set Expansion (SSE) problem [RS10, RST12]. In the SSE problem one is given a graph $G = (V, E)$ such that either (a) for all $S \subseteq V$ of size at most $\gamma|V|$ it holds that $\Phi(S) \geq 1 - \varepsilon$, or (b) there is $S \subseteq V$ of size at most $\gamma|V|$ such that $\Phi(S) \leq \delta$. The Small-set Expansion hypothesis (SSEH) asserts that for all $\varepsilon, \delta > 0$ this problem is NP-hard for sufficiently small γ, and the papers [RS10, RST12] show that this hypothesis implies the UGC.[1]

   Despite recent progress towards UGC, no new results for the SSEH were proved. In some sense, it seems that to prove hardness results for SSE, one would need to only use gadgets that are small-set expanders or perhaps globally hypercontractive. Otherwise, in the final graph one may create small non-expanding sets that are unintended (namely, that don't correspond to any solution to the initial problem we reduced from). This motivates the study of the role of globally hypercontractive graphs in reductions related to Unique-Games.

The question has been addressed recently in [BBK+21] (with follow-up by [BHKL22a]) for the class of *Affine Unique-Games*, which we define and discuss in the next section.

1.2 Unique-Games over Globally Hypercontractive Graphs

Definition 1.4. *An instance of Affine-UG is an instance of Unique-Games in which the alphabet is the ring of integers modulo q, $\mathbb{Z}_q$, and all of the constraint maps ϕ_e are affine shifts, that is, ϕ_e of the form $\phi_e(\sigma) = \sigma + b_e$ for some $b_e \in \mathbb{Z}_q$.*

[1] To be more precise, it implies a stronger version of the UGC, which states that UG remains hard on graphs that have an expansion profile as the noisy hypercube.

An equivalent but slightly different way to view the Affine-UG problem is as a system of linear equations (X, E) over $\mathbb{Z}_q$. Each equation in E is of the form $x_i - x_j = b$ where $x_i, x_j \in X$ are variables and $b \in \mathbb{Z}_q$ is some constant. Despite looking very restrictive, it is known [KKMO07] that the UGC is true if and only if it holds for the class of Affine UG and furthermore this class captures many interesting optimization problems such as Max-Cut and graph coloring, thus we shall focus our attention on Affine UG henceforth. [2]

In [BBK+21], the authors investigate the complexity of Affine UG on the Johnson graph, albeit with small intersection sizes, the regime in which a characterization theorem for non-expanding sets holds. Namely, they consider the "α-noisy" Johnson graph $J(n, \ell, t)$ in the case that $t = \alpha\ell$, for $\alpha \in (0, 1)$ bounded away from 0 and thought of as a fixed constant independent of ℓ. The result of [BBK+21] asserts that for small enough $\varepsilon > 0$, there is a polynomial time algorithm that given an instance of UG over $J(n, \ell, t)$ with value at least $1 - \varepsilon$, finds an assignment that satisfies at least $C(\ell, \alpha, \varepsilon) > 0$ of the constraints. The most important feature of this result is the lack of dependency on the size of the graph n and the alphabet size, however it has two significant downsides:

1. The completeness guarantee needed for the algorithm to work is close to 1 (ε close to 0). Thus, within the realm of their result, it is still possible that UG over the noisy-Johnson and Grassmann graphs is hard for completeness smaller than 1/2 (wherein general UG is now known to be hard) but suddenly becomes easy when the completeness exceeds 1/2. In fact, UG with completeness $> 1/2$ is easy for the Grassmann graph via the UG algorithm for expanders itself, thus their result does not imply anything non-trivial about the complexity of UG on the Grassmann graph, despite its being globally hypercontractive. Therefore, it is not clear what their algorithm implies about the hard instances of [DKK+18b] which have completeness less than 1/2: (1) are they hard due to the smooth parallel repetition step, or the composition with the Grassmann graph? (2) Is the smoothness of the outer PCP necessary – instead of the Johnson graph with large intersection sizes could one have instead used the noisy-Johnson graph?

2. Their soundness guarantee depends on ℓ, the uniformity of the sets in the Johnson graph, which in most PCP constructions grows with the alphabet size of the instance. Indeed, such graphs are often used to encode a global function $f\colon [n] \to \Gamma$ (often from an error correcting code) where $|\Gamma| = O(1)$. In that case, the Johnson graph corresponds to the restrictions encoding $F[A] = f|_A$, in which case each vertex A of the Johnson graph is assigned a label from Γ^ℓ corresponding to the restriction of f to A. Thus, in the hard instances of UG obtained via PCP constructions (in particular the one above) the parameter ℓ grows with the alphabet size of the UG instance, hence we would like the performance of our algorithm to not depend on it.

   We remark though, that it is okay for the runtime of our algorithm to depend on it (just like an algorithm running in time n^k that solves UG would refute the conjecture, but the trivial random algorithm that satisfies at least a $1/k$ fraction of the constraints does not).

[2] We remark that the reduction of [KKMO07] does not preserve the topology of the graph. We are therefore not able to translate our results directly to the class of general UG, and believe this is an interesting direction for further study.

1.3 Our Results

Our main result asserts that there is a polynomial time algorithm for solving Affine UG over globally hypercontractive graphs that bypasses these two barriers thus addressing the motivations presented in Section 1.1. As the term globally hypercontractive graph is not formally defined, below are some concrete instances of graphs on which this applies. In Section 5.1 we give a semi-formal definition of globally hypercontractive graphs and also show how our algorithm and analysis can be abstracted to solve UG on such graphs, as long as one is provided with an SoS certificate of global hypercontractivity.

We first consider the noisy-Johnson graph, for which we have the following two results. The first result addresses the case that the completeness of the instance is close to 1, in which case our algorithm matches the guarantee of the algorithm of [BBK+21] for certifiable small-set expanders, and in particular the α-noisy hypercube graph:

Theorem 1.5. *There is $\varepsilon_0 > 0$ such that for all $\alpha \in (0,1)$ the following holds for all $0 < \varepsilon \leq \varepsilon_0$. There exists an algorithm whose running time is $n^{\mathrm{poly}(\ell,|\Sigma|,1/\varepsilon)}$ which, on input Ψ which is an affine UG instance over $J(n,\ell,\alpha\ell)$ promised to be at least $(1-\varepsilon)$-satisfiable, finds an assignment that satisfies at least $2^{-O\left(\frac{\sqrt{\varepsilon}}{\alpha}\right)}$-fraction of the constraints in Ψ.*

The second result addresses the case of UG instances with arbitrarily small (but bounded away from 0) completeness, in which case our algorithm satisfies a constant fraction of the constraints:

Theorem 1.6. *For all $\alpha \in (0,1)$ and $c > 0$, there is $\delta > 0$ such that the following holds. There exists an algorithm whose running time is n^D with $D = \ell^{\mathrm{poly}(|\Sigma|\ell^{1/c})}$[3] which on input Ψ, an affine UG instance over $J(n,\ell,\alpha\ell)$ promised to be at least c-satisfiable, finds an assignment that satisfies at least δ-fraction of the constraints in Ψ.*

We note that the soundness guarantee in the theorems above does not depend on ℓ when $\alpha = \Omega(1)$, but degrades as α approaches 0. In Section 1.3 below we discuss why one cannot get a polytime algorithm that obtains an $\Omega(1)$-valued solution when $\alpha = o(1)$ (e.g. $1/\sqrt{\ell}$) therefore showing that this dependence on α is necessary.

We can get similar results given any of the globally hypercontractive graphs mentioned earlier. Below we give a corollary for the Grassmann graph. We show that there is a polynomial time algorithm solving affine UG over the Grassmann graph, even on instances with small completeness:

Theorem 1.7. *For all $c > 0$ there exists $\delta > 0$ such that the following holds. There exists an algorithm whose running time is n^D with $D = \ell^{\mathrm{poly}(|\Sigma|\ell^{1/c})}$ which on input Ψ, an affine UG instance over $\mathsf{Grass}(n,\ell)$ promised to be at least c-satisfiable, finds an assignment that satisfies at least δ-fraction of the constraints in Ψ.*

As discussed before, since the spectral gap of the Grassmann graph is 1/2, UG algorithms over expanders already imply Theorem 1.7 for $c \gg 1/2$. Thus, the main contribution of Theorem 1.7 is the algorithm on Grassmann graphs that works for arbitrarily small completeness.

Below we state our result for random walks on high dimensional expanders (HDX), a large class of graphs that generalize the Johnson graphs but do not necessarily possess its strong symmetries.

[3] We note that we have not optimized for D and the $\exp(\ell)$-dependence arises due to the degree of the SoS proofs in Section 4. We used a blackbox statement to convert some of the proofs therein to SoS proofs, and we conjecture that one can in fact improve the SoS degree to $O(\ell)$ when done carefully.

These include graphs stemming from cut-offs of [LSV05]'s construction of Ramanujan complexes, or [KO18]'s construction of coset complex expanders. These graphs exhibit the nice high-dimensional expansion properties (e.g. global hypercontractivity) of the Johnson graphs yet are substantially different in other aspects, such as being of bounded degree.

Theorem 1.8. *For all $\alpha \in (0,1)$ and $c > 0$, there exists $\delta > 0$ such that the following holds. Let X be any d-dimensional two-sided γ-local-spectral expander with $\gamma \ll o_\ell(1)$ and $d > \ell$. There exists an algorithm whose running time is n^D with $D = \ell^{\mathrm{poly}(|\Sigma|\ell^{1/c})}$ which on input Ψ, an affine UG instance over the canonical walk M on $X(\ell)$ of depth α, promised to be at least c-satisfiable, finds an assignment that satisfies at least δ-fraction of the constraints in Ψ.*

Since we have not defined any of the HDX terminology, let us note that this is indeed a generalization of Theorem 1.6. The Johnson graph corresponds to the complete complex X (which is the simplest instantiation of a two-sided local spectral expander), and the α-noisy Johnson graph $J(n, \ell, \alpha\ell)$ corresponds to a "canonical" random-walk on $X(\ell)$ that goes down $\alpha\ell$-levels and comes back up randomly to $X(\ell)$ while ensuring that it changes exactly $\alpha\ell$ elements in a vertex. In fact, in the above theorem we can allow M to be any complete random walk on $X(\ell)$ and our soundness guarantee will only depend on c and certain parameters of M that are inherently independent of ℓ [4].

For the sake of simplicity of presentation, in the first part of the paper we focus on the Johnson graph. In Section 5 we provide an abstraction of our techniques and discuss the algorithm for general globally hypercontractive graphs, in particular for the Grassmann graph and random-walks over HDXs.

An Interpretation of Our Results

Our results suggest that the hardness in the instances of UG obtained via the reduction of [KMS17, DKK+18a, DKK+18b, KMS18] does not come from the Grassmann graph (which is globally hypercontractive), but rather from the smooth parallel repetition step. Recall that this step uses a Johnson graph with a large intersection parameter ($J(n, \ell, \alpha\ell)$ with $\alpha \approx 0$), that is not globally-hypercontractive. Therefore combining the knowledge from the reduction and our algorithm we get that the α-noisy-Johnson graphs are hard for UG when $\alpha = o(1)$ and become easy when α is bounded away from 0, thus also explaining why our soundness guarantee must necessarily decay with α (under $P \neq NP$). Indeed, we would be able to make such an assertion provided that our results held for general UG (as opposed to only affine UG) or if the reduction above produced instances of Affine UG. Though we believe an algorithm for general UG should exist along the lines of our algorithm, we do not know how to prove so and leave this is an interesting direction to investigate.

Albeit, ignoring the subtlety between general and affine UG, this means that any future progress on UGC will have to use graphs that are *not* globally hypercontractive, possibly again via the technique of *smooth* parallel repetition.

[4] Concretely it depends on the stripped threshold rank of M above a certain threshold as defined in [BHKL22a]. For example, when M is the canonical random walk with depth α on $X(\ell)$, and the completeness is $c = 1 - \varepsilon$, this quantity is $r(M) = O(\sqrt{\varepsilon}/\alpha)$ and our soundness guarantee is $\exp(-r)$, matching that of Theorem 1.5.

2 Preliminaries

Notations. For a (weighted) graph $G = (V, E)$, we denote by $(u, v) \sim E$ the distribution over edges proportional to their weight, and by A_G the transition matrix related to the random walk on G. We define the Laplacian of G by $L_G = I - A_G$, and denote the stationary distribution of G by π_G. In this paper we will mostly be working with regular graphs and therefore the uniform distribution over G. We use $\langle f, g \rangle_G$ to denote $\mathbb{E}_{u \sim \pi_G}[f(u)g(u)]$. We often drop the subscript of G when it is clear from context.

If A is some probabilistic event or condition, we use $\mathbb{I}(A)$ to denote the indicator random variable of A (i.e., $\mathbb{I}(A) = 1$ if A occurs and $\mathbb{I}(A) = 0$ otherwise).

2.1 The Sum of Squares Relaxation of Unique-Games

Our algorithm is based on the SoS semidefinite programming (SDP) relaxation, and in particular its view as optimizing over *pseudo expectation* operators. In this section, we briefly present the necessary background as well as several basic facts that we will use, and we refer the reader to the surveys [BS14, RSS18, FKP19] for a more systematic presentation.

Given a Unique-Games instance $I = (G = (V, E), \Sigma, \Pi)$ the value of I can be computed by the following integer program over zero-one variables $\{X_{u,a}\}_{u \in V, a \in \Sigma}$:

$$\begin{aligned} \max_{X} \quad & \mathop{\mathbb{E}}_{(u,v) \in E} \sum_{a \in \Sigma} X_{u,a} X_{v,\pi_{uv}(a)} & (1) \\ s.t. \quad & X_{u,a}^2 = X_{u,a} && \forall u \in V, a \in \Sigma, \\ & X_{u,a} X_{u,b} = 0 && \forall u \in V, a \neq b \in \Sigma, \\ & \sum_{a} X_{u,a} = 1 && \forall u \in V. \end{aligned}$$

Indeed, an assignment to the variables $X_{i,a}$ represent an assignment to the vertices of G, in which a vertex $v \in V$ takes label $a \in \Sigma$ if $X_{v,a} = 1$, and the objective function counts the fraction of constraints that are satisfied satisfied.

The degree D SoS relaxation of the above program is a convex optimization program that relaxes the above program. This program can be written as an optimization problem in which the goal is to find a vector-valued assignment to the variables $X_{S,\sigma}$, where we have such variable for each set of vertices S of size at most D and labels for them $\sigma \in \Sigma^S$. The constraints of the program address the inner products between these vectors, attempting to ensure that for any two sets of vertices S and T such that $|S \cup T| \leq D$, the inner products $\langle X_{S,\sigma}, X_{T,\sigma'} \rangle$ represent a distribution over assignments to $S \cup T$, and that these distributions are locally consistent. It will be more convenient for us to take a different but equivalent view on the solutions to the SoS relaxation, in the language of pseudoexpectations as presented next.

2.2 Pseudoexpectations, Pseudodistributions and Pseudoprobabilities

Pseudoexpectations. A degree D pseudoexpectation $\widetilde{\mathbb{E}}$ is an operator $\widetilde{\mathbb{E}} : X^{\leq D} \to \mathbb{R}$, where $X^{\leq D}$ is the set of all monomials in the X variables up to degree D, and $\widetilde{\mathbb{E}}$ satisfies the above equality constraints and the Booleanity constraints $\{X_{u,a}^2 = X_{u,a}\}$ as axioms. For brevity, we will refer to this set of axioms as $\mathcal{A}_I$, dropping the subscript when I is clear from context. More

generally, given a polynomial optimization program $P = \{\max_x p(x) \ s.t. \ q_i(x) = 0, \forall i \in [m]\}$, the degree-$D$ sum-of-squares semidefinite programming relaxation of P is a semidefinite program of size $n^{O(D)}$ that returns a *pseudoexpectation operator* $\widetilde{\mathbb{E}} : x^{\leq D} \to \mathbb{R}$. This operator can be uniquely extended to give a pseudoexpectation operator on the set of all polynomials of degree at most D by linearity (defined precisely below). This operator satisfies four properties:

- Scaling: $\widetilde{\mathbb{E}}[1] = 1$.
- Linearity: $\widetilde{\mathbb{E}}[a \cdot f(x) + b \cdot g(x)] = a \cdot \widetilde{\mathbb{E}}[f(x)] + b \cdot \widetilde{\mathbb{E}}[g(x)]$, for all $a, b \in \mathbb{R}$ and all degree $\leq D$ polynomials f, g.
- Non-negativity of low-degree squares: $\widetilde{\mathbb{E}}[s(x)^2] \geq 0$ for all polynomials $s(x)$ with $\deg(s) \leq \frac{D}{2}$.
- Program constraints: $\widetilde{\mathbb{E}}[f(x) \cdot q_i(x)] = 0$ for all $i \in [m]$ and polynomials $f(x)$ such that $\deg(fq_i) \leq D$.

Depending on the problem we are trying to solve, we also discuss the value achieves by the pseudoexpectation $\widetilde{\mathbb{E}}$, which is defined to be the pseudoexpectation of the objective function. In our case of interest, namely the case of Affine Unique-Games, the *value* of $\widetilde{\mathbb{E}}$ on the instance I is denoted by $\mathrm{val}_\mu(I)$ and is defined to be

$$\widetilde{\mathbb{E}}[\mathrm{val}_I(X)] = \widetilde{\mathbb{E}}\left[\mathop{\mathbb{E}}_{(u,v)\in E} \sum_{a\in\Sigma} X_{u,a} X_{v,\pi_{uv}(a)}\right].$$

Our pseudoexpectation will be guaranteed to achieve a value which matches the completeness guarantee (for example, in the context of Theorems 1.6, 1.7 it will satisfy that $\widetilde{\mathbb{E}}[\mathrm{val}_I(X)] \geq c$).

Pseudodistribution. We often refer to a pseudoexpectation operator $\widetilde{\mathbb{E}}$ as an operator corresponding to a pseudodistribution μ over variables X. This is analogous to the case where we have an actual distribution μ and its corresponding expectation operator $\mathbb{E}_\mu[\cdot]$. This notation makes our analysis using $\widetilde{\mathbb{E}}$ operators more intuitive, since many properties that are true of actual distributions also hold for pseudodistributions. Hence when we say that we are given a degree D pseudodistribution μ we are referring to the degree D pseudoexpectation operator $\widetilde{\mathbb{E}}_\mu$.

Pseudoprobabilities.

Definition 2.1 (Pseudoprobability of an event)**.** *Let μ be a pseudodistribution of degree D. For an event $\mathcal{E}(X, X')$ such that $\mathbb{I}[\mathcal{E}(X)]$ can be expressed as a degree-D function of X, we define the* pseudoprobability of $\mathcal{E}(X)$ *to be*

$$\widetilde{\Pr_\mu}[\mathcal{E}(X)] = \widetilde{\mathbb{E}}_\mu[\mathbb{I}(\mathcal{E}(X)].$$

Similarly, if $\mathcal{F}(X)$ is an event and $\deg(\mathbb{I}[\mathcal{F}(X)])+\deg(\mathbb{I}[\mathcal{E}(X)]) \leq D$*, we define the* pseudoprobability of $\mathcal{E}(X)$ conditioned on $\mathcal{F}(X)$ *to be*

$$\widetilde{\Pr_\mu}[\mathcal{E}(X) \mid \mathcal{F}(X)] = \widetilde{\mathbb{E}}_\mu[\mathbb{I}(\mathcal{E}(X)) \mid \mathbb{I}(\mathcal{F}(X))] = \frac{\widetilde{\mathbb{E}}[\mathbb{I}(\mathcal{E}(X)) \cdot \mathbb{I}(\mathcal{F}(X))]}{\widetilde{\mathbb{E}}[\mathbb{I}(\mathcal{F}(X))]}.$$

2.3 SoS-ing Mathematical Proofs

Our argument will use several mathematical statements (such as global hypercontractivity), and we will need to be able to argue that these statements are also satisfied in the context of the SoS program and its variables. Thus, we will need to be able to ensure that we use tools that can be proved via sum of squares inequalities of low-degree (and axioms), and below we collect a few such standard tools that we will use. We use the following notation for SoS prooofs.

Sum of squares proofs notations. Given a set of axioms $\mathcal{A} = \{q_i = 0\}_i \cup \{g_j \geq 0\}_j$ for polynomials $q_i, g_j \in \mathbb{R}[X]$, we say that "there is a degree-d sum-of-squares proof that $f \geq h$ modulo $\mathcal{A}$" if: $f = h + s + \sum_k c_k \cdot Q_k + \sum_t r_t \cdot G_t$ where each polynomial Q_k and G_t is a product of some polynomials from $\{q_i\}$ and $\{g_j\}_j$ respectively, $s, \{c_k\}_k, \{r_t\}_t \in \mathbb{R}[X]$ are real polynomials such that s and $\{r_t\}_t$ are sums of squares, and the maximum degree among $s, \{c_k Q_k\}_k, \{r_t G_t\}_t$ is at most d. We will use the notation $\mathcal{A} \vdash_d f(x) \geq h(x)$ to denote the existence of such an equality. We also sometimes use $f(x) \succeq h(x)$ to denote that the inequality is a SoS inequality.

2.3.1 Basic Inequalities

The first of which is the following basic forms of the Cauchy-Schwarz and Hölder inequalities. The proofs are by now standard and can be found for example in [BKS14, OZ13].

Lemma 2.2 (Cauchy Schwarz)**.** *For for all $\epsilon \in \mathbb{R}_+$,*

$$\vdash_2 YZ \leq \frac{\epsilon}{2} Y^2 + \frac{1}{2\epsilon} Z^2.$$

Lemma 2.3 (Cauchy Schwarz)**.** *A degree-D pseudoexpectation operator where $D \geq 2\max(\deg(f), \deg(g))$ satisfies that*

$$\widetilde{\mathbb{E}}[fg]^2 \leq \widetilde{\mathbb{E}}[f^2]\,\widetilde{\mathbb{E}}[g^2].$$

Fact 2.4 (Hölder's Inequality)**.** *For all real $\nu > 0$ we have that,*

$$\vdash_4 Y^3 Z \leq \frac{3\nu}{4} Y^4 + \frac{1}{4\nu^3} Z^4.$$

The theory of univariate sum-of-squares (in particular, Lukács Theorem) says that if a univariate polynomial is non-negative on an interval, this fact is also SoS-certifiable. The following corollary of Lukács theorem is well-known.

Corollary 2.5 (Corollary of Lukács Theorem)**.** *Let q be a degree-d polynomial which is non-negative on $[a, b]$. Then given the axioms $\mathcal{A} = \{x \geq a\} \cup \{x \leq b\}$, there is a degree-$2d$ SoS proof that q is non-negative, $\mathcal{A} \vdash_{2d} q(x) \geq 0$.*

We will use the above multiple times to convert univariate inequalities into SoS inequalities in a blackbox manner.

Given multivariate inequalities the theorem above no longer holds. Nevertheless given a strictly positive polynomial f that is bounded away from 0, one can get an SoS proof of degree which is exponential in $\deg(f)$ using Theorem 3 in [Sch04]. We state a corollary of [Sch04] that is sufficient for our purposes.

Theorem 2.6 (Corollary of [Sch04])**.** *Let S be the set $[0,1]^k$. Let $f(x)$ be a polynomial of degree d with $\nu = \min\{f(x) \mid x \in S\} > 0$ and $||f||$ denoting the maximum absolute value of f's coefficients. Then f has an SoS certificate of bounded degree:*

$$\{x_i \in [0,1] | i \in [k]\} \vdash_D f(x) \geq 0,$$

with $D = \text{poly}(k^{\deg(f)}, ||f||/\nu)$.

We will use the above fact in Section 4 to convert inequalities involving polynomials of constant degree and on constantly many variables into SoS inequalities. We believe it should be possible to get a degree bound above which is $\text{poly}(\deg(f))$ in cases where we have reasonable polynomials f, though as far as we know this has not been proved in generality.

2.3.2 Approximating Indicators via Low-degree Polynomials

Our argument will involve indicators of events such as $f(X) \geq \beta$ where f is a low-degree polynomial, and we will want to condition on such events. Strictly speaking, the function $\mathbb{I}[f(X) \geq \beta]$ is not a low-degree polynomial and therefore we cannot condition on it. However, it is not difficult to show that such indicators can be approximated by low-degree polynomials, and we will need to use such ideas. Indeed, in this section we present such an approximation theorem that will be used throughout our proofs.

The following theorem, due to [DGJ+10], provides a low-degree approximation to a step function.

Theorem 2.7. *Let $\mathbb{I}[x \geq \beta]$ be the step function at $\beta \in (0,1)$. Then for each $0 < \nu < \beta$ there is a univariate polynomial $p_{\beta,\nu}$ of degree $O(\frac{1}{\nu}\log^2 \frac{1}{\nu})$ such that:*

1. *$|p_{\beta,\nu}(x) - \mathbb{I}[x \geq \beta]| \leq \nu$ for all $x \in [0,\beta] \cup [\beta+\nu, 1]$.*
2. *$p_{\beta,\nu}$ is monotonically increasing on $(\beta, \beta+\nu)$.*
3. *$0 \leq p_{\beta,\nu}(x) \leq 1$ for all $x \in [0,1]$.*
4. *All coefficients of $p_{\beta,\nu}$ are at most $2^{O(\log(1/\nu)/\nu)}$ in absolute value.*

Further the first three facts are SoS-certifiable in degree $2\deg(p_{\beta,\nu})$.

The first three items follow from [DGJ+10, Theorem 4.5], the fourth item is immediate by Markov brothers' inequality, and the SoS certifiability follows from Corollary 2.5.

The following fact provides convenient point-wise bounds on the approximating polynomials of indicators from above.

Fact 2.8 (Markov Inequality for Bounded Polynomials)**.** *Let $p := p_{\beta,\nu}$ be the degree-$D = \widetilde{O}(1/\nu)$ polynomial guaranteed by Theorem 2.7. Then p satisfies Markov's inequality:*

$$\{0 \leq x \leq 1\} \vdash_{\deg(p)} \{p(x) \geq 1 - \frac{1-x}{1-\beta-\nu} - \nu\} \cup \{p(x) \leq \frac{x}{\beta-\nu} + \nu\}$$

Proof. We perform case analysis on x and then use Corollary 2.5 to conclude the proof is SoS. For the first inequality, for $x \in [0, \beta + \nu)$ we have

$$p(x) \succeq 0 \succeq 1 - \frac{1-x}{1-\beta-\nu},$$

where we have used that $p(x) \succeq 0$ and $\frac{1-x}{1-\beta-\nu} \succeq 1$. For $x \in [\beta+\nu, 1]$,

$$p(x) \succeq 1 - \varepsilon \succeq 1 - \varepsilon - \frac{1-x}{1-\alpha-\delta},$$

where we have used that $x \in [0,1]$ so that we are subtracting a positive quantity. Combining these claims concludes the proof of the first claim.

To see the second claim, notice that for $x \in [0, \beta-\nu]$, $p(x) \leq \nu$, and for $x \in (\beta-\nu, 1]$, $p(x) \leq 1 \leq \frac{x}{\beta-\nu}$. This concludes the proof. □

2.4 Manipulating pseudoexpectations

2.4.1 Reweighing and conditioning:

We will sometimes *reweigh* or *condition* our degree-D pseudodistribution by a polynomial $s(x)$ where $s(x)$ is non-negative under the program axioms, i.e. $\mathcal{A} \vdash_d s(x)$ for $d < D$. Technically, this operation amounts to defining a new pseudoexpectation operator $\widetilde{\mathbb{E}}'$ of degree $D-d$ by taking,

$$\widetilde{E}'[x^\alpha] = \frac{\widetilde{\mathbb{E}}[x^\alpha \cdot s(x)]}{\widetilde{\mathbb{E}}[s(x)]},$$

for every monomial x^α of degree at most $D-d$. As an example, under the unique games axioms presented in 1 one can prove that a variable X_i is in $[0,1]$, hence one can reweigh the pseudodistribution by X_i. One can show that reweighing preserves the four properties of the pseudodistribution up to degree $D-d$. Thus, we will also refer to this operation as "conditioning", and denote $\widetilde{\mathbb{E}}'$ by $\widetilde{\mathbb{E}}[\cdot \mid s(x)]$. Often times, the polynomial $s(x)$ we will "condition" on will be a smooth approximation of some event E, in which case the above operation takes the interpretation of conditioning our sample from the pseudodistribution to satisfy some properties specified by the event E. We refer the reader to [BRS11, BKS17] for further discussion of reweighting.

2.4.2 Independent Samples

Recall that a given pseudoexpectation operator $\widetilde{\mathbb{E}} : X^{\leq D} \to \mathbb{R}$ has the interpretation as averaging of functions $f(X)$ over a pseudodistribution $X \sim \mu$. We will need to be able to mimic averaging over two independently chosen samples $X, X' \sim \mu$,[5] for that we define the product pseudoexpectation $\widetilde{\mathbb{E}}_{X,X'}$ as follows: let $X^\alpha (X')^\beta$ be a monomial of degree at most D in variables X, X'; we define $\widetilde{\mathbb{E}}_{X,X'}[X^\alpha (X')^\beta] := \widetilde{\mathbb{E}}_X[X^\alpha] \cdot \widetilde{\mathbb{E}}_X[X^\beta]$. It is easy to check that $\widetilde{\mathbb{E}}_{X,X'}$ is also a pseudoexpectation operator corresponding to two independent samples from the pseudodistribution μ; see Fact 2.9.

Given two independent samples $X, X' \sim \mu$, we will often be interested in the variables $\{Z_{v,s}\}_{v \in V, s \in \Sigma}$ that are used to define the shift-partition discussed in the introduction. Formally, we define them as

$$Z_{u,s} = \sum_{a \in \Sigma} X_{u,a} X'_{u,a+s}.$$

[5] Similar constructs have been used in the literature, see e.g. [BKS14].

In the rest of this section we present a few facts about polynomials in independent samples, and the pseudodistribution $\widetilde{\mathbb{E}}_{X,X'}$ on them. Some of these facts will be general, and some of which will be specific to the shift partition variables $Z_{u,s}$.

The first fact asserts that $\widetilde{\mathbb{E}}_{X,X'}$ defined above is a legitimate pseudodistribution that inherits all of the constraints that $\widetilde{\mathbb{E}}$ satisfies.

Fact 2.9. *If $\widetilde{\mathbb{E}}_X$ is a valid pseudodistribution of degree D in variables X, then $\widetilde{\mathbb{E}}_{X,X'}$ is a valid pseudodistribution of degree D. Furthermore, if there are additional SoS inequalities that are true for $\widetilde{\mathbb{E}}_X$, they also hold for $\widetilde{\mathbb{E}}_{X,X'}$.*

Now, we prove some properties specific to the Z variables. The following fact asserts that the shift partition variables $Z_{v,s}$ indeed behave like a partition, in the sense that if X satisfy the constraints in the program 1, then the variables $Z_{v,s}$ indeed define a partition.

Fact 2.10. *Define the shift variable $Z_{u,s} = \sum_{a\in\Sigma} X_{u,a}X'_{u,a+s}$ to be the indicator that $X_u - X'_u = s$, for X, X' degree-8 solutions to the SoS relaxation of the UG integer program (1), and for each edge (u,v) the variable $Y_{(u,v)} = \sum_a X_{u,a}X_{v,\pi_{uv}(a)}$ to be the indicator that the constraint on the edge (u,v) is satisfied.*

Then the Z variables satisfy:

1. *Booleanity: $Z_{v,s}^2 = Z_{v,a}$.*
2. *Partition constraints: $Z_{v,s}Z_{v,s'} = 0$ for $s \neq s'$, and $\sum_s Z_{u,s} = 1$.*
3. *Crossing edges violate an assignment: $Z_{u,s}Z_{v,s'}Y_{(u,v)}Y'_{(u,v)} = 0$ for every edge $(u,v) \in E$ and $s \neq s'$.*

Proof. The first two items are easily verified via direct computation, using properties of the variables $X_{u,a}$. We prove that the final property holds. Since our UG instance is affine, we have that for each $i,j \in E$, $\pi_{ij}(a) = a + h_{ij}$ for some $h_{ij} \in \Sigma$. Therefore,

$$\begin{aligned} Z_{i,s}Z_{j,t}Y_{(i,j)}Y'_{(i,j)} &= \sum_{a,b,c,d\in\Sigma} X_{i,a}X'_{i,a+s}\cdot X_{j,b}X'_{j,b+t}\cdot X_{i,c}X_{j,c+h_{ij}}\cdot X'_{i,d}X'_{j,d+h_{ij}} \\ &= 0, \end{aligned}$$

where we derive the final equality from the disjointness constraints (i.e. that $X_{i,a}X_{i,b} = 0$ whenever $a \neq b$), as for the above term to be nonzero we require $a = c$, $d = a+s$, $b = d-t+h_{ij} = a+s-t+h_{ij}$, and also $b = c + h_{ij}$, which implies $a + s - t = c$, a contradiction since $t \neq s$. This establishes the final property. □

2.4.3 Shift-Symmetry

Next, we define the notion of shift-symmetric functions and pseudodistributions and establish some properties that they satisfies.

Definition 2.11 (Shift-Symmetry)**.** *We say that a pseudodistribution μ is shift symmetric if for any monomial $\prod_{i=1}^{D} X_{u_i,a_i}$ and $s \in \Sigma$ it holds that*

$$\widetilde{\mathop{\mathbb{E}}_{\mu}}\left[\prod_{i=1}^{D} X_{u_i,a_i}\right] = \widetilde{\mathop{\mathbb{E}}_{\mu}}\left[\prod_{i=1}^{D} X_{u_i,a_i+s}\right].$$

We say that a polynomial $p(X_{u_1,\sigma_1},\ldots)$ equivalently also thought of as a function $f:\Sigma^n\to\mathbb{R}$ is shift-symmetric if $f(X)=f(X+s)$ for all $s\in\Sigma$.

Given a pseudodistribution obtained by the SoS relaxation of sum of squares of an Affine Unique-Game, we are able to transform it into a shift-symmetric pseudodistributions with the same value by defining a new pseudodistribution μ^{sym} as follows

$$\widetilde{E}_{\mu^{\mathrm{sym}}}\left[\prod_{i=1}^{D}X_{u_i,a_i}\right]=\frac{1}{|\Sigma|}\sum_{s\in\Sigma}\widetilde{\mathbb{E}}_{\mu}\left[\prod_{i=1}^{D}X_{u_i,a_i+s}\right].$$

It is easy to check that $\widetilde{\mathbb{E}}_{\mu^{\mathrm{sym}}}$ is a valid pseudoexpectation, and that it's value is the same; indeed, this follows since if $(a,b)\in\Sigma$ satisfies a constraint in an affine Unique-Games instance, then $(a+s,b+s)$ also satisfy that constraint for any $s\in\Sigma$. More generally one can check that $\widetilde{\mathbb{E}}_{\mu^{\mathrm{sym}}}[p(X)]=\widetilde{\mathbb{E}}_{\mu}[p(X)]$ for all shift-symmetric polynomials p. Thus, we will assume henceforth that our pseudoexpectation and the pseudodistribution can be made shift symmetric without losing the value and in general preserving the pseudoexpectation of shift-symmetric polynomials.

2.5 Information Theory

We will use $\mu|_R$ to denote the marginal distribution of a random variable $R\sim\mu$. We use $TV(A,B)$ to denote the total-variation distance between two distributions A,B.

Definition 2.12 (Mutual Information)**.** *Given a distribution μ over (X,Y), the mutual information between X,Y is defined as:*

$$I_\mu(X;Y)=D_{KL}(\mu\;||\;\mu|_X\times\mu|_Y),$$

where D_{KL} is the Kullback-Leibler divergence. The conditional mutual information between (X,Y) with respect to a random variable Z is defined as:

$$I(X;Y|Z)=\mathop{\mathbb{E}}_{z\sim Z}[I_{\mu|Z=z}(X;Y)].$$

Lemma 2.13 (Pinsker's inequality)**.** *Given any two distributions D_1,D_2:*

$$TV(D_1,D_2)\le\sqrt{\frac{1}{2}D_{KL}(D_1||D_2)}.$$

Using this we get that for a distribution μ over (Y,Y'):

$$TV(\mu,\mu_Y\times\mu_{Y'})\le O(\sqrt{I(Y;Y')}).$$

Lemma 2.14 (Data processing inequality)**.** *Let X,Y,A,B be random variables such that $H(A|X)=0$ and $H(B|Y)=0$, i.e. A is fully determined by X and B is fully determined by Y. Then:*

$$I(A;B)\le I(X;Y).$$

3 Proof of Theorem 1.5

In this section we prove Theorem 1.6 in the case that $c = 1 - \varepsilon$ where $\varepsilon > 0$ is small so as to isolate the "conditioning on an event" challenge as explained in the introduction. In the next section, we explain the modifications that are necessary to prove Theorem 1.6 in full generality.

We begin by formally defining the Johnson graph.

Definition 3.1 (Johnson Graph)**.** *For any $\alpha \in (0,1)$ and $n, \ell \in \mathbb{N}$ with $\alpha\ell \in \mathbb{N}$ and $n > \ell$, we define the (n, ℓ, α)-*Johnson graph *$J(n, \ell, \alpha\ell)$ to be the graph whose vertex set is $\binom{[n]}{\ell}$ and where edges are between pairs of vertices $U, V \in \binom{[n]}{\ell}$ if and only if $|U \cap V| = (1-\alpha)\ell$. We will drop the $(n, \ell, \alpha\ell)$ when clear from context, and use $[N]$ to denote the set of vertices of J.*

We will often refer to α as the noise parameter of the graph.

Notation: The $(n, \ell, \alpha\ell)$-Johnson graph contains other Johnson graphs as subgraphs, corresponding to the basic sets from the introduction. Indeed, for any $S \subset [n]$ of size smaller than ℓ we may consider the induced subgraph on all vertices A such that $A \supseteq S$. We will often refer to such subgraphs as $|S|$-restrictions and denote it by $J|_S$. The motivation for this name is that for given a function $F \colon V(J) \to \mathbb{R}$ and S, we can define the restricted function $F|_S \colon V(J|_S) \to \mathbb{R}$ by $F|_S[A] = F[A \cup S]$. We will use the notation $\delta(F|_S)$ to denote $\mathbb{E}_{u \in J|_S}[F(u)]$.

3.1 The Algorithm for Affine Unique Games over the Johnson Graph

In this section, we describe our algorithm for Affine Unique-Games on $J(n, \ell, \alpha)$. Throguhout, when we say $q = \exp(r)$ we mean that there exist universal positive constants c_1, c_2 such that $2^{c_1 r} \leq q \leq 2^{c_2 r}$. Further if we set a parameter q to be $\exp(r)$ we mean that we set $q = 2^{cr}$ for some universal constant c. We also have an error parameter $\nu > 0$ which we determine in the end but should be thought of as small but bounded away from 0.

High level description of the algorithm

1. First, our algorithm solves the degree D sum of squares relaxation of program (1) to find a pseudo-expectation $\widetilde{\mathbb{E}}_\mu$ corresponding to a pseudo-distribution whose value is at least $1 - \varepsilon$.

2. Next, using the manipulations that were described above, we produce a new one corresponding to two independent samples, as in Sections 2.4.2.

3. In the new pseudo-distribution, it holds that the expected objective function for both the samples X and X' is at least $1-\varepsilon$, hence morally the fraction of edges that are simultaneously satisfied by X and X' is $1 - 2\varepsilon$. We argue that in this case, there exists a subcube C corresponding to a restriction of constant size on which we have a large *shift-partition potential* [6]. We formally define this notion soon, but remark for now that it guarantees that a simple rounding procedure as described in the introduction manages to satisfy a constant fraction of the constraints inside the subcube.

[6] The existence of such a subcube is argued via global hypercontractivity, and to ensure this applies for our pseudodistribution we have to make sure that the global hypercontractivity result we used is proved within the sum-of-squares system.

4. We use a subprocedure, that we refer to as SubRound, to find a subcube C with large shift-partition potential and then an $\Omega(1)$-valued assignment for the vertices in the subcube C.

5. We randomize the edges incident on the vertices of C to get a new instance I' (with possibly lower value) and iterate the algorithm again. The goal of this step is that in the next iteration, we will find a subcube C' which only has very small overlap with the vertices we have assigned thus far. Satisfying a large fraction of C' in the same manner as we did for C would then ensure that we have satisfied new edges and made progress. Iterating thus, we manage to satisfy an $\Omega(1)$-fraction of the graph. This kind of iteration was also used in [BBK+21], but at each step they only managed to satisfy an $\Omega(1/\ell^r)$-fraction of the constraints in an r-subcube.

3.1.1 Shift-partition Potential [BBK+21]

Our analysis uses the definition of the shift-partition potential from [BBK+21], but we explain it here for completeness. Recall that given two assignments X and X' to an affine Unique-Games instance I, one may define the shift partition F_s which consists of vertices v for which $X(v)-X'(v) = s$. As explained in the introduction we will heavily use the shift partition, but for technical reasons we only want to work with vertices u on which many of the constraints adjacent to them are satisfied. For an assignment X and a vertex $v \in V(G)$, let $\mathsf{val}_v^G(X)$ denote the value of v, i.e. the fraction of the edges of G that are incident on v and are also satisfied by X. We drop the superscript G when clear from context.

Naturally, this means that we want to include v in F_s only if $X(v)-X'(v) = s$ and $\mathsf{val}_v^G(X), \mathsf{val}_v^G(X')$ are at least somewhat large. To stay within the realm of SoS though, we replace these indicator constraints with approximating polynomials as in Theorem 2.7.

Definition 3.2 (Shift-Partition [BBK+21])**.** *Given two affine UG assignments X and X' to G define the following functions $\{F_s : V(G) \to \mathbb{R}\}_{s\in\Sigma}$*

$$F_s^G(u) = \mathbb{I}(X_u - X'_u = s) \cdot p_{\beta,\nu}(\mathrm{val}_u^G(X))p_{\beta,\nu}(\mathrm{val}_u^G(X')), \tag{2}$$

where $p_{\beta,\nu}(x)$ is the degree-$\tilde{O}(1/\nu)$ polynomial which is a polynomial approximation to $\mathbb{I}[x \geq \beta]$ with accuracy parameter ν as in Theorem 2.7. We drop the superscript G in F_s^G when the graph is clear from context.

Equipped with the definition of shift partition, we can now define the shift-partition potential that governs our most basic rounding procedures inside subcubes.

Definition 3.3 (Shift-Partition Potential [BBK+21])**.** *The shift-partition size given two UG assignments X and X' on a graph G is defined as:*

$$\Phi_{\beta,\nu}^G(X, X') = \sum_{s\in\Sigma} \mathop{\mathbb{E}}_{u\in G}[F_s^G(u)]^2.$$

The shift-partition potential for a pseudodistribution $\mathcal{D}$ over (X, X') is defined as:

$$\Phi_{\beta,\nu}^G(\mathcal{D}) = \mathop{\widetilde{\mathbb{E}}}_{(X,X')\sim\mathcal{D}}[\Phi_{\beta,\nu}^G(X, X')].$$

Intuitively, $\mathbb{E}_{u\in G}[F_s^G(u)]$ measures the fractional size of F_s in the shift partition, hence the shift partition potential measures the collision probability of the shift partition. In particular, it can be $\Omega(1)$ if and inly if there is s such that the part F_s in the partition has constant density.

Throughout our arguments, we will consider the shift-partition potential with respect to the whole Johnson graph, as well as with respect to subcubes inside it. When we consider the shift-partition potential with respect to a subcube $J|_a$ we abbreviate it as $\Phi^a_{\beta,\nu}(\mathcal{D})$.

3.1.2 The Main Routine

We now give a formal description of our algorithm, modulo the procedure SubRound which we present later.

Algorithm 3.4 (Unique Games on the Johnson Graph)**.** The input to the algorithm is an instance I of affine Unique-Games over a Johnson graph $J(n,\ell,\alpha)$ such that $\mathsf{val}(I) \geq 1-\varepsilon$. The output of the algorithm is an assignment to $V(J)$.

1. Let $r = \left\lfloor \frac{64\sqrt{\eta}}{\alpha} \right\rfloor$, $\delta := \frac{\varepsilon}{\exp(r)}$, $\gamma = \varepsilon$ and $D = \frac{|\Sigma|^3 \ell^{O(r)}}{\varepsilon^3}$. Fix $\mathcal{A}_I$ to be the set of unique games axioms/ integer program over the instance I (Program (1)).
2. Set $j = 1, I_0 = I, R_0 = \phi$.
3. While $|R_{j-1}| < \frac{\gamma}{2}|V(J)|$ do the following:
   (a) Solve the degree-D SoS SDP relaxation for the integer program $\mathcal{A}_{I_{j-1}}$ and apply the transformation from Sections 2.4.2 to get a pseudo-expectation corresponding to two independent samples. Let $\mu_j \times \mu_j$ be the corresponding pseudo-distribution.
   (b) Find $r' \leq r$ and an r'-restricted subcube C_j with high SubRound-value[7], namely such that SubRound-val$_{\mu_j}(C_j) \geq \delta$.
   (c) Run SubRound (Algorithm 3.8) on C_j with pseudodistribution μ_j to get an assignment f_j to $V(C_j)$.
   (d) Let S_j be a subgraph of C_j induced by the set of vertices that have not been previously assigned by any partial assignment $f_k, k < j$ and assign them using f_j. Set $R_j = R_{j-1} \cup H_j$.
   (e) Choose random affine-constraints on edges incident on $V(R_j)$ and let the new instance by I_j.
   (f) Increment j.
4. Output any assignment to $V(J)$ that agrees with all partial assignments f_j (assigned to S_j) considered above.

3.1.3 The Subroutine SubRound and Condition&Round

Before describing the subroutine SubRound, we present several notations that are necessary for the analysis. For a vertex $u \in V(G)$ we will introduce new auxiliary Boolean variables p_u and p'_u. Morally speaking, p_u are the indicators that many of the constraints adjacent to u are satisfied by the assignment X, but formally we approximate it by polynomials.

[7] This is the value of the assignment returned by the subroutine SubRound in (3.8).

First note that given a valid degree D pseudodistribution $\mathcal{D}$ one can extract valid and consistent local distributions over any D variables from X, for e.g. looking at the marginal of (X_u, X_v) in $\mathcal{D}$ we get a distribution over $\Sigma \times \Sigma$. Similarly for any indicator variables R in $\mathcal{D}$ (where $R_u = R_u^2$ under the axioms of $\mathcal{D}$) one could analogously define local distributions over D variables from X, X', R, R'. Unfortunately the indicator that the value of u is large is not a polynomial. Nevertheless using $\mathcal{D}$ we can extract the following collection of local distributions over X, X', p, p', where p is a new set of variables we introduce and are not present in μ. We denote this collection by $\mathcal{L}(\mathcal{D}, p_{\beta,\nu})$.

Formally, given a pseudodistribution $\mathcal{D}$ of degree $D > \widetilde{O}(1/\nu)$ over (X, X') assignments to $\mathcal{A}_I$ so that the corresponding pseudo-expectation has a high value, we define the local distributions $\mathcal{L}(\mathcal{D}, p_{\beta,\nu})$ over the variables of p, p', X, X' in the following way:

Definition 3.5 (Collection of Local Distributions $\mathcal{L}(\mathcal{D}, p_{\beta,\nu})$)**.** *Let $\mathcal{D}$ be a pseudodistribution of degree $D \geq \tilde{\Omega}(d/\nu)$ over (X, X') assignments to $\mathcal{A}_I$, and let the polynomial $p_{\beta,\nu}$ be from Theorem 2.7 for $\varepsilon = \delta = \nu$ and $\alpha = \beta$. We define joint distributions over collections of d variables from p, p', X, X' as:*

$$\widetilde{\Pr_{\mathcal{L}}}[p_u = 1] := \widetilde{\mathbb{E}}_{\mathcal{D}}[p_{\beta,\nu}(\mathrm{val}_u(X))], \quad \widetilde{\Pr_{\mathcal{L}}}[p_u = 0] = 1 - \widetilde{\mathbb{E}}_{\mathcal{D}}[p_{\beta,\nu}(\mathrm{val}_u(X))], \quad \widetilde{\Pr_{\mathcal{L}}}[X_u = \sigma] = \widetilde{\mathbb{E}}[X_{u,\sigma}],$$

and analogously for X'_u, p'_u. More generally we can extend the above definition to define the probability of conjunction of events in the variables X, X', p, p'. For all subsets $S, T, A, B \subseteq [N]$ with $|S| + |T| + |A| + |B| \leq d$, $b_1, \ldots, b_{|S|}, b'_1, \ldots, b'_{|T|} \in \{0, 1\}$ and $\sigma_1, \ldots, \sigma_{|A|}, \sigma'_1, \ldots, \sigma'_{|B|} \in \Sigma$ define:

$$\widetilde{\Pr_{\mathcal{L}}}\left[\bigcap_{u\in S} p_u = b_u, \bigcap_{v\in T} p'_v = b'_v, \bigcap_{a\in A} X_a = \sigma_a, \bigcap_{b\in B} X'_b = \sigma'_b\right]$$
$$= \widetilde{\mathbb{E}}_{\mathcal{D}}\left[\prod_{u\in S}(1 - b_u + (-1)^{1-b_u} p_{\beta,\nu}(\mathrm{val}_u(X))) \prod_{v\in T}(1 - b_v + (-1)^{1-b_v} p_{\beta,\nu}(\mathrm{val}_v(X'))) \prod_{a\in A} X_{a,\sigma_a} \prod_{b\in B} X'_{b,\sigma_b}\right].$$

Remark 3.6. *Note that since $\mathcal{D}$ has large enough degree and $\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} p_{\beta,\nu}(\mathrm{val}_u(X)) \in [0, 1]$, it is easy to check that the above collection of local distributions form valid probability distributions over d variables at a time, and are consistent with each other.*

Using the variables p, p' we further define auxiliary random variables $Y_{u,v}$ and $Y'_{u,v}$ for $u \neq v \in [N]$ that will be particularly useful in the description and analysis of our algorithm:

$$Y_{u,v} = (X_u, X_v, p_u, p_v), \quad Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v). \tag{3}$$

Additionally, analogous to the notation $Y_{u,v}$, for all $S = \{i_1, \ldots, i_{|S|}\} \subseteq [N]$, define the random variables Y_S, Y'_S as:

$$Y_S = (X_{i_1}, \ldots, X_{i_{|S|}}, p_{i_1}, \ldots, p_{i_{|S|}}), \quad Y'_S = (X'_{i_1}, \ldots, X'_{i_{|S|}}, p'_{i_1}, \ldots, p'_{i_{|S|}}).$$

From Definition 3.5 we know that the event $Y_S = y_S$ naturally corresponds to an SoS polynomial Q_S such that $\widetilde{\Pr}_{\mathcal{L}(\mu\times\mu, p_{\beta,\nu})}[Y_S = y_S] = \widetilde{\mathbb{E}}_\mu[Q_S]$ and similarly for events on Y'_S. Therefore we will use the notation "condition μ on $Y_S = y_S$" to mean reweighting μ on the corresponding polynomial Q_S. We will frequently look at quantities like $\widetilde{\Pr}_{\mathcal{L}}[\mathbb{I}[Y_S = y_S] \wedge \mathbb{I}[Y_T = y_T]]$ and $I(Y_S; Y_T)$ where $|S \cup T| \leq d$. In such places we emphasize that we're thinking of the local distribution on $Y_{S\cup T}$ and computing probabilities or mutual information on this distribution.

The Subroutine Condition&Round. To formally state SubRound, we first need to present the subroutine Condition&Round from [BBK+21] which is similar to the basic rounding procedure from the introduction.

Algorithm 3.7 (Condition&Round [BBK+21])**.**
Input: A degree-D (for $D \geq 2$) shift-symmetric pseudodistribution μ for a UG instance $I' = (G = (V, E), \Pi)$ over alphabet Σ.
Output: Returns an assignment $x \in \Sigma^V$.

Sample a random solution Z:

1. Sample a vertex $u \sim \pi$ and condition on $X_u = 0$ to obtain the new marginals $\widetilde{\mathbb{E}}_\mu[\cdot \mid X_u = 0]$.
2. Sample a solution Z by choosing each collapsed variable's labels independently according to its marginals: $Z_v \sim \widetilde{\mathbb{E}}_\mu[X_v \mid X_u = 0]$.

Derandomized Condition&Round. For future reference, it will be more convenient for us to analyze a derandomized version of Condition&Round, which makes the above procedure deterministic using standard methods such as the method of conditional expectations. We omit the straightforward details, and for us it suffices that this derandomized procedure always achieves value which is at least the expected value of what Condition&Round gives.

We can now describe SubRound that obtains a partial assignment for the Johnson graph.

High level description of SubRound. In this routine, we are given a pseudoexpectation and a subcube $J|_a$ (on which we hope the value is large), and we wish to find a good assignment to the nodes in the given subcube, or quit. This method successfully assigns values to $V(J|_a)$ in the case that the shift-partition potential function on the subcube is large.

Towards this end, we find an appropriate conditioning on the random variables $Y_{u,v}$ so that after performing it, they are nearly pairwise independent (we use mutual information to denote this; this is the step that we referred to in the introduction as eliminating global correlations). Denote the new distribution obtained as $\mu_1 \times \mu_2$.

Simultaneously we find an event (which we state as a polynomial to stay within the realm of SoS) $P_a(X, X')$, which roughly says that the part $\{v \ : \ X(v) - X'(v) = s\}$ is dense inside $J|_a$, such that this event has significant probability and conditioned on it the shift-partition inside $J|_a$ is large, where both these facts are measured with respect to $\mu_1 \times \mu_2$. We give a formal definition of $P_a(X, X')$ in Lemma 3.9.

Finally, we run the procedure Condition&Round using either the symmetrized versions of μ_1 or μ_2 which returns a good assignment to the given subcube provided that the previous steps succeeded.

We now move on to the formal description of SubRound.

Algorithm 3.8 (SubRound: Rounding for a subgraph)**.**
Input: Takes as input an affine Unique-Games instance $I = (G, \Pi, \Sigma)$ over an $(n, \ell, \alpha\ell)$-Johnson graph G and alphabet Σ, a pseudo-distribution μ with $\mathrm{val}_\mu(I) \geq 1 - \eta$, a j-restricted subcube $J|_a$, with $j \leq r = \left\lfloor \frac{32\sqrt{\eta}}{\alpha} \right\rfloor$.
Output: Returns an assignment in $\Sigma^{|V(J|_a)|}$ to the vertices of $J|_a$.

1. Fix $t = \frac{|\Sigma|^3 \ell^{\Theta(r)}}{\eta^3}, \tau = \Theta(\frac{\eta}{|\Sigma|^2 \ell^{2r} \exp(r)}), \beta = \sqrt{\eta}, \nu = \eta \exp(-r)$.

2. Find a polynomial $P_a(X, X')$ from the set of polynomials given by Lemma 3.9, subsets $A, B \subseteq V(J|_a)$ of size at most t and strings y_A, y_B so that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives pseudodistributions μ_1, μ_2 satisfying:

   (a) Shift partition potential inside $J|_a$ is significant:
   $$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}[\Phi^a_{\beta,\nu}(X, X') P_a(X, X')] \geq \exp(-r)\, \widetilde{\mathbb{E}}[P_a(X, X')].$$

   (b) The probability of P_a is significant:
   $$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}[P_a(X, X')] \geq \Omega\left(\frac{\sqrt{\eta}}{|\Sigma| \ell^j \exp(r)}\right).$$

   (c) The random variables $Y_{u,v}$ have small global correlation inside $J|_a$:
   $$\mathbb{E}_{\substack{u_1, v_1 \sim S : u_1 \neq v_1 \\ u_2, v_2 \sim S : u_2 \neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau.$$

   (d) The random variables $Y'_{u,v}$ have small global correlation inside $J|_a$:
   $$\mathbb{E}_{\substack{u_1, v_1 \sim S : u_1 \neq v_1 \\ u_2, v_2 \sim S : u_2 \neq v_2}}[I(Y'_{u_1,v_1}; Y'_{u_2,v_2})] \leq \tau.$$

   Here $S = V(J|_a)$, and the mutual information is taken with respect to the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$. If no such polynomial or conditioning exists then output the all zeros assignment on $V(J|_a)$ and exit.

3. Symmetrize μ_1, μ_2 using the procedure in Section 2.4.3 to obtain $\mu_1^{\text{sym}}, \mu_2^{\text{sym}}$. Perform derandomized Condition&Round on μ_1^{sym} or μ_2^{sym} and output the higher valued assignment obtained for $V(J|_a)$.

3.2 Analysis of Algorithm 3.4

3.2.1 High Level Description of the Analysis

We prove that Algorithm 3.4 returns a solution with value independent of the alphabet size. We begin by explaining the ideas of the analysis [BBK+21] for small-set expanders, and then explain how this analysis is adapted to our case.

Small-set expanders [BBK+21]: Consider the shift-partition potential, and suppose for simplicity of presentation that the p's are replaced with the actual indicator functions. Then the functions F_s cover almost all the vertices of the graph, and any edge that goes across parts in the shift-partition must be violated by either X or X'. Hence if X, X' were $(1-\varepsilon)$-satisfying assignments then the shift-partition is non-expanding, in the sense that at most 2ε-fraction of the edges go across the parts (those that are violated by either X or X').

In the case of $(2\varepsilon, \delta)$-small-set expanders, [BBK+21] then conclude that the fractional-size of one of the parts must be at least $\Omega(\delta)$, and in fact the shift-partition size: $\sum_s \mathbb{E}_{u \sim G}[F_s(u)]^2$, must be $\geq \Omega(\delta)$. On the other hand they prove that if the shift-partition potential is large on $\mu \times \mu$ then Condition&Round on μ succeeds in rounding to an $\Omega_{\varepsilon,\delta}(1)$-assignment, and formally they show:

1. Given a degree D certificate of $(O(\varepsilon), \delta)$-small-set expansion for a graph G and a degree $\Omega(D)$ pseudodistribution μ with value $1-\varepsilon$ on any affine UG instance $I = (G, \Pi)$, conclude that $\Phi^G_{\beta,\nu}(\mu \times \mu) \geq \delta$.

2. Show that if $\Phi^G_{\beta,\nu}(\mu \times \mu)$ is large then Condition&Round outputs an assignment with large value.

Our analysis for Johnson graphs: Let us see how to adapt this analysis to the case of Johnson graphs. Step (1) is far from being true since Johnson graphs are not small-set expanders. But we know that all non-expanding sets in these graphs must have large size when restricted to some subcube, although the set might not be large in the whole graph. We use this property to conclude that the shift-partition potential must be large on a subcube. This part of the analysis turns out to be much more non-trivial than the certifiable-SSE case and is where the bulk of our technical work lies.

Once we have this we use a suitably modified version of step (2) to conclude that Condition&Round when applied to the subcube outputs an $\Omega(1)$-assignment to the subcube. We then use an iteration lemma to run this algorithm multiple times to output an $\Omega(1)$-satisfying assignment to the whole graph.

3.2.2 Lemmas to be Proven Later

In this section, we state a few lemmas that are necessary for the analysis of Algorithm 3.4, whose proofs are deferred to later sections. In Section 3.2.3 we use these lemmas to analyze the performance of Algorithm 3.4.

The analysis of Algorithm 3.4 proceeds by showing that given a pseudodistribution for I with high value, there exists a subcube with high SubRound value, i.e. showing the success of Step 3(a). To do so, we first prove that the structure theorem for Johnson graphs (Theorem 3.17) implies that given a pseudodistribution μ with large value, there exists a subcube $J|_a$ with large shift-partition potential denoted by $\Phi^a_{\beta,\nu}$. Since the Johnson graph is not a small-set expander though, we have the following more subtle statement: There exists an $r = O(\sqrt{\varepsilon}/\alpha)$-restricted subcube a and an event P_a that has large probability, such that if we condition $\mu \times \mu$ on P_a, the induced shift-partition potential is large on the subcube $J|_a$. The event P_a roughly corresponds to the indicator that the size of one of the shift-partition components is large inside the subcube $J|_a$, i.e. $\mathbb{I}(\delta(F_s|_a) > \exp(-r))$ and has probability $\geq \ell^{-O(r)}$.

Lemma 3.9. *There exists a constant $\varepsilon_0 \in (0,1)$, such that for all positive constants $\varepsilon \leq \varepsilon_0$, $\alpha, \tau \leq 1$, $\nu \leq \varepsilon \exp(-r)$, all integers $\ell \geq \Omega(r), n \geq \ell$, where $r = \lfloor \frac{32\sqrt{\varepsilon}}{\alpha} \rfloor$ the following holds.*

Let I be an affine UG instance on $J(n, \ell, \alpha\ell)$ and μ be a pseudodistribution over assignments for I with $\mathrm{val}_\mu(I) \geq 1-\varepsilon$ and degree at least $D = \widetilde{\Omega}\left(\frac{|\Sigma|\ell^r \exp(r)}{\nu\tau\sqrt{\varepsilon}}\right)$. Then there exists a restriction $a \subseteq [n]$ of size $j \leq r$, a degree $\widetilde{O}(1/\nu)$ polynomial $P_a(X, X')$ from a fixed set of $|\Sigma|$ polynomials, subsets $A, B \subseteq V(J|_a)$ of size at most $\widetilde{O}(\frac{|\Sigma|\ell^j \exp(r)}{\tau\sqrt{\varepsilon}})$ and strings y_A, y_B such that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives degree $\widetilde{\Omega}(1/\nu)$ pseudodistributions μ_1 and μ_2 such that:

1. $\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} P_a(X, X') \in [0,1]$.

2. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\Phi^a_{\sqrt{\varepsilon},\nu}(X,X')P_a(X,X')] \geq \exp(-r)\,\widetilde{\mathbb{E}}[P_a(X,X')]$.

3. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a(X,X')] \geq \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j\exp(r)}\right)$.

4. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y_{u_1,v_1};Y_{u_2,v_2})] \leq \tau$.

5. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y'_{u_1,v_1};Y'_{u_2,v_2})] \leq \tau$,

where $S = V(J|_a)$, $Y_{u,v} = (X_u, X_v, p_u, p_v)$, $Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ *and the mutual information is taken with respect to the collection of local distributions* $\mathcal{L}(\mu_1\times\mu_2, p_{\sqrt{\varepsilon},\nu})$.

The next lemma captures the intuition that given a collection of random variables $Z_{u,v}$ that are roughly pairwise independent (that are functions of assignments X and X'), and an event $E(X,X')$ with significant probability, then typically $Z_{u,v} \mid E$ is close to $Z_{u,v}$ in statistical distance. Informally, this asserts that conditioning on the event E does not change the distributions of (X,X') locally.

Lemma 3.10. *For all* $\tau, p, \beta, \nu, \delta \in (0,1)$, *integers* D, N, $m \leq N$ *the following holds: Let* $\mu_1\times\mu_2$ *be a degree* $D + \widetilde{\Omega}(1/\nu)$ *pseudodistribution over* (X,X') *satisfying* $\mathcal{A}_I(X,X')$ *and* $E(X,X')$ *be a polynomial such that,* $\mathcal{A}_I \vdash_D E(X,X') \in [0,1]$ *and* $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[E(X,X')] \geq p$. *Suppose for* $S \subseteq [N]$ *we have that,*

1. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y_{u_1,v_1};Y_{u_2,v_2})] \leq \tau$.

2. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y'_{u_1,v_1};Y'_{u_2,v_2})] \leq \tau$,

where $Y_{u,v} = (X_u, X_v, p_u, p_v), Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ *and the mutual information is with respect to the collection of local distributions* $\mathcal{L}(\mu_1\times\mu_2, p_{\beta,\nu})$ *(Definition 3.5). Then we have that,*

$$\Pr_{u,v\sim S:u\neq v}[TV((Y_{u,v},Y'_{u,v})|E,(Y_{u,v},Y'_{u,v})) \geq \delta] \leq O\left(\frac{\sqrt{\tau}+1/|S|}{p\delta^2}\right),$$

where the distribution $(Y_{u,v},Y'_{u,v})|E$ *refers to the joint distribution on these variables defined by the collection of local distributions* $\mathcal{L}(\mu_1\times\mu_2|E, p_{\beta,\nu})$ *and similarly* $(Y_{u,v},Y'_{u,v})$ *refers to the distribution defined by* $\mathcal{L}(\mu_1\times\mu_2, p_{\beta,\nu})$.

We now restrict our attention to the nice subcube $J|_a$ obtained from Lemma 3.9 and we henceforth only care about the relevant pseudodistributions (e.g. $\mu_1\times\mu_2|P_a$) when marginalized to $J|_a$. For simplicity of notation we still refer to the marginalized pseudodistributions using their original notation.

We show that large shift-potential $\Phi^a(\mu_1\times\mu_2|P_a) \geq \delta$ with the additional property that an average $(Y_{u,v},Y'_{u,v})$-pair when drawn from $\mu_1\times\mu_2|P_a$ is close to its distribution when drawn from $\mu_1\times\mu_2$, implies that Condition&Round (applied on $J|_a$) succeeds on either μ_1 or μ_2.

Lemma 3.11. *Let* $I = (G,\Pi)$ *be an affine instance of Unique Games over the alphabet* Σ *and* $\mu_1\times\mu_2$ *be a degree* $D + \widetilde{O}(1/\nu)$ *pseudodistribution over assignments* (X,X') *to* I. *Let* $E(X,X')$ *be a polynomial such that* $\mathcal{A}_I(X,X') \vdash_D E(X,X') \in [0,1]$. *Suppose we have that:*

$$\Pr_{u,v\sim V(G):u\neq v}[TV((Y_{u,v},Y'_{u,v})|E,(Y_{u,v},Y'_{u,v})) > \delta] \leq \zeta,$$

where the distribution $(Y_{u,v}, Y'_{u,v})|E$ *refers to the joint distribution on these variables defined by the collection of local distributions* $\mathcal{L}(\mu_1 \times \mu_2|E, p_{\beta,\nu})$ *and similarly* $(Y_{u,v}, Y'_{u,v})$ *refers to the distribution defined by* $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ *(Definition 3.5).*

If $\Phi^G_{\beta,\nu}(\mu_1 \times \mu_2|E(X,X')) \geq \gamma$*, then on at least one of the pseudodistributions* μ_1^{sym} *or* μ_2^{sym} *Algorithm 3.7 runs in time* $\mathrm{poly}(|V(G)|)$ *and returns an assignment of expected value at least*

$$(\beta-\nu)^2\left(\gamma - O(\delta+\zeta) - \frac{1}{|V(G)|}\right) - 3\nu(\beta-\nu)$$

for I*.*

Using the lemmas above for $G = J|_a$ one can analyze a single iteration of SubRound and show that it manages to assign a subcube of I and within it satisfy a constant fraction of the edges. This subcube might be of size $o(1)$ of the whole graph though, therefore to complete the analysis of our algorithm we need the following lemma, asserting that we can iterate this procedure (as done in Algorithm 3.4) to satisfy a constant fraction of constraints of I.

Lemma 3.12. *Let* $c, \gamma, \delta \in (0,1]$ *and* $r \in \mathbb{N}$*. Let* I *be an affine UG instance on alphabet* Σ *on* $J_{n,\ell,\alpha}$ *with* ℓ, n *large enough and value at least* c*. Suppose we have a subroutine* $\mathcal{A}$ *which given as input any affine UG instance* I' *on* $J_{n,\ell,\alpha}$ *with* $\mathrm{val}_\mu(I') \geq c - \gamma$*, returns an* $\leq r$*-restricted subcube* H *on* J *and a partial assignment* f *such that,* $val_f(H) \geq \delta$*. Then if* $\mathcal{A}$ *runs in time* $T(\mathcal{A})$*, there is a* $O(|V(J)|T(\mathcal{A}) + |V(J)|^3)$*-time algorithm which finds a solution for* I *that satisfies an* $\Omega(\delta\gamma(1-\alpha)^r)$*-fraction of the edges of* J*.*

Proof. Deferred to Section A. □

3.2.3 The Analysis of Algorithm 3.4

Theorem 3.13. *There exists a constant* $\varepsilon_0 \in (0,1)$*, such that for all positive constants* $\varepsilon \leq \varepsilon_0$*,* $\alpha \in \mathbb{Q}$ *with* $\alpha \leq 1/2$*, all integers* $\ell \geq \Omega(r)$ *with* $\alpha\ell \in \mathbb{N}$ *and* $r = \Theta(\sqrt{\varepsilon}/\alpha)$ *and* n *large enough, Algorithm 3.4 has the following guarantee: If* I *is an instance of affine Unique Games on the* (n,ℓ,α)*-Johnson graph* G *with alphabet* Σ *and* $\mathrm{val}(I) = 1-\varepsilon$*, then in time* $|V(J)|^{\mathrm{poly}(\ell^r, |\Sigma|, 1/\varepsilon)}$ *Algorithm 3.4 returns an* $\frac{\mathrm{poly}(\varepsilon)}{\exp(r)}$*-satisfying assignment for* I*.*

Proof. Let $r = \lfloor 32\varepsilon/\alpha \rfloor$. Given the $\leq r$-restricted subcube $a \subseteq [n], P_a(X,X')$ and $\mu_1 \times \mu_2$ from Lemma 3.9 we have that $\Phi^a_{\sqrt{\varepsilon},\nu}(\mu_1 \times \mu_2|P_a) \geq \exp(-r)$, therefore we apply Lemma 3.10 with $S = J|_a$, the polynomial $E(X,X') = P_a(X,X')$ and the parameters $\beta = \sqrt{\varepsilon}, \nu = \exp(-r)\varepsilon, \delta = \exp(-r), p = \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^r \exp(r)}\right)$ and $\tau = \Theta(\frac{\varepsilon}{|\Sigma|^2\ell^{2r}\exp(r)})$. The parameter τ has been chosen so that $O(\frac{\sqrt{\tau}+1/|S|}{p\delta^2}) = \exp(-r)$ and therefore we can apply the rounding lemma (Lemma 3.11) with $\zeta = \exp(-r)$ and $\gamma = \exp(-r)$ and the same settings of β, ν, δ. This shows that there exists a subcube $a \in \binom{[n]}{j}$ with SubRound value that is at least $\Omega(\varepsilon/\exp(r))$ if $\deg(\mu) \geq \widetilde{\Omega}\left(\frac{|\Sigma|\ell^r\exp(r)}{\nu\tau\sqrt{\varepsilon}}\right)$. Hence it suffices to have degree of μ equal to $\frac{|\Sigma|^3\ell^{O(r)}}{\varepsilon^3}$.

By using SubRound as a subroutine, we finish the proof of this theorem by applying the iteration Lemma 3.12 with $c = 1-\varepsilon$, $\gamma = \varepsilon$, $\delta = \Omega(\varepsilon/\exp(r))$ and $r = \lfloor 64\sqrt{\varepsilon}/\alpha \rfloor$ so that $(1-\alpha)^r = O(\sqrt{\varepsilon})$ (Claim 3.21). We can check that SubRound satisfies the hypotheses of the lemma: it finds a $\leq r$-restricted subcube and an assignment to it with value $\Omega(\delta)$ in time $|V(J)|^{\frac{|\Sigma|^3\ell^{O(r)}}{\varepsilon^3}}$ (this follows from

the degree upper bound on μ). This gives us that Algorithm 3.4 outputs an assignment of value at least $\Omega(\varepsilon^3/\exp(r))$ in time $|V(G)|^{\frac{|\Sigma|^3 \ell^{O(r)}}{\varepsilon^3}}$. □

3.3 Proof of Lemma 3.9: Finding Subcube with Large Shift-partition

In this section we use the structure theorem for Johnson graphs to prove the existence of a restriction $a \subseteq [n]$ of constant size (corresponding to the induced subgraph $J|_a$) with large induced shift-partition potential when conditioned on a non-negative polynomial P_a, i.e. $\Phi^a_{\beta,\nu}(\mu \times \mu | P_a)$ is large. Recall that to analyze our final Condition&Round step though we need to show conditioning $\mu \times \mu$ on P_a does not introduce too many correlations between X, X'. Therefore we first perform a global correlation reduction procedure on $\mu \times \mu$ to get the product pseudodistribution $\mu_1 \times \mu_2$ while preserving the property that $\Phi^a_{\beta,\nu}(\mu_1 \times \mu_2 | P_a)$ is large. The exact quantity that aids our rounding analysis is:

$$I(Y_{u_1,v_1}; Y_{u_2,v_2}) = I(X_{u_1}, X_{v_1}, p_{u_1}, p_{v_1}; X_{u_2}, X_{v_2}, p_{u_2}, p_{v_2}),$$

defined according to the collection of local distributions in Definition 3.5. We upper bound this by using the global correlation reduction procedure of Raghavendra-Tan [RT12].

3.3.1 Restricting the Shift Partition to a Subgraph

To prove Lemma 3.9 we will need the notion of *global shift-partition potential restricted to subcube* $J|_a$, which is almost the same as shift-partition potential on $J|_a$ except for one key difference:

Definition 3.14 (Global shift-potential restricted to Subgraphs [BBK+21])**.** *For any $\nu, \beta \in (0,1)$ and subgraph H of G, define the* global shift-partition size restricted to the subgraph H *to be the quantity:*

$$\Phi_{\beta,\nu}(X, X')|_H = \sum_{s \in \Sigma} \mathop{\mathbb{E}}_{u \in H} [F^G_s(u)]^2,$$

where F_s's are the functions defining the shift-partition (Definition 3.3). Let the global shift-partition potential with respect to a pseudodistribution $\mathcal{D}$ over pairs of assignments (X, X') to G be:

$$\Phi_{\beta,\nu}(\mathcal{D})|_H = \mathop{\widetilde{\mathbb{E}}}_{(X,X') \sim \mathcal{D}} [\Phi_{\beta,\nu}(X, X')|_H].$$

Note the difference between the global potential and the shift-partition potential on H: the global shift-partition potential $\Phi_{\beta,\nu}(X, X')|_H$ measures the size of the global partition inside H, i.e. $\mathrm{val}^G_u(X)$ is a function of all the edges of the graph that are incident on u, not just the edges in the subgraph H. The potential $\Phi^H_{\beta,\nu}(X, X')$, on the other hand, measures the value of a vertex u only *inside* the subgraph H.

3.3.2 Global Hypercontractivity

We will need the following definition to describe the structure theorem for Johnson graphs:

Definition 3.15 (Restrictions of Functions)**.** *For the (n, ℓ, α)-Johnson graph J, given a function $F : V(J) \to \mathbb{R}$ and a set $a \subseteq [n]$ with $a = r$, such that $0 \le r \le \ell - 1$, we define the restricted function $F|_a : \binom{[n] \setminus a}{\ell - r} \to \mathbb{R}$ as,*

$$F|_a(X) = F(a \cup X).$$

Further, let $\delta(F|_a)$ denote the fractional size of the function restricted to the subcube $J|_a$, that is,

$$\delta(F|_a) := \mathop{\mathbb{E}}_{X \sim \binom{[n]\setminus a}{\ell - r}} [F|_a(X)].$$

When $a = \phi$ and $r = 0$, we have that $F|_a(X) = F(X)$ for all $X \in \binom{[n]}{\ell}$ and $\delta(F|_a) = \delta(F) = \mathbb{E}[F]$.

In [KMMS18] it is shown that pseudorandom sets expand. Formally, we define pseudorandom sets and in general pseudorandom functions as follows:

Definition 3.16 (Pseudorandom functions)**.** *A set S is called (r, γ)-pseudorandom if for all $\leq r$-restrictions a, $\delta(S|_a) \leq \gamma$. Similarly a function F bounded in $[0, 1]$ is called (r, γ)-pseudorandom if for all $\leq r$-restrictions a, $\delta(F^2|_a) \leq \gamma$.*

We will need a version of this result for general Boolean and pseudorandom functions F, and moreover to show that it is proved in the SoS proof system of constant degree. We get an SoS proof that shows that if $\delta(F^2|_a) \leq \gamma$ for all a of size $\leq r$ and F is Boolean, then $\langle F, LF \rangle$ is large, with $q_a(F)$ being the SoS multipliers of the axioms $\{\gamma - \delta(F^2|_a)\}_a$ and $(\Pi_{\geq \lambda_r} F)(X)$ being the multiplier for the axioms $\{F(X)^3 = F(X)\}$. We have the following statement:

Theorem 3.17 (Expansion Theorem for Johnson Graphs)**.** *For all $\alpha \in (0, 1)$, all integers $\ell \geq 1/\alpha$ and $n \geq \ell$, the following holds: Let $J(n, \ell, \alpha\ell)$ be the α-noisy Johnson graph. For every constant $\gamma \in (0, 1)$ and positive integer $r \leq O(\ell)$, every function $F : V(J) \to \mathbb{R}$ that is (r, γ)-pseudorandom has high expansion:*

$$\{F(X) \in [0, 1]\}_{X \in V(J)} \vdash_{O(1)}$$

$$\langle F, LF \rangle \geq \delta(F)(1 - (1 - \alpha)^{r+1})(1 - \gamma^{1/3} \exp(r)) - \sum_{j=0}^{r} \frac{c_j \ell^j}{\gamma} \mathop{\mathbb{E}}_{a \sim \binom{[n]}{j}} [q_a(F)(\delta(F^2|_a) - \gamma)] + B(F) \tag{4}$$

$$\{F(X) \in [0, 1]\} \quad \vdash_2 \quad 0 \leq q_a(F) \leq 1 \tag{5}$$

where for all $j \leq r$, c_j's are positive constants of size at most $\exp(r)$, for all size j subsets $a \subseteq [n]$, $q_a(F)$ are degree 2 polynomials and $B(F) = \frac{4}{3} \mathbb{E}_X[(F^3 - F)\Pi_{\geq \lambda_r} F]$ for $\lambda_r = (1 - \alpha)^r$ and $\Pi_{\geq \lambda}$ denoting the projection operator to the top-eigenspace of J of eigenvalues $\geq \lambda$.

Proof. Deferred to Section C. □

One can equivalently view the expansion result in [KMMS18] as asserting that if F is a Boolean function which is the indicator of a set $\mathcal{F} \subseteq \binom{[n]}{\ell}$ whose edge expansion is bounded away from 1, then $\delta(F|_a) \geq \Omega(1)$ for some r-restricted subcube a with constant r. In fact, if the expansion of F is at most $1 - \varepsilon$ on $J_{n,\ell,\alpha}$, then $\delta(F|_a) \geq \exp(-r)$ for $r = O(\varepsilon/\alpha)$. Similarly, we can use the above SoS statement to conclude that even if F is "almost-Boolean" and non-expanding, then $q_a(F)(\delta(F^2|_a) - \gamma) > 0$ for some $O(1)$-restriction a. In the context of a pseudodistribution μ over non-expanding sets F, one can conclude that conditioning μ on $q_a(F)$ results in a new pseudodistribution μ' where $\delta(F^2|_a) > \gamma$ on average, and therefore $q_a(F)$ roughly corresponds to the "event" that F is dense on $J|_a$.

We use the observation that the shift-partition defined with respect to $(1 - \varepsilon)$-satisfying assignments X and X' has expansion at most 2ε, therefore is non-expanding. Therefore as above, using

the structure theorem we conclude that at least one of the sets in the shift-partition is not (r,γ)-pseudorandom for $r = O(\varepsilon/\alpha)$ and $\gamma = \exp(-r)$, i.e. $\delta(F_s|_a) \geq \gamma$ for some $s \in \Sigma$ and $\leq r$-restriction a. Let $\mathrm{viol}(X)$ denote the fraction of edges that an assignment X violates in the instance I. We frame this fact in SoS using the following lemma:

Lemma 3.18. *Under the conditions of Lemma 3.9, for all $\eta > 0$ and $r \leq \ell$ we get:*

$$\mathcal{A}_I \vdash_{\tilde{O}(1/\nu)} \tag{6}$$

$$\sum_{j=0}^{r} c_j \exp(r)\ell^j \mathop{\mathbb{E}}_{a\in\binom{[n]}{j}}[\sum_s q_a(F_s)(\delta(F_s|_a) - \exp(-r))]$$

$$\geq \frac{1-\lambda_{r+1}}{2} - (\mathrm{viol}(X) + \mathrm{viol}(X'))\left(1 + \frac{3(1-\lambda_{r+1}) + 12 + 8\eta}{6(1-\beta-\nu)}\right) - \frac{2}{3\eta} - \nu\left(5 - \lambda_{r+1} + \frac{8\eta}{3}\right), \tag{7}$$

where $\lambda_{r+1} = (1-\alpha)^{r+1}$, q_a are the polynomials in Theorem 3.17, $\mathcal{A}_I$ is the set of axioms defined for I by program (1).

Proof. This proof proceeds exactly as the proof of Lemma 4.4 for certifiable small-set expanders in [BBK+21]: we apply Theorem 3.17 to each function F_s and sum up over s. Doing so, we get the following inequality:

$$\sum_s \langle F_s, LF_s\rangle \geq (1-\lambda_{r+1})(1-\gamma^{1/3}\exp(r))\sum_s \delta(F_s) - \sum_{j=0}^{r}\frac{c_j\ell^j}{\gamma}\mathop{\mathbb{E}}_{a\in[n]^j}[\sum_s q_a(F_s)(\delta(F_s^2|_a) - \gamma)]$$

$$+ \frac{4}{3}\sum_s \mathbb{E}[(F_s^3 - F_s)\Pi_{\geq\lambda_r}F_s] \tag{8}$$

We now set $\gamma = \exp(-r)$ such that $(1-\gamma^{1/3}\exp(r)) = 1/2$. Further we bound each of the terms $\sum_s\langle F_s, LF_s\rangle, \sum_s \delta(F_s)$ and the Booleanity error: $\frac{4}{3}\sum_s \mathbb{E}[(F_s^3 - F_s)\Pi_{\geq\lambda_r}F_s]$.

First note that the fraction of low-valued vertices is small and in particular is at most $\frac{\mathrm{viol}(X)+\mathrm{viol}(X')}{1-\beta}$. Since the functions cover all of the high-valued vertices we get $\sum\delta(F_s) \gtrsim 1 - \frac{\mathrm{viol}(X)+\mathrm{viol}(X')}{1-\beta}$. Next, $\sum_s\langle F_s, LF_s\rangle$ counts the fraction of edges crossing the shift-partition. Every such edge must be violated by either X or X' or must be incident on a vertex with low value, therefore we get: $\sum_s\langle F_s, LF_s\rangle \lesssim \mathrm{viol}(X) + \mathrm{viol}(X') + \frac{2\mathrm{viol}(X)+2\mathrm{viol}(X')}{1-\beta}$. We have put approximate inequalities here since there are some error terms generated because F_s's are not exact indicator functions. The Booleanity term $B(F)$ in Theorem 3.17 is also small (after summing up) because F_s's are approximate-indicators (note that it is 0 for 0/1 functions since $F^3 = F$)[8]. These statements have been made formal in the claims from Section 4.2 of [BBK+21], specifically Claim 4.2, 4.7 and 4.9 therein. They are simple to prove given the properties of the approximate-indicator polynomial $p_{\beta,\nu}$ hence we omit them here.

Using the above claims to bound each sum, plugging in the bounds in (8) and rearranging we get the desired inequality.

□

[8]Note that if the functions F_s were defined using $\mathbb{I}[\mathrm{val}_u(X) \geq \beta]$ then we would obtain the lemma statement without the terms involving η and with $\nu = 0$

3.3.3 Reducing Global Correlation

The last ingredient we need for the proof of Lemma 3.9 is that given a pseudo-expectation of sufficiently high degree, one can construct different pseudo-expectations (which are conditionals of the initial pseudo-expectation) that have no global correlations. More precisely:

Lemma 3.19. *For all $\tau, p, \beta, \nu \in (0,1)$ and $|\Sigma|, D, D' \in \mathbb{Z}$ such that $D' \geq \max(D, \widetilde{\Omega}(1/\nu))$ the following holds.*

Suppose there is a degree $D' + \widetilde{\Omega}(\frac{\log|\Sigma|}{p\nu\tau})$ pseudodistribution μ over UG assignments that satisfies $\mathcal{A}_I$, and a polynomial $E(X, X')$ satisfying $\widetilde{\mathbb{E}}_{(X,X')\sim\mu\times\mu}[E(X,X')] \geq p$ and $\mathcal{A}_I \vdash_D E(X,X') \leq 1$. Then for all subsets $S \subseteq [N]$, there exist subsets $A, B \subseteq S$ of size at most $O(\frac{\log|\Sigma|}{p\tau})$ and strings y_A, y_B such that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives pseudodistributions μ_1 and μ_2 of degree at least D' such that:

1. $\widetilde{\mathbb{E}}_{X,X'\sim\mu_1\times\mu_2}[E(X,X')] \geq \frac{p}{2}$.
2. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau$.
3. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y'_{u_1,v_1}; Y'_{u_2,v_2})] \leq \tau$,

where for $A \subseteq [N]$, $Y_A = (X_{i_1}, \ldots, X_{i_{|A|}}, p_{i_1}, \ldots, p_{i_{|A|}})$ (same for Y') and the mutual information is with respect to the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$.

Proof. Deferred to Section 3.4. □

3.3.4 Proof of Lemma 3.9

We combine the lemmas stated in the previous section and set parameters to complete the proof of Lemma 3.9.

Lemma 3.20 (Restatement of Lemma 3.9)**.** *There exists a constant $\varepsilon_0 \in (0,1)$, such that for all positive constants $\varepsilon \leq \varepsilon_0$, $\alpha, \tau \leq 1$, $\nu \leq \varepsilon\exp(-r)$, all integers $\ell \geq \Omega(r), n \geq \ell$, where $r = \lfloor \frac{32\sqrt{\varepsilon}}{\alpha} \rfloor$ the following holds: Let I be an affine UG instance on $J_{n,\ell,\alpha}$ and μ be a pseudodistribution over assignments for I with $\mathrm{val}_\mu(I) \geq 1-\varepsilon$ and degree at least $\widetilde{\Omega}\left(\frac{|\Sigma|\ell^r\exp(r)}{\nu\tau\sqrt{\varepsilon}}\right)$. There exists a restriction $a \subseteq [n]$ of size $j \leq r$, a degree $\widetilde{O}(1/\nu)$ polynomial $P_a(X,X')$ in a fixed set of $|\Sigma|$ polynomials, subsets $A, B \subseteq V(J|_a)$ of size at most $\widetilde{O}(\frac{|\Sigma|\ell^j\exp(r)}{\tau\sqrt{\varepsilon}})$ and strings y_A, y_B such that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives degree $\widetilde{\Omega}(1/\nu)$ pseudodistributions μ_1 and μ_2 such that:*

1. $\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} P_a(X,X') \in [0,1]$.
2. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\Phi^a_{\sqrt{\varepsilon},\nu}(X,X')P_a(X,X')] \geq \exp(-r)\,\widetilde{\mathbb{E}}[P_a(X,X')]$.
3. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a(X,X')] \geq \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j\exp(r)}\right)$.
4. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau$.

5. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y'_{u_1,v_1};Y'_{u_2,v_2})] \leq \tau$,

where $S = V(J|_a)$, $Y_{u,v} = (X_u, X_v, p_u, p_v)$, $Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ *and the mutual information is taken with respect to the collection of local distributions* $\mathcal{L}(\mu_1 \times \mu_2, p_{\sqrt{\varepsilon},\nu})$.

Proof. We will apply Lemma 3.18 with the following parameters: $r = \lfloor 32\sqrt{\varepsilon}/\alpha \rfloor$, so that $1 - \lambda_{r+1} \geq 16\sqrt{\varepsilon}$, $\eta = 1/\sqrt{\varepsilon}$ and $\beta = 201\sqrt{\varepsilon}$. Since ε is sufficiently small and $\nu \leq \varepsilon/2$ we get that $1/(1-\beta-\nu) < 2$. By assumption $\widetilde{\mathbb{E}}_\mu[\mathrm{viol}(X)] = \widetilde{\mathbb{E}}_\mu[\mathrm{viol}(X')] \leq \varepsilon$, $\widetilde{\mathbb{E}}_\mu$ has degree $\widetilde{O}(1/\nu)$ and $\widetilde{\mathbb{E}}_\mu$ satisfies $\mathcal{A}_I$, therefore taking the pseudoexpectation of equation 7 with respect to the pseudodistribution $\mu \times \mu$ we get:

$$\sum_{j=0}^{r} c_j \exp(r)\ell^j \mathop{\mathbb{E}}_{a\in\binom{[n]}{j}} \widetilde{\mathbb{E}}_{\mu\times\mu}[\sum_s q_a(F_s)(\delta(F_s|_a) - \exp(-r))] \geq \frac{\sqrt{\varepsilon}}{3}. \tag{9}$$

Since c_j's are smaller than $\exp(r)$, an averaging argument gives us a size $j \leq r$ restriction $a \subseteq [n]$ and an $s \in \Sigma$ such that:

$$\widetilde{\mathbb{E}}_{\mu\times\mu}[q_a(F_s)(\delta(F_s|_a) - \exp(-r))] \geq \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j \exp(r)}\right). \tag{10}$$

Let $P_a(X, X')$ be the polynomial $q_a(F_s(X, X'))$ from above, abbreviated henceforth as P_a. First note that since q_a is a degree $O(1)$-SoS polynomial we immediately get that P_a is a degree $\widetilde{O}(1/\nu)$ SoS polynomial such that:

$$\mathcal{A}_I \quad \vdash_{\widetilde{O}(1/\nu)} F_s(u) \in [0,1] \quad \vdash_{O(1)} \quad q_a(F_s(X,X')) = P_a(X,X') \in [0,1],$$

thus proving the first statement in the theorem. We now reduce the global correlations using Lemma 3.19. Namely, we take μ with S being the set of variables corresponding to the subcube a, the polynomial $E(X, X') = P_a(\delta(F_s|_a) - \exp(-r))$ (for which we know that $\mathcal{A} \vdash_{\widetilde{O}(1/\nu)} E(X, X') \leq 1$), $\beta = \sqrt{\varepsilon}$ and τ, ν as in the lemma statement. Thus, we get a pseudodistribution $\mu_1 \times \mu_2$ over UG assignments to the subcube a, such that:

$$\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a(\delta(F_s|_a) - \exp(-r))] \geq \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j \exp(r)}\right), \tag{11}$$

along with the conditions on the mutual information of μ_1, μ_2. This proves point (4),(5) from the lemma statement.

Establishing point (2) of the lemma. Let us now derive the fact that the global shift-partition potential $\Phi(\mu_1 \times \mu_2)|_a$ is large conditioned on P_a. Rearranging (11) we get that:

$$\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a\delta(F_s|_a)] \geq \exp(-r) \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a] + \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j \exp(r)}\right) \geq \exp(-r) \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a]. \tag{12}$$

Since P_a is an SoS polynomial we can reweight $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\cdot]$ by P_a to get $\widetilde{\mathbb{E}}'$ and apply Cauchy-Schwarz to get:

$$\exp(-r) \leq \widetilde{\mathbb{E}}'[\delta(F_s|_a)] \leq \sqrt{\widetilde{\mathbb{E}}'[\delta(F_s|_a)^2]} = \sqrt{\frac{\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\delta(F_s|_a)^2 P_a]}{\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a]}},$$

rearranging which we get that: $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\delta(F_s|_a)^2 P_a] \geq \exp(-r)\,\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a]$. By definition, the global potential restricted to a is equal to $\sum_{s\in\Sigma}\delta(F_s|_a)^2$. Therefore adding the terms $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\delta(F_{s'}|_a)^2 P_a]$ for $s'\in\Sigma, s'\neq s$ (which are all non-negative) to the LHS of (12) we get:

$$\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[\Phi_{\beta,\nu}(X,X')|_a P_a] \geq \exp(-r)\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[P_a]. \tag{13}$$

We will now relate the global potential to the shift-partition potential on a, where the only difference between the quantities is that in the former the value of a vertex is the fraction of edges satisfied in the whole graph, whereas in the latter it is the value calculated according to only the edges inside the subcube a (in the terms $p_{\beta,\nu}(\mathrm{val}(u))$).

First note that that an r-restricted subcube has bounded expansion when r is not too large: $\phi(J|_a)\leq O(\sqrt{\varepsilon})$ (Claim 3.21). Using this we get that,

$$\Phi^a_{\beta-200\sqrt{\varepsilon},\nu}(X,X') \geq \Phi_{\beta,\nu}(X,X')|_a - 4\nu,$$

which we prove formally in Claim 3.22.

Overall, we get:

$$\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[\Phi^a_{\sqrt{\varepsilon},\nu}(X,X')P_a] \geq \widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[(\Phi_{201\sqrt{\varepsilon},\nu}(X,X')|_a - 4\nu)P_a] \geq \exp(-r)\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[P_a],$$

since $\nu\leq\exp(-r)/8$, which establishes point (2) of the lemma.

Point (3) follows from (11). Indeed, since $\delta(F_s|_a)\leq 1$ and $P_a(X,X')\geq 0$ (both facts are certifiable in degree $\widetilde{O}(1/\nu)$) we get that: $\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} P_a(X,X')(\delta(F_s|_a)-\exp(-r))] \leq (1-\exp(-r))P_a(X,X')$. Taking $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}$ and plugging this into equation (11) we get that:

$$\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[P_a] \geq \Omega\left(\frac{\sqrt{\varepsilon}}{|\Sigma|\ell^j\exp(r)}\right) \qquad \square$$

3.3.5 Auxiliary Claims

We end this section by giving the proofs of Claims 3.21 and 3.22. These were also used in [BBK+21] but we include the proofs here for completeness.

Claim 3.21 (Claim 6.10 [BBK+21])**.** *If $r=\left\lfloor\frac{32\sqrt{\varepsilon}}{\alpha}\right\rfloor<\frac{\ell}{4}$ and $s<r$, an s-restricted subcube of $J(n,\ell,\alpha\ell)$ has expansion at most $200\sqrt{\varepsilon}$.*

Proof. Let $J|_Y$ be an s-restricted subcube. We have that,

$$1-\phi(J|_Y) = \frac{\binom{\ell-|Y|}{\alpha\ell}}{\binom{\ell}{\alpha\ell}} \geq \frac{\binom{\ell-r}{\alpha\ell}}{\binom{\ell}{\alpha\ell}} = \left(\frac{\ell-\alpha\ell}{\ell}\right)\left(\frac{\ell-\alpha\ell-1}{\ell-1}\right)\cdots\left(\frac{\ell-\alpha\ell-r+1}{\ell-r+1}\right).$$

Now since $r\leq\ell/4$ by assumption, each of the parenthesized terms is at least $\left(\frac{3\ell/4-\alpha\ell}{3\ell/4}\right)=(1-4\alpha/3)$, so

$$1-\phi(J|_Y)\geq\left(1-\frac{4\alpha}{3}\right)^r\geq 1-\frac{4r\alpha}{3}.$$

Since $r=\left\lfloor\frac{32\sqrt{\varepsilon}}{\alpha}\right\rfloor<\frac{75\sqrt{\varepsilon}}{\alpha}$, we get that $\phi(J|_Y)<200\sqrt{\varepsilon}$ as desired. $\square$

Claim 3.22 (Claim 6.11 [BBK+21]). *Suppose that C is an r-restricted subcube of $J_{n,\ell,\alpha}$ with $r = \left\lfloor \frac{32\sqrt{\varepsilon}}{\alpha} \right\rfloor$. Then if Φ^C is the shift-partition potential restricted to C, for any $\beta \geq 201\sqrt{\varepsilon}$ and $\nu < \varepsilon$,*

$$\Phi^C_{\beta - 200\sqrt{\varepsilon},\nu}(X, X') \geq \Phi_{\beta,\nu}(X, X')|_C - 4\nu,$$

and furthermore this is certifiable in degree $\widetilde{O}(1/\nu)$ SoS.

Proof. When $r = \left\lfloor \frac{32\sqrt{\varepsilon}}{\alpha} \right\rfloor$, the expansion of C is at most $1 - (1 - 4\alpha/3)^r \leq 200\sqrt{\varepsilon}$ by Claim 3.21. Furthermore, from the definition of the Johnson graph this holds vertex-by-vertex; every $v \in C$ has at most a $200\sqrt{\varepsilon}$-fraction of its neighbors outgoing. Therefore,

$$\mathbb{I}[\mathrm{val}^C_u(X) \geq \beta - 200\sqrt{\varepsilon}] \geq \mathbb{I}[\mathrm{val}_u(X) \geq \beta],$$

and furthermore since $\nu < \varepsilon$,

$$p_{\beta - 200\sqrt{\varepsilon},\nu}(\mathrm{val}^C_u(X)) + \nu \geq p_{\beta,\nu}(\mathrm{val}_u(X)) - \nu.$$

The same holds for $p(\mathrm{val}_u(X'))$. Let $Z_{u,s} = \mathbb{I}[X_u - X'_u = s]$. Therefore, by definition,

$$\begin{aligned}
\Phi_{\beta - 200\sqrt{\varepsilon},\nu}(X|_C, X'|_C) &= \sum_{s \in \Sigma} \mathop{\mathbb{E}}_{u \in C} \left[Z_{u,s} \cdot p_{\beta - 200\sqrt{\varepsilon},\nu}(\mathrm{val}^C_u(X)) \cdot p_{\beta - 200\sqrt{\varepsilon},\nu}(\mathrm{val}^C_u(X')) \right]^2 \\
&\geq \sum_{s \in \Sigma} \mathop{\mathbb{E}}_{u \in C} \left[Z_{u,s} \cdot (p_{\beta,\nu}(\mathrm{val}_u(X)) - 2\nu) \cdot (p_{\beta,\nu}(\mathrm{val}_u(X')) - 2\nu) \right]^2 \\
&\geq \Phi_{\beta,\nu}(X, X')|_C - 4\nu,
\end{aligned}$$

where each inequality is a sum-of-squares inequality of degree at most $2\deg(p)$. □

3.4 Proof of Lemma 3.19: Reducing Global Correlation

We will use the following lemma from [RT12].

Lemma 3.23. *There exists $t \leq r$ such that:*

$$\mathop{\mathbb{E}}_{i_1,\ldots,i_t \sim [M]} \mathop{\mathbb{E}}_{i,j \sim [M]} [I(Y_i; Y_j \mid Y_{i_1}, \ldots, Y_{i_t})] \leq \frac{\log q}{r - 1},$$

where q is the size of the domain of Y_i.

Note that the above lemma holds as long as there is a local collection of distributions over the variables $(Y_1, \ldots, Y_M)$ that are valid probability distributions over all collections of $t + 2$ variables and are consistent with each other. We will apply the above lemma to reduce the global correlation between $(Y_{u_1,v_1}; Y_{u_2,v_2})$ (similarly for Y'), for the local collection of distributions $\mathcal{L}$ (Definition 3.5).

Lemma 3.24 (Lemma 3.19 restated). *For all $\tau, p, \beta, \nu \in (0, 1)$ and $|\Sigma|, D, D' \in \mathbb{Z}$, $D' \geq \max(D, \widetilde{\Omega}(1/\nu))$ the following holds: Suppose there is a degree $D' + \widetilde{\Omega}(\frac{\log|\Sigma|}{p\nu\tau})$ pseudodistribution μ over UG assignments that satisfies $\mathcal{A}_I$, and a polynomial $E(X, X')$ satisfying $\widetilde{\mathbb{E}}_{(X,X') \sim \mu \times \mu}[E(X, X')] \geq p$ and $\mathcal{A}_I \vdash_D E(X, X') \leq 1$. Then for all subsets $S \subseteq [N]$, there exist subsets $A, B \subseteq S$ of size at most $O(\frac{\log|\Sigma|}{p\tau})$ and strings y_A, y_B such that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives pseudodistributions μ_1 and μ_2 of degree at least D' such that:*

1. $\widetilde{\mathbb{E}}_{X,X' \sim \mu_1 \times \mu_2}[E(X,X')] \geq \frac{p}{2}$.
2. $\mathbb{E}_{\substack{u_1,v_1 \sim S: u_1 \neq v_1 \\ u_2,v_2 \sim S: u_2 \neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau$.
3. $\mathbb{E}_{\substack{u_1,v_1 \sim S: u_1 \neq v_1 \\ u_2,v_2 \sim S: u_2 \neq v_2}}[I(Y'_{u_1,v_1}; Y'_{u_2,v_2})] \leq \tau$,

where for $A \subseteq [N]$, $Y_A = (X_{i_1}, \ldots, X_{i_{|A|}}, p_{i_1}, \ldots, p_{i_{|A|}})$ (same for Y') and the mutual information is with respect to the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$.

Proof. Without loss of generality suppose that $S \subseteq [N]$ is $[m]$. Let $\binom{[m]}{r}$ denote the set of r-sized subsets of $[m]$ and m_r denote $\binom{m}{r}$. Let $Y = (Y_{a_1}, \ldots, Y_{a_{m_2}})$ for $a_t = (u,v) \in \binom{[m]}{2}$. For brevity we will use $[m_2]$ for $\binom{[m]}{2}$. Given the pseudodistribution $\mu \times \mu$ over assignments (X, X') we will consider the local collection of distributions $\mathcal{L}(\mu \times \mu, p_{\beta,\nu})$ over the variables of X, X', p, p' and let this also denote the induced collection of distributions over (Y, Y') (Definition 3.5). Recall the notation for random variable Y_A, for any $A \subseteq [m]$: $Y_A = (X_{i_1}, \ldots, X_{i_{|A|}}, p_{i_1}, \ldots, p_{i_{|A|}})$. We will use y_S to denote instantiations of Y_S.

Let $\mathcal{L}$ denote $\mathcal{L}(\mu \times \mu, p_{\beta,\nu})$. Applying Lemma 3.23 to the collection of distributions $\mathcal{L}$ over Y we get that for all $\tau' > 0$, conditioning on $t = O(\frac{\log|\Sigma|}{\tau'})$ variables gives:

$$\mathop{\mathbb{E}}_{i_1,\ldots,i_t \sim [m_2]} [\mathop{\mathbb{E}}_{a,b \sim [m_2]} I(Y_a; Y_b \mid Y_{i_1}, \ldots, Y_{i_t})] \leq \tau'. \tag{14}$$

Let $\mathcal{L}|_R$ denote the distribution that $\mathcal{L}$ induces on the random variable R. We know that the distribution $\mathcal{L}|_{(Y_{i_1},\ldots,Y_{i_t})}$ can be reduced to the distribution $\mathcal{L}|_{Y_{i_1 \cup \ldots \cup i_t}}$ by discarding repeating indices in $i_1, \ldots, i_t$. This is because $\mathcal{L}$ is consistent on the value it assigns to a vertex u when it occurs in $Y_{u,v}$ or $Y_{u,v'}$. We will drop the subscript R from $\mathcal{L}|_R$ when the random variable is clear from context. Expanding the definition of conditional mutual information we therefore get:

$$\begin{aligned} I(Y_a; Y_b \mid Y_{i_1}, \ldots, Y_{i_t}) &= \mathop{\mathbb{E}}_{(y_{i_1},\ldots,y_{i_t}) \sim \mathcal{L}|_{(Y_{i_1},\ldots,Y_{i_t})}} [I(Y_a; Y_b \mid Y_{i_1} = y_{i_1}, \ldots, Y_{i_t} = y_{i_t})] \\ &= \mathop{\mathbb{E}}_{y_{i_1 \cup \ldots \cup i_t} \sim \mathcal{L}|_{Y_{i_1 \cup \ldots \cup i_t}}} [I(Y_a; Y_b \mid Y_{i_1 \cup \ldots \cup i_t} = y_{i_1 \cup \ldots \cup i_t}]. \end{aligned}$$

Plugging the above into equation (14) and applying Markov's inequality we get that for all $\alpha \in (0,1)$:

$$\Pr_{\substack{i_1,\ldots,i_t \sim [m_2] \\ y_{i_1 \cup \ldots \cup i_t} \sim \mathcal{L}}} \left[\mathop{\mathbb{E}}_{a,b \sim [m_2]} [I(Y_a; Y_b \mid Y_{i_1 \cup \ldots \cup i_t} = y_{i_1 \cup \ldots \cup i_t})] \geq \frac{\tau'}{\alpha} \right] \leq \alpha. \tag{15}$$

Repeating the above analysis for Y' we get:

$$\Pr_{\substack{j_1,\ldots,j_t \sim [m_2] \\ y'_{j_1 \cup \ldots \cup j_t} \sim \mathcal{L}}} \left[\mathop{\mathbb{E}}_{a,b \sim [m_2]} [I(Y_a; Y_b \mid Y'_{j_1 \cup \ldots \cup j_t} = y_{j_1 \cup \ldots \cup j_t})] \geq \frac{\tau'}{\alpha} \right] \leq \alpha. \tag{16}$$

Now recall that (Definition 3.5) the probability of the event $(X_u, p_u) = (\sigma, b)$ under $\mathcal{L}$, for some $\sigma \in \Sigma$ and $b \in \{0,1\}$, corresponds to the pseudoexpectation under μ of an appropriate polynomial we will denote by $Q_{u,(\sigma,b)}$:

$$\widetilde{\Pr_{\mathcal{L}}}[(X_u, p_u) = (\sigma, b)] = \widetilde{\mathbb{E}}[X_{u,\sigma}(1 - b + (-1)^{1-b} p(\mathrm{val}_u(X)))]$$

$$:= \widetilde{\mathbb{E}}_{\mu}[Q_{u,(\sigma,b)}]. \tag{17}$$

We can extend the above definition by letting $Q_{S,\cup_{u\in S}(\sigma_u,b_u)}$ denote the polynomial $\prod_{u\in S} Q_{u,(\sigma_u,b_u)}$. For $y_S = (\cup_{u\in S}\sigma_u, \cup_{u\in S} b_u)$ we get that:

$$\widetilde{\Pr}_{\mathcal{L}}[(\cup_{u\in S}X_u, \cup_{u\in S}p_u) = Y_S = y_S] = \widetilde{\mathbb{E}}_{\mu}[Q_{S,\cup_{u\in S}(\sigma_u,b_u)}]. \tag{18}$$

One can define analogous notation for Y' (e.g. the polynomials $Q'_{u,(\sigma,b)}$, etc) and derive the statements above. We know that $\mathcal{A}_I \vdash_{t\ \widetilde{O}(1/\nu)} Q_{S,\cup_{u\in S}(\sigma_u,b_u)} \in [0,1]$. For ease of notation let us define the following expression:

$$\begin{aligned}\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X') \mid Y_S = y_S, Y'_T = y'_T] &:= \widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X') \mid Q_{S,y_S}\cdot Q'_{T,y'_T}] \\ &= \frac{\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X')Q_{S,y_S}Q'_{T,y'_T}]}{\widetilde{\mathbb{E}}_{\mu\times\mu}[Q_{S,y_S}Q'_{T,y'_T}]} \\ &= \frac{\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X')Q_{S,y_S}Q'_{T,y'_T}]}{\widetilde{\Pr}_{\mathcal{L}}[Y_S = y_S, Y'_T = y'_T]},\end{aligned}$$

where the last equality follows from equation (18). Analogous to the definition of conditional expectation we can check that:

$$\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X')] = \mathop{\mathbb{E}}_{\substack{i_1,\ldots,i_t\sim[m]\\ j_1,\ldots,j_t\sim[m]\\ y_{i_1\cup\ldots\cup i_t}, y'_{j_1\cup\ldots\cup j_t}\sim\mathcal{L}}} [\widetilde{\mathbb{E}}_{X,X'}[E(X,X') \mid Y_{i_1\cup\ldots\cup i_t} = y_{i_1\cup\ldots\cup i_t}, Y'_{j_1\cup\ldots\cup j_t} = y'_{j_1\cup\ldots\cup j_t}]].$$

Since $\mathcal{A}_I \vdash_D E(X,X') \le 1$ and $\deg(\mu) \ge D + \widetilde{\Omega}(t/\nu)$ we get that $E(X,X') \le 1$ even after conditioning on a non-negative event Q of degree $\widetilde{O}(t/\nu)$: $\widetilde{\mathbb{E}}[E(X,X') \mid Q] \le 1$. An averaging argument implies that:

$$\Pr_{\substack{i_1,\ldots,j_1,\ldots\sim[m_2]\\ y_{i_1\cup\ldots\cup i_t}, y'_{j_1\cup\ldots\cup j_t}\sim\mathcal{L}}} \left[\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X') \mid Y_{i_1\cup\ldots\cup i_t} = y_{i_1\cup\ldots\cup i_t}, Y'_{j_1\cup\ldots\cup j_t} = y'_{j_1\cup\ldots\cup j_t}] \ge \frac{p}{2}\right] \ge \frac{p}{2}. \tag{19}$$

Choosing $\alpha = p/8$ and $\tau' < p\tau/8$ we can take a union bound over the events in equations (15), (16), (19) to get that there exist sets $i_1,\ldots,i_t,j_1,\ldots,j_t \in [m_2]$ and $y_{i_1\cup\ldots\cup i_t}, y'_{j_1\cup\ldots\cup j_t} \sim \mathcal{L}$ such that,

$$\mathop{\mathbb{E}}_{a,b\sim[m_2]}[I(Y_a; Y_b \mid Y_{i_1\cup\ldots\cup i_t} = y_{i_1\cup\ldots\cup i_t}] \le \frac{\tau'}{\alpha} < \tau.$$

$$\mathop{\mathbb{E}}_{a,b\sim[m_2]}[I(Y'_a; Y'_b \mid Y'_{j_1\cup\ldots\cup j_t} = y'_{j_1\cup\ldots\cup j_t})] \le \frac{\tau'}{\alpha} < \tau.$$

$$\widetilde{\mathbb{E}}_{\mu\times\mu}[E(X,X') \mid Y_{i_1\cup\ldots\cup i_t} = y_{i_1\cup\ldots\cup i_t}, Y'_{j_1\cup\ldots\cup j_t} = y'_{j_1\cup\ldots\cup j_t}] \ge \frac{p}{2}.$$

Let μ_1 be the pseudodistribution on X that we get by conditioning μ on $Q_{i_1\cup\ldots\cup i_t, y_{i_1\cup\ldots\cup i_t}}$ and let μ_2 be the pseudodistribution on X' that we get by conditioning μ on $Q_{j_1\cup\ldots\cup j_t, y'_{j_1\cup\ldots\cup j_t}}$. It is easy to check that:

$$\mathcal{L}(\mu\times\mu, p_{\beta,\nu}) | (Y_{i_1\cup\ldots\cup i_t} = y_{i_1\cup\ldots\cup i_t}, Y'_{j_1\cup\ldots\cup j_t} = y'_{j_1\cup\ldots\cup j_t}) = \mathcal{L}(\mu_1\times\mu_2, p_{\beta,\nu}),$$

thus giving us the three properties we need in the lemma. It remains to check the degree bounds on μ that we require. We have conditioned on $t = O(\log|\Sigma|/p\tau)$ variables from (Y, Y'), each having degree $\widetilde{O}(1/\nu)$ in X, X', therefore it suffices to have degree of μ to be $\widetilde{\Omega}\left(\frac{\log|\Sigma|}{p\tau\nu}\right)$. After conditioning we get that $\deg(\mu_1 \times \mu_2) \geq D'$ if $\deg(\mu) \geq D' + \widetilde{\Omega}\left(\frac{\log|\Sigma|}{p\tau\nu}\right)$ thus completing the proof. □

3.5 Proof of Lemma 3.10: Conditioning Does Not Introduce Correlations

In this section we prove that given a product distribution over (X, X') that satisfies low global correlation with respect to the variables (Y, Y'), where $Y_{u,v} = (X_u, X_v, p_u, p_v)$, and an event $E(X, X')$ that holds with large probability, conditioning on E cannot correlate too many pairs (Y_i, Y'_i). This is a general lemma that holds if the variables (Y, Y') are local functions of the underlying variables (X, X') and satisfy low global correlation, but here we state it for our specific application only.

Lemma 3.25 (Restatement of Lemma 3.10)**.** *For all $\tau, p, \beta, \nu, \delta \in (0,1)$, integers D, N, $m \leq N$ the following holds: Let $\mu_1 \times \mu_2$ be a degree $D + \widetilde{\Omega}(1/\nu)$ pseudodistribution over (X, X') satisfying $\mathcal{A}_I(X, X')$ and $E(X, X')$ be a polynomial such that, $\mathcal{A}_I \vdash_D E(X, X') \in [0,1]$ and $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[E(X, X')] \geq p$. Suppose for $S \subseteq [N]$ we have that,*

1. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau$.
2. $\mathbb{E}_{\substack{u_1,v_1\sim S:u_1\neq v_1\\ u_2,v_2\sim S:u_2\neq v_2}}[I(Y'_{u_1,v_1}; Y'_{u_2,v_2})] \leq \tau$,

where $Y_{u,v} = (X_u, X_v, p_u, p_v), Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ and the mutual information is with respect to the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ (Definition 3.5). Then we have that,

$$\Pr_{u,v\sim S:u\neq v}[TV((Y_{u,v}, Y'_{u,v})|E, (Y_{u,v}, Y'_{u,v})) \geq \delta] \leq O\left(\frac{\sqrt{\tau} + 1/|S|}{p\delta^2}\right),$$

where the distribution $(Y_{u,v}, Y'_{u,v})|E$ refers to the joint distribution on these variables defined by the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2|E, p_{\beta,\nu})$ and similarly $(Y_{u,v}, Y'_{u,v})$ refers to the distribution defined by $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$.

Proof. As in the proof of Lemma 3.19, without loss of generality suppose $S = [m]$, let $[m_2]$ denote 2-sized subsets of m and $Y = (Y_{a_1}, \ldots, Y_{a_M})$ for $a_t = (u, v), (u, v) \in [m_2]$. Given the pseudodistribution $\mathcal{D}$ over assignments (X, X') we will consider the local collection of distributions $\mathcal{L}(\mathcal{D}, p_{\beta,\nu})$ over the variables X, X', p, p' and therefore the induced collection of distributions over $(Y, Y') = (Y_{a_1}, \ldots, Y_{a_M}, Y'_{a_1}, \ldots, Y'_{a_M})$ (Definition 3.5).

Throughout the proof we will consider random variables (Y, Y') drawn from the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ and from the conditioned collection of distributions $\mathcal{L}(\mu_1 \times \mu_2|E, p_{\beta,\nu})$. For brevity of notation we will use $(Y_a, Y'_a)|E$ to be the joint distribution induced by $\mathcal{L}(\mu_1 \times \mu_2|E, p_{\beta,\nu})$ on (Y_a, Y'_a) and use (Y_a, Y'_a) to be the distribution induced by $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$. We will abbreviate $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ to $\mathcal{L}$. Let U be the set of variables $a \in [m_2]$ for which $TV((Y_a, Y'_a)|E, (Y_a, Y'_a)) \geq \delta$ and let the fractional-size of U be γ. If Ω denotes the domain of Y_u, Y'_u then for every $a \in U$ there exists a set $S_a \subseteq \Omega \times \Omega$ such that:

$$\widetilde{\Pr_{\mathcal{L}}}[(Y_a, Y'_a) \in S_a \mid E] - \widetilde{\Pr_{\mathcal{L}}}[(Y_a, Y'_a) \in S_a] \geq \delta. \tag{20}$$

Let e_a denote $\widetilde{\Pr}_{\mathcal{L}}[(Y_a, Y'_a) \in S_a]$. Define the random variables $Z_a = \mathbb{1}((Y_a, Y'_a) \in S_a) - e_a$. Define:

$$Z = \mathop{\mathbb{E}}_{a\sim U}[Z_a] = \mathop{\mathbb{E}}_{a\in U}[\mathbb{1}((Y_a, Y'_a) \in S_a) - e_a].$$

Let $\widetilde{\mathbb{E}}_{\mathcal{L}}$ denote the natural expectation operator corresponding to the local distributions $\mathcal{L}$. One can check that $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z] = \widetilde{\mathbb{E}}_{\mathcal{L}}[Z_a] = 0$, and we now calculate its variance. For two events A, B on the variables Y, Y' let $\widetilde{\mathbb{C}\mathrm{ov}}(A, B)$ denote $\widetilde{\mathbb{E}}_{\mathcal{L}}[AB] - \widetilde{\mathbb{E}}_{\mathcal{L}}[A]\,\widetilde{\mathbb{E}}_{\mathcal{L}}[B]$. Firstly for all $a, b \in [m_2]$ using Pinsker's inequality and data processing inequality we have that,

$$\begin{aligned}
\widetilde{\mathbb{C}\mathrm{ov}}(\mathbb{1}((Y_a, Y'_a) \in S_a), \mathbb{1}((Y_b, Y'_b) \in S_b)) &\le TV((Y_a, Y'_a, Y_b, Y'_b), (Y_a, Y'_a) \times (Y_b, Y'_b)) \\
&\le O(\sqrt{I(Y_a, Y'_a; Y_b, Y'_b)}) \\
&\le O(\sqrt{I(Y_a; Y_b) + I(Y'_a; Y'_b)})
\end{aligned}$$

The proof will proceed by proving upper and lower bounds on $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2]$, where the upper bound uses low global correlation properties of Y and Y' and the lower bound uses the large deviation we have by equation 20.

Upper bound for $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2]$: We have the following upper bound:

$$\begin{aligned}
\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[Z^2] &= \mathop{\mathbb{E}}_{a,b\sim U}[\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[(\mathbb{1}((Y_a, Y'_a) \in S_a) - e_a)(\mathbb{1}((Y_b, Y'_b) \in S_b) - e_b)]] \\
&= \mathop{\mathbb{E}}_{a,b\sim U}[\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[\mathbb{1}((Y_a, Y'_a) \in S_a) - e_a]\,\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[\mathbb{1}((Y_b, Y'_b) \in S_b) - e_b] + \widetilde{\mathbb{C}\mathrm{ov}}(\mathbb{1}((Y_a, Y'_a) \in S_a), \mathbb{1}((Y_b, Y'_b) \in S_b))] \\
&\le \mathop{\mathbb{E}}_{a,b\sim U}[O(\sqrt{I(Y_a; Y_b) + I(Y'_a; Y'_b)})] \\
&\le O(\sqrt{\mathop{\mathbb{E}}_{a,b\sim U}[I(Y_a; Y_b) + I(Y'_a; Y'_b)]}) \\
&\le O\left(\frac{\sqrt{\tau}}{\gamma}\right),
\end{aligned}$$

where the last inequality follows because $\mathbb{E}_{a,b\sim[m_2]}[I(Y_a; Y_b)] \ge \gamma^2\,\mathbb{E}_{a,b\sim U}[I(Y_a; Y_b)]$, and by assumption $\mathbb{E}_{a,b\sim[M]}[I(Y_a; Y_b)] \le \tau$ (similarly for Y').

Lower bound for $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2]$: Let Q_a be the polynomial corresponding to the event $\mathbb{I}[(Y_a, Y'_a) \in S_a]$, i.e. $\widetilde{\Pr}_{\mathcal{L}}[(Y_a, Y'_a) \in S_a] = \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[Q_a]$. Let $P(X, X')$ abbreviated as P denote the polynomial $\mathbb{E}_a[Q_a - e_a]$. Let $Q_{a\cup b}$ be the polynomial such that $\widetilde{\Pr}_{\mathcal{L}}[(Y_a, Y'_a) \in S_a \wedge (Y_b, Y'_b) \in S_b] = \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[Q_{a\cup b}]$. If $a \cap b = \phi$ we have that $Q_{a\cup b} = Q_a \cdot Q_b$, whereas this may not hold if they intersect in one or two variables. But we have that $\Pr_{a,b\sim U}[a \cap b \ne \phi] \le O(1/|U|)$. Using these facts we first show that, $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2] + O(1/|U|) \ge \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P^2]$ via the following two equations:

$$\begin{aligned}
\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[Z^2] &= \mathop{\mathbb{E}}_{a,b\sim U}[\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[(\mathbb{1}((Y_a, Y'_a) \in S_a) - e_a)(\mathbb{1}((Y_b, Y'_b) \in S_b) - e_b)]] \\
&= \Pr_{a,b\sim U}[a \cap b = \phi] \mathop{\mathbb{E}}_{a,b\sim U: a\cap b=\phi}[\widetilde{\mathop{\mathbb{E}}_{\mu_1\times\mu_2}}[(Q_a - e_a)(Q_b - e_b)]] \\
&+ \Pr_{a,b\sim U}[a \cap b \ne \phi] \mathop{\mathbb{E}}_{a,b\sim U: a\cap b\ne\phi}[\widetilde{\mathop{\mathbb{E}}_{\mathcal{L}}}[(\mathbb{1}((Y_a, Y'_a) \in S_a) - e_a)(\mathbb{1}((Y_b, Y'_b) \in S_b) - e_b)]]
\end{aligned}$$

$$\geq \Pr_{a,b\sim U}[a\cap b=\phi] \mathop{\mathbb{E}}_{a,b\sim U:a\cap b=\phi}[\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[(Q_a-e_a)(Q_b-e_b)]] - O\left(\frac{1}{|U|}\right). \tag{21}$$

On the other hand we have that:

$$\begin{aligned}
\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P^2] &= \mathop{\mathbb{E}}_{a,b\sim U}[\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[(Q_a-e_a)(Q_b-e_b)]] \\
&= \Pr_{a,b\sim U}[a\cap b=\phi] \mathop{\mathbb{E}}_{a,b\sim U:a\cap b=\phi}[\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[(Q_a-e_a)(Q_b-e_b)]] \\
&+ \Pr_{a,b\sim U}[a\cap b\neq\phi] \mathop{\mathbb{E}}_{a,b\sim U:a\cap b=\phi}[\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[(Q_a-e_a)(Q_b-e_b)]] \\
&\leq \Pr_{a,b\sim U}[a\cap b=\phi] \mathop{\mathbb{E}}_{a,b\sim U:a\cap b\neq\phi}[\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[(Q_a-e_a)(Q_b-e_b)]] + O\left(\frac{1}{|U|}\right)
\end{aligned} \tag{22}$$

Combining (21) and (22) we get that:

$$\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2] + O\left(\frac{1}{|U|}\right) \geq \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P^2].$$

Recall that $|U|=\gamma|S|$. Since $\mathcal{A}_I \vdash_D E(X,X')\in[0,1]$ we get that:

$$\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2] + O\left(\frac{1}{\gamma|S|}\right) \geq \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P^2] \geq \widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[E]\,\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P^2|E] \geq p\,\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P|E]^2 = p\,\widetilde{\mathbb{E}}_{\mathcal{L}}[Z|E]^2,$$

where the last inequality is by Cauchy-Schwarz on the pseudodistribution $\mu_1\times\mu_2|E$ and the last equality follows by the definition of P and Z. Using equation (20) we know that for all $a\in U$:

$$\widetilde{\mathbb{E}}_{\mathcal{L}}[Z_a \mid E] = \widetilde{\Pr_{\mathcal{L}}}[(Y_a,Y'_a)\in S_a \mid E] - e_a \geq \delta,$$

which implies that $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z\mid E]\geq\delta$.

Combining the upper and lower bounds on $\widetilde{\mathbb{E}}_{\mathcal{L}}[Z^2]$ we get that $\gamma\leq O\left(\frac{\sqrt{\tau}+1/|S|}{p\delta^2}\right)$, completing the proof of the lemma.

□

3.6 Proof of Lemma 3.11: Rounding Subgraphs with Large Shift Potential

In this section, we will show that when the shift-partition potential $\Phi^G_{\beta,\nu}$ (Definition 3.3) is large with respect to a pseudodistribution $\mathcal{D}$ (with certain nice properties), then the Condition&Round Algorithm (Algorithm 3.7) succeeds in returning a good assignment for the unique games instance. The proofs in this section follow along the lines of the analysis of Condition&Round given in [BBK+21], albeit instead of independence between X,X' we only have approximate local independence. Since we will always be working with the graph G we will henceforth drop the superscript G from the shift-partition potential Φ. We will prove the following theorem in this section:

Lemma 3.26 (Restatement of Lemma 3.11)**.** *Let $I=(G,\Pi)$ be an affine instance of Unique Games over the alphabet Σ and $\mu_1\times\mu_2$ be a degree $D+\widetilde{O}(1/\nu)$ pseudodistribution over assignments (X,X') to I. Let $E(X,X')$ be a polynomial such that $\mathcal{A}_I(X,X')\vdash_D E(X,X')\in[0,1]$. Suppose we have that:*

$$\Pr_{u,v\sim V(G):u\neq v}[TV((Y_{u,v},Y'_{u,v})|E,(Y_{u,v},Y'_{u,v}))>\delta]\leq\zeta,$$

where the distribution $(Y_{u,v}, Y'_{u,v})|E$ *refers to the joint distribution on these variables defined by the collection of local distributions* $\mathcal{L}(\mu_1 \times \mu_2|E, p_{\beta,\nu})$ *and similarly* $(Y_{u,v}, Y'_{u,v})$ *refers to the distribution defined by* $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ *(Definition 3.5).*

If $\Phi^G_{\beta,\nu}(\mu_1 \times \mu_2|E(X,X')) \geq \gamma$*, then on at least one of the pseudodistributions* μ_1^{sym} *or* μ_2^{sym} *Algorithm 3.7 runs in time* $\mathrm{poly}(|V(G)|)$ *and returns an assignment of expected value at least*

$$(\beta-\nu)^2\left(\gamma - O(\delta+\zeta) - \frac{1}{|V(G)|}\right) - 3\nu(\beta-\nu)$$

for I*.*

Proof of Lemma 3.26. Following the proof strategy of [BBK+21] we define the alternate shift potential $\Psi(\mu)$:

Definition 3.27 (Alternate shift potential [BBK+21])**.** *The* alternate shift potential *of a degree-*$D \geq 4$ *pseudodistribution* μ *is given by*

$$\Psi(\mu) := \mathop{\mathbb{E}}_{u,v\sim\pi(G)}\left[\sum_{s\in\Sigma} \widetilde{\Pr_{\mu}}[X_v - X_u = s]^2 \cdot \widetilde{\mathbb{E}}[\mathrm{val}_v(X) \mid X_v - X_u = s]\right],$$

where $\pi(G)$ *is the stationary measure on* G *and* $\mathrm{val}_v(X)$ *denotes the value of vertex* v*.*

We will show that if $\Phi(\mu_1 \times \mu_2|E)$ is large and E does not introduce too many correlations, then either $\Psi(\mu_1)$ or $\Psi(\mu_2)$ must be large:

Lemma 3.28. *Let* $\mu_1 \times \mu_2$ *be a degree* $\widetilde{O}(D+1/\nu)$ *pseudodistribution over* (X,X') *and let* $E(X,X')$ *be a polynomial such that* $\mathcal{A}_I \vdash_D E(X,X') \in [0,1]$*. Suppose we have that:*

$$\Pr_{u,v\sim V(G):u\neq v}[TV((Y_{u,v},Y'_{u,v})|E,(Y_{u,v},Y'_{u,v})) > \delta] \leq \zeta.$$

If the shift-partition potential of $\mu_1 \times \mu_2|E(X,X')$ *is large, then the alternate shift-potential of* μ_1 *or* μ_2 *must be large as well:*

$$\Phi_{\beta,\nu}(\mu_1 \times \mu_2|E(X,X')) \leq \frac{\Psi(\mu_1)}{2(\beta-\nu)^2} + \frac{\Psi(\mu_2)}{2(\beta-\nu)^2} + \frac{3\nu}{\beta-\nu} + 2\delta + 2\zeta + \frac{1}{|V(G)|}.$$

Proof. Deferred to Section 3.6.1. □

Given a pseudodistribution μ, we recall that μ^{sym} is the shift symmetrized pseudodistribution as in Section 2.4.3. We note that $\Psi(\mu) = \Psi(\mu^{\mathrm{sym}})$ because Ψ as $\widetilde{\Pr_{\mu^{\mathrm{sym}}}}[X_u - X_v = s] = \widetilde{\Pr_{\mu}}[X_u - X_v = s]$ and $\widetilde{\mathbb{E}}_{\mu^{\mathrm{sym}}}[\mathrm{val}_v(X)|X_v - X_u = s] = \widetilde{\mathbb{E}}_{\mu}[\mathrm{val}_v(X)|X_v - X_u = s]$ for all s. Therefore we can use the following lemma from [BBK+21] that shows that when the alternate shift potential of a shift-symmetric pseudodistribution is large, a single step of conditioning and rounding returns a solution of high objective value:

Lemma 3.29 (Lemma 3.6 [BBK+21])**.** *Let* $I = (G,\Pi)$ *be an affine instance of Unique Games over the alphabet* Σ*, let* $\kappa > 0$ *and let* μ *be a degree-4 shift-symmetric pseudodistribution for* I*. If* $\Psi(\mu) \geq \kappa$*, then Algorithm 3.7 returns a solution of expected value at least* κ*.*

The statement now follows by applying the last lemma on either μ_1^{sym} or μ_2^{sym} (depending which one has a higher Ψ). □

3.6.1 Relating the potentials: Proof of Lemma 3.28

Proof of Lemma 3.28. We begin by recalling that in the definition of $\Phi_{\beta,\eta}$, we used an ν-additive polynomial approximation $p(x)$ of degree $\widetilde{O}(1/\nu)$ to the indicator function $\mathbb{I}[x \geq \beta]$ on the interval $x \in [0,1]$, guaranteed by Theorem 2.7. We will use E to denote the polynomial $E(X, X')$, $\mathcal{D}$ to denote the pseudodistribution $\mu_1 \times \mu_2 | E$ and π to denote the uniform (in general stationary) measure over $V(G)$. Let p_u denote the polynomial $p_{\beta,\nu}(\mathrm{val}_u(X))$ and $p'_u = p_{\beta,\nu}(\mathrm{val}_u(X'))$. Recall also the collection of local distributions $\mathcal{L}(\cdot, p_{\beta,\nu})$ (Definition 3.5). We will overload the notation p_u, p'_u to mean the variables from Definition 3.5 and the polynomial $p_{\beta,\nu}(\mathrm{val}_u(X))$, and the use should be clear from context. Expanding the definition of $\Phi_{\beta,\nu}(\mathcal{D})$ we get:

$$\begin{aligned}
\Phi_{\beta,\nu}(\mathcal{D}) &= \widetilde{\mathbb{E}}_{\mathcal{D}} \left[\sum_{s \in \Sigma} \left(\mathop{\mathbb{E}}_{u \sim \pi} \mathbb{I}[X_u - X'_u = s] p_u p'_u \right)^2 \right] \\
&= \widetilde{\mathbb{E}}_{\mathcal{D}} \left[\sum_{s \in \Sigma} \mathop{\mathbb{E}}_{u,v \sim \pi} \mathbb{I}[X_u - X'_u = X_v - X'_v = s] p_u p_v p'_u p'_v \right] \\
&= \mathop{\mathbb{E}}_{u,v \sim \pi} \widetilde{\mathbb{E}}_{\mathcal{D}} \left[\mathbb{I}[X_u - X_v = X'_u - X'_v] \cdot p_u p_v p'_u p'_v \right], \\
&= \mathop{\mathbb{E}}_{u,v \sim \pi} [\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2} \left[\mathbb{I}[X_u - X_v = X'_u - X'_v] \cdot p_u p_v p'_u p'_v \right] + \mathsf{err}(u,v)], \qquad (23)
\end{aligned}$$

for $\mathsf{err}(u,v)$ defined accordingly for each pair (u,v) (note that we have switched from the pseudo-distribution $\mathcal{D}$ to the pseudo-distribution $\mu_1 \times \mu_2$ in the last transition).

Bounding the error term in (23). For $u = v$, we get that $\mathsf{err}(u,v) = \widetilde{\mathbb{E}}_{\mathcal{D}}[p_u^2 (p'_u)^2] - \widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}[p_u^2 (p'_u)^2] \leq 1$. Let us bound $\mathsf{err}(u,v)$ for $u \neq v$. Let $R(X_u, X_v, X'_u, X'_v, p_u, p_v, p'_u, p'_v)$ be the polynomial $\mathbb{I}[X_u - X_v = X'_u - X'_v] p_u p_v p'_u p'_v$ (p_u denotes the polynomial $p_{\beta,\nu}(\mathrm{val}_u(X))$). Here we will overload the notation of p_u to also denote the indicator variables from Definition 3.5 and let $R(a_1, \ldots, a_4, b_1, \ldots, b_4)$ for $a_1, \ldots, a_4 \in \Sigma$ and $b_1, \ldots, b_4 \in \{0,1\}$ also denote the evaluation of $R(X_u, \ldots, p'_v)$ on the values $a_1, \ldots, b_4$: $\mathbb{I}[a_1 - a_2 = a_3 - a_4] b_1 b_2 b_3 b_4$. Using Definition 3.5 we have that when $u \neq v$: $\widetilde{\mathbb{E}}_{\mathcal{D}}[R] = \widetilde{\Pr}_{\mathcal{L}(\mathcal{D}, p_{\beta,\nu})}[X_u - X_v = X'_u - X'_v, p_u = p_v = p'_u = p'_v = 1]$ and similarly for the distribution $\mu_1 \times \mu_2$ in place of $\mathcal{D}$. Therefore we get that for $u \neq v$:

$$\begin{aligned}
&\mathsf{err}(u,v) \\
&= \widetilde{\mathbb{E}}_{\mathcal{D}}[R(X_u, \ldots, p'_v)] - \widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}[R(X_u, \ldots, p'_v)] \\
&= \widetilde{\Pr}_{\mathcal{L}(\mathcal{D}, p_{\beta,\nu})}[X_u - X_v = X'_u - X'_v, p_u = \ldots = p'_v = 1] - \widetilde{\Pr}_{\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})}[X_u - X_v = X'_u - X'_v, p_u = \ldots = p'_v = 1] \\
&= \sum_{a_1, \ldots, a_4 \in \Sigma} R(a_1, \ldots, a_4, 1, \ldots, 1) \left(\widetilde{\Pr}_{\mathcal{L}(\mathcal{D}, p_{\beta,\nu})}[X_u = a_1, \ldots, p_u = 1, \ldots] - \widetilde{\Pr}_{\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})}[X_u = a_1, \ldots, p_u = 1 \ldots] \right) \\
&\leq \sum_{a_1, \ldots, a_4 \in \Sigma} \left| \widetilde{\Pr}_{\mathcal{L}(\mathcal{D}, p_{\beta,\nu})}[X_u = a_1, \ldots, p_u = 1, \ldots] - \widetilde{\Pr}_{\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})}[X_u = a_1, \ldots, p_u = 1, \ldots] \right| \\
&= 2TV((Y_{u,v}, Y'_{u,v}) | E, (Y_{u,v}, Y'_{u,v})),
\end{aligned}$$

where $Y_{u,v} = (X_u, X_v, p_u, p_v)$, $Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ as in the lemma, the distribution $(Y_{u,v}, Y'_{u,v}) | E$ refers to the joint distribution on these variables defined by the collection of local distributions

$\mathcal{L}(\mu_1 \times \mu_2 | E, p_{\beta,\nu})$ and similarly $(Y_{u,v}, Y'_{u,v})$ refers to the distribution defined by $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$ (Definition 3.5).

Combining, we get that:

$$\begin{aligned}
\mathop{\mathbb{E}}_{u,v\sim\pi}[\mathsf{err}(u,v)] &\leq \mathop{\mathbb{E}}_{u,v\sim\pi:u\neq v}[\mathsf{err}(u,v)] + \Pr_{u,v\sim\pi}[u=v]\,\mathop{\mathbb{E}}_{u}[\mathsf{err}(u,u)] \\
&\leq \mathop{\mathbb{E}}_{u,v\sim\pi:u\neq v}[2TV((Y_{u,v},Y'_{u,v})|E,(Y_{u,v},Y'_{u,v}))] + \frac{1}{|V(G)|} \\
&\leq 2\delta + 2\zeta + \frac{1}{|V(G)|}, \qquad (24)
\end{aligned}$$

by the assumption in the lemma statement.

Bounding the main term in (23). We now upper bound the first term in (23).

$$\begin{aligned}
&\mathop{\mathbb{E}}_{u,v\sim\pi}[\widetilde{\mathop{\mathbb{E}}}_{\mu_1\times\mu_2}[\mathbb{I}[X_u - X_v = X'_u - X'_v]\cdot p_u p_v p'_u p'_v] \\
&= \mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}[\mathbb{I}[X_u - X_v = s]\cdot p_u p_v]\,\widetilde{\mathop{\mathbb{E}}}_{\mu_2}[\mathbb{I}[X'_u - X'_v = s]\cdot p'_u p'_v]\right].
\end{aligned}$$

We next use that $p(x), \mathrm{val}_u(X), \mathrm{val}_u(X') \in [0,1]$ and Fact 2.8 asserting that $p(x) \leq \frac{x}{\beta-\nu} + \nu$ for all $x \in [0,1]$, and furthermore this is SoS-certifiable. Thus, pulling out a factor of p_v, p'_v and applying Fact 2.8 to bound p_u, p'_u we get that the first term on the right hand side of (23) is at most

$$\begin{aligned}
&\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}\left[\mathbb{I}[X_u - X_v = s]\cdot\frac{\mathrm{val}_u(X)}{\beta-\nu}\right]\widetilde{\mathop{\mathbb{E}}}_{\mu_2}\left[\mathbb{I}[X'_u - X'_v = s]\cdot\frac{\mathrm{val}_u(X')}{\beta-\nu}\right]\right] + \frac{3\nu}{\beta-\nu} \\
&\leq \frac{1}{2}\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}\left[\mathbb{I}[X_u - X_v = s]\cdot\frac{\mathrm{val}_u(X)}{\beta-\nu}\right]^2\right] + \frac{1}{2}\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_2}\left[\mathbb{I}[X'_u - X'_v = s]\cdot\frac{\mathrm{val}_u(X')}{\beta-\nu}\right]^2\right] \\
&+ \frac{3\nu}{\beta-\nu},
\end{aligned}$$

where we have used that the AM-GM inequality. The first term may be bounded as

$$\begin{aligned}
&\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}\left[\mathbb{I}[X_u - X_v = s]\cdot\frac{\mathrm{val}_u(X)}{\beta-\nu}\right]^2\right] \\
\leq &\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}\left[\mathbb{I}[X_u - X_v = s]\cdot\frac{1}{\beta-\nu}\right]\widetilde{\mathop{\mathbb{E}}}_{\mu_1}\left[\mathbb{I}[X_u - X_v = s]\frac{\mathrm{val}_u(X)}{\beta-\nu}\right]\right] \\
= &\frac{1}{(\beta-\nu)^2}\mathop{\mathbb{E}}_{u,v\sim\pi}\left[\sum_{s\in\Sigma}\widetilde{\mathop{\mathbb{E}}}_{\mu_1}[\mathbb{I}[X_u - X_v = s]]^2\cdot\widetilde{\mathop{\mathbb{E}}}_{\mu_1}[\mathrm{val}_u(X)\mid X_u - X_v = s]\right] \\
= &\frac{\Psi(\mu_1)}{(\beta-\nu)^2},
\end{aligned}$$

where we have used that $\mathrm{val}_u(X) \leq 1$ and applied the definition of conditional pseudoexpectation, and similarly the second term is upper bounded by $\frac{\Psi(\mu_2)}{(\beta-\nu)^2}$. Plugging this and (24) into (23) finishes the proof. □

4 Unique Games with low completeness

In this section we give an analysis which works for UG instances with arbitrary small completeness (but bounded away from 0). The only step that changes is concluding that there is a subcube with large shift-potential (after conditioning).

Recall that in the proof of the analogous lemma for large completeness (Lemma 3.9) we used Claims 3.21 and 3.22 to conclude that if a subcube has large global shift-partition potential it also has large shift-partition potential. This was possible because all the vertices with high global value will also have high value inside the subcube as the expansion of the subcube is small ($O(\varepsilon)$ if the completeness was $1-\varepsilon$). This fact is no longer true in the low completeness regime, as subcubes now may have expansion close to 1. To circumvent this issue, we no longer go via the intermediate global shift-partition route and directly try to conclude that the shift-partition potential is large on a subcube. Towards this end we need a stronger conclusion of global hypercontractivity which we refer to as an "edge-covering" statement: the subcubes on which some part of the shift partition is large cover nearly all of the the internal edges of the shift-partition.

As we shall see, if the value of our pseudodistribution is originally c, the fraction of internal edges is at least c^2 (in particular c^2-fraction of edges are satisfied by both X and X'), and as we are able to cover almost all internal edges we get that there exists a subcube with large completeness and in fact large shift-partition potential. Given this version of Lemma 3.9, the rest of the analysis remains the same.

Notation: For the purposes of this section we first consider the simpler shift-partition defined by functions $\{G_s : V(G) \to \mathbb{R}[X, X']\}_{s\in\Sigma}$:

$$G_s(u) = \mathbb{1}(X_u - X'_u = s).$$

Note that G_s's cover all the vertices, that is $\sum_s \delta(G_s) = 1$ and satisfy Booleanity: $G_s(u)^2 = G_s(u)$ for all $u \in V(G)$.

Given two assignments X, X', let $\mathrm{val}_u(X \wedge X')$ denote the fraction of edges incident on u that are satisfied by *both* X and X'. Note that $\mathrm{val}_u(X \wedge X')$ is a degree four polynomial in (X, X'):

$$\mathrm{val}_u(X \wedge X') = \mathop{\mathbb{E}}_{v:(u,v)\in E}[(\sum_{\sigma\in\Sigma} X_{u,\sigma}X_{v,\pi_{u,v}(\sigma)})(\sum_{\sigma\in\Sigma} X'_{u,\sigma}X'_{v,\pi_{u,v}(\sigma)})].$$

Let $\mathrm{val}_I(X \wedge X') = \mathbb{E}_u[\mathrm{val}_u(X \wedge X')]$ denote the fraction of edges in the graph that are satisfied by both X and X'. Further for an i-restriction $a \in \binom{[n]}{i}$ and a vertex $u \in J|_a$ let $\mathrm{val}^a_u(X \wedge X')$ denote the fraction of edges incident on u in $J|_a$ that are satisfied by both X and X'. This is also a degree 4 polynomial in (X, X').

Definition 4.1 (Approximate Indicator for Dense Subcubes)**.** *Let $\varepsilon_0, \ldots, \varepsilon_r \in (0, 1)$ to be determined later. For any i-restriction a let $R_{s,a}$ denote the (approximate) indicator that G_s is ε_i-dense in $J|_a$, but not $\varepsilon_{|b|}$-dense inside any subcube $J|_b$ where $b \subsetneq a$. To define this as a polynomial let us first define the approximate indicator $p_{<\beta,\nu}(x)$ as $1 - p_{\beta,\nu}(x)$. One can check that $p_{<\beta,\nu}$ approximates $\mathbb{1}[x < \beta]$ when $x \in [0, 1]$ with similar properties as $p_{\beta,\nu}$ (Theorem 2.7) that approximates $\mathbb{1}[x \geq \beta]$ for $x \in [0, 1]$. Formally using the polynomial approximation for an indicator define*

$$R_{s,a} = p_{\varepsilon_i,\nu}(\delta(G_s|_a)) \prod_{j<i} \prod_{b\subset a:|b|=j} p_{<\varepsilon_j,\nu}(\delta(G_s|_b)).$$

4.1 The Edge Covering Theorem

Our argument will need an upper bound on the number of events $R_{s,a}$ that can occur simultaneously, which roughly speaking asks how many i-restricted subcubes can a given set F be dense on. As stated, there is no good upper bound for this: if F is dense on an $i-1$-restricted subcube then it would be quite dense on many i-restricted subcubes containing it. This is the reason that in the event $R_{s,a}$, we required that the i-restricted subcube is dense but there is no j-restricted subcube ($j < i$) containing it on which our set is still somewhat dense.

Claim 4.2. *If $\varepsilon_0, \ldots, \varepsilon_r$ satisfy that $\varepsilon_{i-1} \leq \varepsilon_i/(2^{i+1}i)$ then we get that:*

$$\vdash_{\ell^{O(\ell^i/\nu)}} \mathop{\mathbb{E}}_{a \sim \binom{[n]}{i}} [R_{s,a}] \leq \frac{4\delta(G_s)}{\varepsilon_i^2 \ell^i} + O(\nu).$$

Proof. We first present an argument when the polynomials $p_{\beta,\nu}$ in $R_{s,a}$ are replaced by indicators. We then give a sketch of how to convert the proof into an SoS proof.

Converting the $p_{\beta,\nu}$'s to indicators, we get that $R_{s,a}$ corresponds to $T_{s,a}$:

$$T_{s,a} = \mathbb{1}(\delta(G_s|_a) \geq \varepsilon_i) \prod_{j<i} \prod_{b \subset a : |b|=j} \mathbb{1}(\delta(G_s|_b) < \varepsilon_j)$$

that is the event that G_s is ε_i-dense in a, but for all subsets b of a, it is at most ε_j-dense. We will show that the fraction of such restrictions a must be small.

We consider the Fourier-analytic function $f_{i,F}$ defined in Section C, Definition C.6 and use its alternative formula from Lemma C.10:

$$f_{i,F}(a) = \sum_{b \subseteq a} (-1)^{i-|b|} \delta(F|_b) \geq \delta(F|_a) - \sum_{b \subsetneq a} \delta(F|_b).$$

Technically we have only defined these functions for the Cayley-version of the Johnson graph $C_{n,\ell,\alpha}$, but one can use the above definition for Johnson graphs and derive the same properties that we use here upto $o(1)$ error terms. We will ignore these $o(1)$-error terms in this proof.

If $T_{s,a} = 1$ we get that, $\delta(F|_a) \geq \varepsilon_i$, but each $\delta(F|_b)$ is at most $\varepsilon_{|b|} < \varepsilon_i/(2^{i+1}i)$ so we get:

$$f_{i,G_s}(a) \geq \varepsilon_i - \sum_{j<i} \binom{i}{j} \varepsilon_j \geq \varepsilon_i - \sum_{j<i} 2^i \varepsilon_j \geq \frac{\varepsilon_i}{2}.$$

This immediately gives:

$$T_{s,a} \leq 1 \leq 4 f_{i,G_s}(a)^2/\varepsilon_i^2, \tag{25}$$

if $T_{s,a} = 1$. If $T_{s,a} = 0$ then the above continues to hold.

On the other hand by the definition of f_i Lemma C.11 gives that,

$$\mathbb{E}[f_{i,G_s}(a)^2] = \frac{W^i[G_s]}{\ell^i} \leq \frac{\mathbb{E}[G_s^2]}{\ell^i}.$$

Combining with equation 25 we get:

$$\mathop{\mathbb{E}}_{a \sim \binom{[n]}{i}} [T_{s,a}] \leq \mathop{\mathbb{E}}_{a} \left[\frac{4 f_{i,G_s}(a)^2}{\varepsilon_i^2} \right] \leq \frac{4\delta(G_s)}{\varepsilon_i^2 \ell^i}.$$

SoS-ing the proof: We sketch an SoS proof for the statement for $R_{s,a}$ with $a \in \binom{[n]}{1}$. The full statement follows analogously.

Let δ_0 denote $\delta(G_s)$ and δ_1 denote $\delta(G_s|_a)$. Analogous to (25) we will first show that:

$$\frac{4(\delta_1-\delta_0)^2}{\varepsilon_1^2} + 2\nu - R_{s,a} > \nu \tag{26}$$

if $\delta_1, \delta_0 \in [0,1]$. This will not be an SoS proof, but after showing this we can use a blackbox theorem to convert it into an SoS proof by the properties of our domain $[0,1]^2$. (26) follows by case analysis.

1. If $R_{s,a} > \nu$ it implies that $p_{\varepsilon_1,\nu}(\delta_1) > \nu$ and $p_{<\varepsilon_0,\nu}(\delta_0) > \nu$. By the definition of the polynomials $p_{\beta,\nu}$ this implies that $\delta_1 > \varepsilon_1$ and $\delta_0 < \varepsilon_0 < \varepsilon_1/2$, which gives that $(\delta_1-\delta_0)^2 \geq \varepsilon_1^2/4$. Since $R_{s,a}$ is always ≤ 1 we get:

$$R_{s,a} \leq 1 \leq \frac{4(\delta_1-\delta_0)^2}{\varepsilon_1^2} < \frac{4(\delta_1-\delta_0)^2}{\varepsilon_1^2} + \nu.$$

2. If $R_{s,a} \leq \nu$ we get:

$$R_{s,a} \leq \frac{4(\delta_1-\delta_0)^2}{\varepsilon_1^2} + \nu.$$

Hence rearranging the above and adding ν on both sides gives (26). We can now apply Theorem 2.6 that shows that positive polynomials bounded away from 0 on the domain $[0,1]^k$ have a bounded degree SoS proof to get:

$$\delta_0, \delta_1 \in [0,1] \vdash_{\exp(1/\nu)} R_{s,a} \leq \frac{4(\delta_1-\delta_0)^2}{\varepsilon_1^2} + 2\nu.$$

One can check that the rest of the proof is already SoS therefore we get:

$$\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} \delta_0, \delta_1 \in [0,1] \vdash_{\exp(1/\nu)} \mathop{\mathbb{E}}_{a \sim \binom{[n]}{1}} [R_{s,a}] \leq \frac{4\delta(G_s)}{\varepsilon_1^2 \ell} + 2\nu.$$

One can do the exact same steps and use the SoS degree bound in Theorem 2.6 to get an SoS proof of the full statement with degree $\ell^{O(\ell^i/\nu)}$. The bound that comes from Theorem 2.6 is exponential in the degree of the polynomial f that we want to prove is SoS, and hence our degree bound is also $\exp(\ell^i/\nu)$. We believe there could be a more direct analysis to get a degree bound of $\mathrm{poly}(1/\nu, \ell^i)$ instead. □

We now move on to stating the edge covering theorem. Below, we inspect the edges that stay inside a part in the shift partition, and show that they can nearly be encapsulated within subcubes on which some part becomes dense.

Theorem 4.3 (Edge-Covering Theorem)**.** *For all constants $r \in \mathbb{N}$, $\varepsilon_0, \ldots, \varepsilon_r \in [0,1]$, where $\varepsilon_r \leq \exp(-r)$ and $\varepsilon_0, \ldots, \varepsilon_r$ satisfy that $\varepsilon_{i-1} \leq \varepsilon_i^5/(2^{6r})$, we get that:*

$$\mathcal{A}_I \vdash_{\ell^{\mathrm{poly}(\ell^r/\nu)}} \mathrm{val}_I(X \wedge X') \leq T_0(X,X') + \ldots + T_r(X,X') + \mathsf{err}, \tag{27}$$

where:

$$T_0(X, X') = \sum_s p_{\varepsilon_0,\nu}(\delta(G_s)) \mathop{\mathbb{E}}_{u\sim G}[G_s(u)\mathrm{val}_u(X \wedge X')],$$

$$T_i(X, X') = \ell^i \mathop{\mathbb{E}}_{a\in\binom{[n]}{i}}[\sum_s R_{s,a} \mathop{\mathbb{E}}_{u\sim J|_a}[G_s(u)\mathrm{val}^a_u(X \wedge X')]]$$

$$\mathsf{err} = 4(1-\alpha)^{r+1} + 2^{6r}\max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4} + |\Sigma|\ell^{O(r)}\sqrt{\nu}.$$

Proof. We will first give a proof of this statement with the definition of $R_{s,a}$ replaced by actual indicator functions in place of the polynomials $p_{\beta,\nu}$. Formally, for an $i \le r$-restriction $a \in \binom{[n]}{i}$ let $R_{s,a}$ be an indicator variable that is 1 if $\delta(G_s|_a) \ge \varepsilon_i$ and $\delta(G_s|_b) \le \varepsilon_{|b|}$ for all $b \subsetneq a$. We will then give a sketch of how to convert this into an SoS proof and work with the true definition of $R_{s,a}$.

For any set S let $\#sat(S)$ denote the number of edges inside S that are satisfied by both X and X'. We first have that:

$$\mathrm{val}_I(X \wedge X') \le \sum_s \frac{\#sat(G_s)}{|E|}, \tag{28}$$

since we know (from Fact 2.10) that any edge that crosses between parts in the shift-partition must be violated by at least one of X or X'. Let us therefore upper bound $\#sat(G_s)$.

Consider the set indicated by G_s for fixed $s \in \Sigma$. Let the set of dense subcubes for G_s be defined as follows:

$$\mathcal{C}_s^0 = \begin{cases} \{J\} & \text{if } \delta(G_s) \ge \varepsilon_0 \\ \phi & \text{otherwise} \end{cases}$$

$$\mathcal{C}_s^{(1)} = \{a \in [n] : R_{s,a} = 1\}$$

$$\vdots$$

$$\mathcal{C}_s^{(r)} = \left\{a \in \binom{[n]}{r} : R_{s,a} = 1\right\}.$$

Let D_s denote the dense part of G_s: $\cup_{i\le r}\cup_{a\in\mathcal{C}_s^{(i)}} G_s \cap J|_a$ and let $H_s = G_s \setminus D_s$ be the pseudorandom part of G_s. It is easy to check that H_s is (r, ε_r)-pseudorandom. The edges inside G_s can be divided into four types of edges: edges that stay inside the dense subcubes, edges that go between two different dense subcubes, edges inside H_s and edges that go between H_s and D_s. Applying this subdivision of edges on $\#sat(G_s)$ we get that:

$$\frac{\#sat(G_s)}{|E(J)|} \le \sum_{i\le r}\sum_{a\in\mathcal{C}_s^{(i)}} \frac{\#sat(G_s\cap J|_a)}{|E(J)|} + \frac{|E(H_s,H_s)|}{|E(J)|} + \frac{|E(H_s,D_s)|}{|E(J)|}$$
$$+ \sum_{i,j\le r}\sum_{h=0}^{\min(i,j)}\sum_{a\in\mathcal{C}_s^{(i)}}\sum_{H\subseteq a,|H|=h}\sum_{\substack{b\in\mathcal{C}_s^{(j)}\\ a\neq b, a\cap b=H}} \frac{|E(J|_a,J|_b)|}{|E(J)|}. \tag{29}$$

We will check that the ith-summand in the first term can be rearranged to give the sth-summand in $T_i(X, X')$. We have that:

$$\sum_{a\in\mathcal{C}_s^{(i)}} \frac{\#sat(G_s\cap J|_a)}{|E(J)|} = \frac{\binom{[n]}{i}|E(J|_a)|}{|E(J)|} \mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}\left[R_{s,a}\frac{\#sat(G_s\cap J|_a)}{|E(J|_a)|}\right]. \tag{30}$$

By a direct computation we get that $\frac{\binom{[n]}{i}|E(J|_a)|}{|E(J)|} \leq \ell^i$. As for the second term above we get that,

$$\frac{\#sat(G_s \cap J|_a)}{|E(J|_a)|} \leq \frac{\sum_{u \in J|_a} \mathbb{I}[u \in G_s] \#sat_a(u)}{|V(J|_a)| d_a} = \mathop{\mathbb{E}}_{u \in J|_a} [\mathbb{I}[u \in G_s] \mathrm{val}_u^a(X \wedge X')],$$

where d_a denotes the degree of a vertex inside the subgraph $J|_a$ and $\#sat_a(u)$ denotes the number of edges in $J|_a$ that are incident on u and are satisfied by both X and X'. So plugging this into (30) and summing up over $i \leq r$ we get that the first term in (29) gives us:

$$\sum_{i \leq r} \sum_{a \in \mathcal{C}_s^{(i)}} \frac{\#sat(G_s \cap J|_a)}{|E(J)|} \leq \sum_{i \leq r} \ell^i \mathop{\mathbb{E}}_{a \sim \binom{[n]}{i}} [R_{s,a} \mathop{\mathbb{E}}_{u \in J|_a} [\mathbb{I}[u \in G_s] \mathrm{val}_u^a(X \wedge X')]]. \tag{31}$$

We will bound the other terms in (29) by some small constant.

Bounding the last term in (29). We focus on $i \leq j$, and in the end multiply the bound we get by factor 2.

If $R_{s,\phi} = 1$, that is, $\delta(G_s) \geq \varepsilon_0$ then we can check that this is 0 since none of the other subcubes will have $R_{s,a} = 1$. So let us assume that this is not the case. Fix i and $H \subseteq i$ of size h. For any $a \in \binom{[n]}{i}, b \in \binom{[n]}{j}$ intersecting in a set of size h we have that $\frac{|E(J|_a, J|_b)|}{|E(J)|} \leq \left(\frac{2\ell}{n}\right)^{i+j-h}$. Indeed, to see this one can think of sampling an edge (x, y) in J uniformly at random, and asking what is the probability that $a \subseteq x$ and $b \subseteq y$. For that, we want $a \cap b$ to be contained in both x and y (which happens with probability at most $(2\ell/n)^h$), and that $a \setminus (a \cap b) \subseteq x$, $b \setminus (a \cap b) \subseteq y$ which happens with probability at most $(2\ell/n)^{i-h} \cdot (2\ell/n)^{j-h}$. Thus, we get that the last term in (29) is at most

$$\sum_{i \leq j \leq r} \sum_{h=0}^{i-1} \sum_{a \in \mathcal{C}_s^{(i)}} \sum_{H \subseteq a, |H|=h} |\{b \mid b \in \mathcal{C}_s^{(j)}, a \cap b = H, a \neq b\}| \left(\frac{2\ell}{n}\right)^{i+j-h}.$$

Here we note that h cannot be i, since then we would be looking at sub-cubes $J|_a \subseteq J|_b$, and so we cannot have both $R_{s,a} = 1$ and $R_{s,b} = 1$.

Applying Claim 4.2 on the restriction $G_s|_H$, we get that

$$|\{b \mid b \in \mathcal{C}_s^{(j)}, a \cap b = H, a \neq b\}| \leq \frac{n^{j-h}}{\ell^{j-h}} \frac{\delta(G_s|_H)}{\varepsilon_j^2} \leq \frac{n^{j-h}}{\ell^{j-h}} \frac{\varepsilon_h}{\varepsilon_j^2}.$$

We used the fact that as $R_{s,b} = 1$, G_s is at most ε_h dense in $J|_h$. Plugging that above we get that the last term on the right hand side of (29) is at most

$$\sum_{i \leq j \leq r} \sum_{h=0}^{i-1} \sum_{a \in \mathcal{C}_s^{(i)}} \sum_{H \subseteq a, |H|=h} \frac{n^{j-h}}{\ell^{j-h}} \frac{\varepsilon_h}{\varepsilon_j^2} \leq \sum_{i,j \leq r} 2^i \cdot i \cdot \frac{n^{j-h}}{\ell^{j-h}} \frac{\varepsilon_h}{\varepsilon_j^2} \cdot |\mathcal{C}_s^{(i)}| \left(\frac{2\ell}{n}\right)^{i+j-h},$$

and as $|\mathcal{C}_s^{(i)}| \leq \frac{\delta(G_s) n^i}{\varepsilon_i^2 \ell^i}$ due to Claim 4.2, we get further upper bound this by

$$\sum_{i,j \leq r} 2^i \cdot i \cdot \frac{n^{i+j-h}}{\ell^{i+j-h}} \frac{\varepsilon_h \delta(G_s)}{\varepsilon_j^2 \varepsilon_i^2} \left(\frac{2\ell}{n}\right)^{i+j-h} \leq 2^{3r+1} r^2 \delta(G_s) \max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4} \leq 2^{6r} \max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4} \delta(G_s).$$

Summing this over s gives that the contribution of the last term in (29) to (28) is at most $2^{6r} \max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4}$.

Bounding the second and third term in (29). Here, we use Theorem 3.17. The point is that the set H_s is pseudo-random, and hence we can get good control over the edges from it to other sets.

More precisely, for the second term note that H_s is (r, ε_r)-pseudorandom as per Definition 3.16, so by Theorem 3.17) with $\gamma = \varepsilon_r$ we get:

$$\frac{|E(H_s)|}{|E(J)|} = \delta(H_s) - \langle H_s, LH_s\rangle \leq 2\delta(H_s)(1-\alpha)^{r+1} \leq 2\delta(G_s)(1-\alpha)^{r+1}, \tag{32}$$

where the first inequality holds as long as $\varepsilon_r \leq (1-\alpha)^{r+1}\exp(-r)$. Note that the second term in the structure theorem $q_a(H_s)(\delta(H_s|_a) - \varepsilon_r)$ is non-positive since q_a is SoS and H_s is pseudorandom. The third term corresponding to Booleanity, $B(H_s)$ is 0 since $H_s^3(u) = H_s(u)$. Summing up over s yields that the contribution of this to the right hand side of (28) is at most $2(1-\alpha)^{r+1}$.

The bound on the third term in (29) is similar. We have that

$$\frac{|E(H_s, D_s)|}{|E(J)|} = \langle D_s, JH_s\rangle \leq \sqrt{\langle D_s, D_s\rangle \cdot \langle H_s, J^2 H_s\rangle} \leq 2\delta(G_s)(1-\alpha)^{r+1}. \tag{33}$$

In the last inequality, we used $\langle D_S, D_s\rangle \leq \delta(G_s)$ and $\langle H_s, J^2 H_s\rangle \leq 2\delta(G_s)(1-\alpha)^{2(r+1)}$ which follows by Theorem 3.17 exactly as above. Summing up over s yields total contribution of at most $2(1-\alpha)^{r+1}$.

SoS-ing the proof: We give a proof sketch of how to convert this into an SoS proof. To avoid Booleanity error we will apply (34) multiple times with $R_{s,a}$ replaced with $p_{\beta,\nu}$'s and take an average. We know that $R_{s,a} = p_1(\delta_1)\cdot\ldots\cdot p_t(\delta_t)$ where p_i's are approximate indicators $p_{\beta_i,\nu}$ (or $p_{<\beta_i,\nu}$) and δ_i's are of the form $\delta(G_s|_b)$ for $b \subseteq a$. Then define $R^c_{s,a}$ as $p^c_1(\delta_1)\cdot\ldots\cdot p^c_t(\delta_t)$ where $p^c_i(\delta_i) = p_{\beta_i + c\nu^2,\nu^2}$ (resp. $p_{<\beta_i+c\nu^2,\nu^2}$) for all $c \in \{0,\ldots,\lfloor 1/\nu\rfloor\}$.

Note that G_s's and the functions $\#sat(G_s \cap J|_a)$ are actual indicators as written since we can express $G_s(u) = \mathbb{I}[X_s - X'_u = s]$ and $\mathbb{I}[e \text{ is satisfied }]$ as a polynomial in (X, X'). Define $H^c_s(u) = G_s(u)\prod_{a\subset u:|a|\leq r}(1 - R^c_{s,a})$ and $D^c_s = G_s(u)(1 - H^c_{s,a})$. Defining dense subcubes with $R^c_{s,a}$ we get the following analogue of (29):

$$\begin{aligned}\frac{\#sat(G_s)}{|E(J)|} \leq \sum_{i\leq r}\sum_{a\in\binom{[n]}{i}} R^c_{s,a}\frac{\#sat(G_s\cap J|_a)}{|E(J)|} + \langle H^c_s, AH^c_s\rangle + \langle D^c_s, JH^c_s\rangle \\ + \sum_{i,j\leq r}\sum_{h=0}^{\min(i,j)}\sum_{a\in\binom{[n]}{i}} R^c_{s,a}\sum_{H\subseteq a,|H|=h}\sum_{\substack{b\in\binom{[n]}{j}\\ a\neq b, a\cap b = H}} R^c_{s,b}\frac{|E(J|_a, J|_b)|}{|E(J)|}.\end{aligned} \tag{34}$$

We then take an expectation over $c \in \{0,\ldots,\lfloor 1/\nu\rfloor\}$ and then a sum over $s \in \Sigma$.

Now we can use the same analysis to upper bound each of these terms. The analysis for the first term follows as is; we get an $\mathbb{E}_c[R^c_{s,a}]$ (instead of $R_{s,a}$) but we will ignore this slight difference as it does not affect our final analysis of the algorithm. For the second term we use the SoS version of Claim 4.2 (as stated) to say that $\mathbb{E}_{a\in\binom{[n]}{i}}[R^c_{s,a}]$ is small and the same tricks like those in the SoS-ization of Claim 4.2. For the third and fourth terms we apply the structure theorem on the almost-Boolean function H_s and bound the Booleanity term $\mathbb{E}_c[\sum_s B(H^c_s)]$ by $|\Sigma|\ell^{O(r)}\sqrt{\nu}$ instead.

Additionally in the fourth term we apply an SoS version of Cauchy-Schwarz. We omit these details as we've carried out similar arguments before, but formally prove the upper bound on $\mathbb{E}_c[\sum_s B(H_s^c)]$ in Claim B.1 in the appendix. □

4.2 The Main Lemma

In this section, we consider UG instances that have completeness bounded away from 1 and possibly very close to 0. The analysis of our algorithm differs only in one step: concluding that the shift potential is large.

The following claim asserts that taking two assignments X, X' from our pseudo-distribution, the expected number of edges that are satisfied by both X and X' is still bounded away from 0:

Claim 4.4. $\widetilde{\mathbb{E}}_{\mu\times\mu}[\mathrm{val}_I(X \wedge X')] \geq c^2$.

$$\begin{aligned}
\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}[\mathrm{val}_I(X \wedge X')] &= \widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}[\mathop{\mathbb{E}}_{e\sim E(G)}[\mathbb{1}(e \text{ satisfied by } X)\mathbb{1}(e \text{ satisfied by } X')]] \\
&= \mathop{\mathbb{E}}_{e\sim E(G)}[\widetilde{\Pr_{\mu}}[e \text{ satisfied by } X]^2] \\
&\geq \mathop{\mathbb{E}}_{e\sim E(G)}[\widetilde{\Pr_{\mu}}[e \text{ satisfied by } X]]^2 \qquad \text{(by Cauchy-Schwarz)} \\
&= \mathrm{val}_\mu(I)^2.
\end{aligned}$$

Lemma 4.5. *For all positive constants $c, \alpha, \tau \leq 1$, $\nu = \frac{c^4}{|\Sigma|^2 \ell^{O(r)}}$, all integers $\ell \geq \Omega(r), n \geq \ell$, where $r = \Theta(\frac{\log c}{\log(1-\alpha)})$ the following holds: Let I be an affine UG instance on $J_{n,\ell,\alpha}$ and μ be a pseudodistribution over assignments for I with $\mathrm{val}_\mu(I) \geq c$ and degree at least $\frac{\ell^{O(r)}\mathrm{poly}(|\Sigma|)}{\mathrm{poly}(c)\tau} + \ell^{\mathrm{poly}(|\Sigma|\ell^r/c)}$. Then there exists a restriction $a \subseteq [n]$ of size $i \leq r$, a degree $\widetilde{O}(1/\nu)$ polynomial $P_a(X, X')$ in a fixed set of $|\Sigma|$ polynomials, subsets $A, B \subseteq V(J|_a)$ of size at most $\frac{\ell^{O(i)}|\Sigma|^2}{c^2\tau}$ and strings y_A, y_B such that conditioning μ on the events $Y_A = y_A$ and $Y_B = y_B$ gives degree $\widetilde{\Omega}(1/\nu)$ pseudodistributions μ_1 and μ_2 such that:*

1. $\mathcal{A}_I \vdash_{\widetilde{O}(1/\nu)} P_a(X, X') \in [0, 1]$.
2. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[\Phi^a_{\beta,\nu}(X, X')P_a(X, X')] \geq \frac{c^{\Omega(r)}}{\exp(r^2)}\widetilde{\mathbb{E}}[P_a(X, X')]$.
3. $\widetilde{\mathbb{E}}_{\mu_1\times\mu_2}[P_a(X, X')] \geq \Omega\left(\frac{c^2}{8r\ell^i|\Sigma|}\right)$.
4. $\mathop{\mathbb{E}}_{\substack{u_1,v_1\sim S: u_1\neq v_1 \\ u_2,v_2\sim S: u_2\neq v_2}}[I(Y_{u_1,v_1}; Y_{u_2,v_2})] \leq \tau$.
5. $\mathop{\mathbb{E}}_{\substack{u_1,v_1\sim S: u_1\neq v_1 \\ u_2,v_2\sim S: u_2\neq v_2}}[I(Y'_{u_1,v_1}; Y'_{u_2,v_2})] \leq \tau$,

where $\beta = \left(\frac{\mathrm{poly}(c)}{\exp(r)}\right)^{r-i+1}$, $S = V(J|_a)$, $Y_{u,v} = (X_u, X_v, p_u, p_v)$, $Y'_{u,v} = (X'_u, X'_v, p'_u, p'_v)$ and the mutual information is taken with respect to the collection of local distributions $\mathcal{L}(\mu_1 \times \mu_2, p_{\beta,\nu})$.

Proof. Our strategy will be to use the edge-covering theorem. We will apply $\widetilde{\mathbb{E}}_{\mu\times\mu}$ on both sides of the inequality and use Claim 4.4 to say that $\widetilde{\mathbb{E}}[\mathrm{val}_I(X \wedge X')]$ is large.

We will set parameters in the end of the proof such that: r is large enough so that $4(1-\alpha)^{r+1} \leq c^2/6$, ν is small enough such that $|\Sigma|\ell^{O(r)}\sqrt{\nu} \leq c^2/6$ and ε_i's satisfy $2^{6r}\max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4} \leq c^2/6$ and further ε_i's satisfy the conditions of Theorem 4.3. So we get that the error term in Theorem 4.3 is small,

$$\mathsf{err} = 4(1-\alpha)^{r+1} + 2^{6r}\max_i \frac{\varepsilon_{i-1}}{\varepsilon_i^4} + |\Sigma|\ell^{O(r)}\sqrt{\nu} \leq \frac{c^2}{2}. \tag{35}$$

Applying $\widetilde{\mathbb{E}}_{\mu\times\mu}$ on both sides of Theorem 4.3 (assuming $\deg(\mu) \geq \ell^{\mathrm{poly}(\ell^r/\nu)}$) and using Claim 4.4 to lower bound pseudoexpectation of the LHS we get:

$$c^2 \leq \widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}[\mathrm{val}_I(X\wedge X')] \leq \sum_{i\in[r]} \widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}[T_i(X,X')] + \mathsf{err} \leq \sum_{i\in[r]} \widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}[T_i(X,X')] + \frac{c^2}{2},$$

Therefore we get that there is an $i\in[r]$ for which $\widetilde{\mathbb{E}}_{\mu\times\mu}[T_i(X,X')] \geq c^2/2r$, and so

$$\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}} \mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}[\sum_s R_{s,a} \mathop{\mathbb{E}}_{u\sim J|_a}[G_s(u)\mathrm{val}_u^a(X\wedge X')]] \geq \frac{c^2}{2r\ell^i}. \tag{36}$$

We aim to condition $\widetilde{\mathbb{E}}_{\mu\times\mu}$ on an appropriate polynomial (as in the proof of Lemma 3.20) so that the resulting pseudoexpectation operator satisfies: $\widetilde{\mathbb{E}}'[\mathbb{E}_{u\sim J|_a}[G_s(u)\mathrm{val}_u^a(X\wedge X')]]] \geq \Omega(c^2/r)$ for some specific i-restriction a. To do so we apply $\widetilde{\mathbb{E}}_{\mu\times\mu}$ to Claim 4.2 and sum up over s, to get

$$\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}\left[\mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}\sum_s R_{s,a}\right] \leq \frac{4}{\varepsilon_i^2\ell^i} + O(|\Sigma|\nu) \leq \frac{5}{\varepsilon_i^2\ell_i},$$

where for the last inequality we have assumed $\nu \leq O(1/\ell^i|\Sigma|)$.

We split the RHS in equation 36 to $c^2/4r\ell^i + c^2/4r\ell^i$, where in the first term we further use the above equation: $1/\ell^i \geq \frac{\varepsilon_i^2}{5}\widetilde{\mathbb{E}}_{\mu\times\mu}[\mathbb{E}_{a\sim\binom{[n]}{i}}[\sum_s R_{s,a}]]$ to get that,

$$\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}} \mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}\left[\sum_s R_{s,a} \mathop{\mathbb{E}}_{u\sim J|_a}[G_s(u)\mathrm{val}_u^a(X\wedge X')]\right] \geq \frac{c^2\varepsilon_i^2}{20r}\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}\left[\mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}[\sum_s R_{s,a}]\right] + \frac{c^2}{4r\ell^i},$$

and by re-arranging

$$\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}} \mathop{\mathbb{E}}_{a\sim\binom{[n]}{i}}\left[\sum_s R_{s,a}\left(\mathop{\mathbb{E}}_{u\sim J|_a}[G_s(u)\mathrm{val}_u^a(X\wedge X')] - \frac{c^2\varepsilon_i^2}{20r}\right)\right] \geq \frac{c^2}{4r\ell^i}.$$

By averaging we get an $a\in\binom{[n]}{i}, s\in\Sigma$ such that:

$$\widetilde{\mathop{\mathbb{E}}_{\mu\times\mu}}\left[R_{s,a}\left(\mathop{\mathbb{E}}_{u\sim J|_a}[G_s(u)\mathrm{val}_u^a(X\wedge X')] - \frac{c^2\varepsilon_i^2}{20r}\right)\right] \geq \frac{c^2}{4r\ell^i|\Sigma|}. \tag{37}$$

The last result is the same as equation (10) in the proof of Lemma 3.20, which is the analogous statement for the completeness s clsoe to 1 case. The rest of the proof is essentially the same as the rest of the proof of Lemma 3.20, and is outlined below.

Applying Lemma 3.19 we get pseudodistributions μ_1, μ_2 with low global correlation (as required in points 4,5 of the Lemma) with:

$$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}\left[R_{s,a}\left(\mathbb{E}_{u \sim J|_a}[G_s(u)\mathrm{val}_u^a(X \wedge X')] - \frac{c^2\varepsilon_i^2}{20r}\right)\right] \geq \frac{c^2}{8r\ell^i|\Sigma|}. \tag{38}$$

Letting $P_a(X, X') = R_{s,a}(X, X')$ it is easy to check that:

$$\mathcal{A}_I \vdash_{\widetilde{O}(\exp(i)/\nu)} P_a(X, X') \in [0, 1], \text{ and } \quad \widetilde{\mathbb{E}}_{\mu_1 \times \mu_2}[P_a(X, X')] \geq \frac{c^2}{8r\ell^i|\Sigma|},$$

which proves points 1 and 3 of the Lemma statement. The only thing remaining to check is that conditioning $\mu_1 \times \mu_2$ on $R_{s,a}$ results in large shift-partition potential on $J|_a$. Recall that the shift-partition potential on $J|_a$ is: $\phi^a_{\beta,\nu}(X, X') = \sum_s \mathbb{E}_{u \sim J|_a}[G_s(u)p_{\beta,\nu}(\mathrm{val}_u^a(X))p_{\beta,\nu}(\mathrm{val}_u^a(X'))]^2$. From equation (38) we have that conditioning $\mu_1 \times \mu_2$ on $R_{s,a}$ gives:

$$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2 | R_{s,a}}[\mathbb{E}_{u \sim J|_a}[G_s(u)\mathrm{val}_u^a(X \wedge X')]] \geq \frac{c^2\varepsilon_i^2}{20r}.$$

We can prove that for $x \in [0, 1] \vdash_{\widetilde{O}(1/\nu)} x \leq p_{\beta,\nu}(x) + \beta + O(\nu)$ (using the standard machinery in Section 2.3.2). Applying this with $\mathrm{val}_u^a(X \wedge X')$ and $\beta = \frac{c^2\varepsilon_i^2}{40r}$, and further noting that $p_{\beta,\nu}(\mathrm{val}_u^a(X \wedge X')) \leq p_{\beta,\nu}(\mathrm{val}_u^a(X))p_{\beta,\nu}(\mathrm{val}_u^a(X')) + O(\nu)$ we get:

$$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2 | R_{s,a}}[\mathbb{E}_{u \sim J|_a}[G_s(u)p_{\beta,\nu}(\mathrm{val}_u^a(X))p_{\beta,\nu}(\mathrm{val}_u^a(X'))]] \geq \frac{c^2\varepsilon_i^2}{40r} - O(\nu).$$

Applying Cauchy-Schwarz we get:

$$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2 | R_{s,a}}[\mathbb{E}_{u \sim J|_a}[G_s(u)p_{\beta,\nu}(\mathrm{val}_u^a(X))p_{\beta,\nu}(\mathrm{val}_u^a(X'))]^2] \geq \Omega\left(\frac{c^4\varepsilon_i^4}{r^2}\right).$$

By adding the terms corresponding to other $s \in \Sigma$ we immediately get:

$$\widetilde{\mathbb{E}}_{\mu_1 \times \mu_2 | R_{s,a}}[\phi^a_{\beta,\nu}(X, X')] \geq \Omega\left(\frac{c^4\varepsilon_i^4}{r^2}\right). \tag{39}$$

We finish the proof by setting r, ε_i, ν according to all the requirements above.

Setting parameters: To make err in (35) small we set $r = \Theta(\frac{\log c}{\log(1-\alpha)})$, $\nu = \frac{c^4}{|\Sigma|^2\ell^{O(r)}}$, $\varepsilon_{j-1} = \varepsilon_j^5/2^{6r}$ with $\varepsilon_r = \Theta(c^2\exp(-r))$. We need $\beta = c^2\varepsilon_i^2/40r$ which equals $(\mathrm{poly}(c)/\exp(r))^{r-i+1}$. We can set $\deg(\mu) = \ell^{\mathrm{poly}(\ell^r/\nu)} = \ell^{\mathrm{poly}(|\Sigma|\ell^r/c)}$ so that it is valid to apply the edge-covering theorem. Additionally when we apply Lemma 3.19 we need $\deg(\mu) \geq \Omega(\log|\Sigma|/p\tau\nu)$ where $p = c^2/4r\ell^i|\Sigma|$ so it suffices to take $\deg(\mu) = \frac{\ell^{O(r)}\mathrm{poly}(|\Sigma|)}{\mathrm{poly}(c)\tau}$. One can check that these parameters are good enough to make the rest of the statements in the proof work out. From (39) we get that the conditioned shift-partition potential is at least $\Omega(c^2\varepsilon_i^4/r^2) \geq \frac{c^{O(r)}}{\exp(r)}$ and this completes the proof. □

4.3 Proof of Theorem 1.6

The algorithm for arbitrary completeness is the same as that for completeness close to 1 albeit with different parameter settings for r, δ, D, γ for the main algorithm and t, τ, β, ν for SubRound. Given Lemma 4.5 the rest of the analysis of the algorithm remains exactly the same. We set these parameters below but omit the description of the algorithm. The guarantees of this algorithm are potentially worse (running time $\approx n^{\ell^{\mathrm{poly}(\ell)}}$ and approximation factor $\approx c^{O(\log 1/c)}$) than the one for completeness close to 1, but the important thing to note is that the running time is polynomial in n (as long as $\ell, |\Sigma|$ are $O(1)$) and the approximation factor only depends on c and not on ℓ, n or $|\Sigma|$.

Theorem 4.6. *For all positive constants $c \in (0,1]$, $\alpha \in \mathbb{Q}$, all integers $\ell \geq \Omega(r)$ with $\alpha\ell \in \mathbb{N}$ and $r = \Theta(\frac{\log c}{\log(1-\alpha)})$ and n large enough, Algorithm 3.4 (with modified parameter settings given below) has the following guarantee: If I is an instance of affine Unique Games on the (n, ℓ, α)-Johnson graph J with alphabet Σ and $\mathrm{val}(I) = c$, then in time $|V(J)|^D$ Algorithm 3.4 returns an $\Omega\left(\frac{c^{\Omega(r)}}{\exp(r^2)}\right)$-satisfying assignment for I, with $D = \frac{\ell^{\mathrm{poly}(|\Sigma|\ell^r/c)}}{c^{O(r)}}$.*

Proof. Let $r = \Theta(\frac{\log c}{\log(1-\alpha)})$. Given the $i \leq r$-restricted subcube $a \subseteq [n], P_a(X, X')$ and $\mu_1 \times \mu_2$ from Lemma 4.5 we have that $\Phi^a_{\beta,\nu}(\mu_1 \times \mu_2 | P_a) \geq \frac{c^{\Omega(r)}}{\exp(r^2)}$, therefore we apply Lemma 3.10 with $S = J|_a$, the polynomial $E(X, X') = P_a(X, X')$ and the parameters $\beta = \left(\frac{\mathrm{poly}(c)}{\exp(r)}\right)^{r-i+1}, \nu = \frac{c^4}{|\Sigma|^2 \ell^{O(r)}}, \delta = \frac{c^{O(r)}}{\exp(r^2)}, p = \Omega\left(\frac{c^2}{8r\ell^i|\Sigma|}\right)$ and $\tau = \frac{c^{O(r)}}{\exp(r^2)\ell^{O(i)}|\Sigma|}$. The parameter τ has been chosen so that $O(\frac{\sqrt{\tau}+1/|S|}{p\delta^2}) = \frac{c^{O(r)}}{\exp(r^2)}$ and therefore we can apply the rounding lemma (Lemma 3.11) with $\zeta = \frac{c^{O(r)}}{\exp(r^2)}$ and $\gamma = \frac{c^{\Omega(r)}}{\exp(r^2)}$ and the same settings of β, ν, δ. This shows that there exists a subcube $a \in \binom{[n]}{i}$ with SubRound value that is at least $\frac{c^{\Omega(r)}}{\exp(r^2)}$ if $\deg(\mu) \geq \frac{\ell^{O(r)}\mathrm{poly}(|\Sigma|)}{\mathrm{poly}(c)\tau} + \ell^{\mathrm{poly}(|\Sigma|\ell^r/c)}$. Hence it suffices to have degree of μ equal to $D = \frac{\ell^{\mathrm{poly}(|\Sigma|\ell^r/c)}}{c^{O(r)}}$.

By using SubRound as a subroutine, we finish the proof of this theorem by applying the iteration Lemma 3.12 with completeness c, $\gamma = c^2$, $\delta = \frac{c^{\Omega(r)}}{\exp(r^2)}$ and $r = \Theta(\frac{\log c}{\log(1-\alpha)})$ so that $(1-\alpha)^r = \mathrm{poly}(c)$. We can check that SubRound (with the parameter settings above) satisfies the hypotheses of the lemma: it finds a $\leq r$-restricted subcube and an assignment to it with value $\Omega(\delta)$ in time $|V(J)|^D$ (where D is the degree bound above). This gives us that Algorithm 3.4 outputs an assignment of value at least $\Omega\left(\frac{c^{\Omega(r)}}{\exp(r^2)}\right)$ in time $|V(G)|^{O(D)}$. □

5 Affine Unique Games on other Globally Hypercontractive Graphs

In this section, we explain how to adapt our techniques from Sections 3, 4 for general globally hypercontractive graphs. We first give a semi-formal definition of global hypercontractivity and go over the steps of our algorithm, noting that even though we stated it for Johnson graphs earlier, it works for general graphs as long as we have an SoS certificate of global hypercontractivity. Later we give the specifics for the proofs of Theorems 1.7 and 1.8: UG algorithms for the Grassmann graph and random walks over HDXs.

5.1 The Components that Go Into the Algorithm

The 4-components that are used in our algorithm proving Theorem 1.6 and the interaction between them can be abstracted as follows.

1. **Global hypercontractivity.** The first component going into our algorithm is a global hypercontractive inequality (or a consequence of it). This is a result asserting that for our underlying graph G, for all $\gamma > 0$ there are $r \in \mathbb{N}$, $\varepsilon_0, \ldots, \varepsilon_r, \zeta, \eta > 0$ and a collection $\mathcal{C} = \mathcal{C}_0 \cup \ldots \cup \mathcal{C}_r$ of sets of vertices (these are the combinations of the basic sets) satisfying the following properties:

   (a) $\mathcal{C}$ **consists of non-expanding sets:** The edge expansion of each $C \in \mathcal{C}$ in G is $1 - \zeta$.

   (b) $\mathcal{C}$ **explains all small sets that do not expand well.** A basic form of this property is that sets $S \subseteq V(G)$ that are not dense in any set in $\mathcal{C}$, in the sense that $\delta(S \cap C) \leq \varepsilon_1 \delta(C)$ for all $C \in \mathcal{C}$, have expansion close to 1, that is, $\Phi(S) \geq 1 - \gamma$.

   A stronger version of this property, that often follows from the above (as is the case of all graphs of interest in this paper) and that is used in the proof of Theorem 1.5, is that one can capture a constant fraction of S using $C \in \mathcal{C}$ in which S is dense. Equivalently, this property says that for a set of vertices $S \subseteq V(G)$, taking $\mathcal{C}' = \{C \in \mathcal{C} \mid C \in \mathcal{C}_i,\ \delta(S \cap C) \geq \varepsilon_i \delta(C)\}$ and defining $C' = \bigcup_{C \in \mathcal{C}'} C$, if $\delta(S \cap C') \leq \eta \delta(S)$ then $\Phi(S) \geq 1 - \gamma$.

   An even stronger form of global hypercontractivity, that again follows from the most basic form in our cases of interest, is the edge covering theorem. We used this form of global hypercontracitivity in the proof of Theorem 1.6. Informally, this result says that not only a substantial chunk of S can be covered by non-expanding sets from $\mathcal{C}$ in which S is dense, but in fact one can capture almost all of the edges staying inside S using $C \in \mathcal{C}$. More precisely, this result asserts that we may take $\mathcal{C}'' \subseteq \mathcal{C}'$ such that the following properties hold:

      i. Edge covering: sampling an edge (u, v) inside S, we have that except with probability ξ it holds that (u, v) is an edge inside some $C'' \subseteq \mathcal{C}''$.

      ii. Maximally dense: in words, this means that every $C'' \in \mathcal{C}''$ has no $C \supsetneq C''$ in $\mathcal{C}$ in which S is somewhat dense. Formally, we require that for all $i = 1, \ldots, r$, $C'' \in \mathcal{C}'' \cap \mathcal{C}_i$, $j < i$ and $C_j \in \mathcal{C}_j$ such that $C_j \supseteq C''$, it holds that $\delta(S \cap C_j) < \varepsilon_j \delta(C_j)$.

   The edge covering theorem allows us to argue that in our shift partition $F_s(X, X') = \{v \in V \mid X(v) - X'(v) = s\}$ we may capture almost all edges that stay within some F_s using the basic sets from $\mathcal{C}$. In particular, since edges that both X and X' satisfy constitute a constant fraction of edges and they all stay within some F_s, it means that we may capture almost all of them using sets from $\mathcal{C}$.

2. By elementary arguments, it follows that there is such a $C \in \mathcal{C}$ and s such that F_s becomes dense in C with noticeable probability and inside C a constant fraction of the edges are satisfied by both X and X'. In our argument we consider this event (this was approximated by the polynomial $P_a(X, X')$ in our algorithm to stay within the realm of SoS). As we explained in the introduction this conditioning may create correlation between X and X' that would cause our rounding technique not to work.

   To circumvent this, we showed that:

(a) **Eliminating global correlations:** using conditioning (as in [RT12]) we are able to make sure that the two collection of random variables $\{Z_{u,v}\}_{u,v\in C}, \{Z'_{u,v}\}_{u,v\in C}$ of interest (that are concerned with local information about the assignments X and X' on vertices u, v) are roughly pairwise independent.

(b) **Conditioning mostly preserves independence:** As the probability of the event that C becomes dense in F_s is noticeable, and $\{Z_{u,v}\}_{u,v\in C}$,$\{Z'_{u,v}\}_{u,v\in C}$ were pairwise-independent to begin with, we argue that it cannot create too many dependencies between $(Z_{u,v}, Z'_{u,v})$.

3. **Rounding.** We then gave a rounding procedure on a subcube C and a pseudodistribution $\mathcal{D}$, where on average (over $\mathcal{D}$) our supposed assignments X, X' satisfy a constant fraction of the edges (formalized via the notion of shift-partition potential), some part F_s of the shift partition is large, and on which we have near independence between certain random variables $\{(Z_{u,v}, Z'_{u,v})\}_{u,v}$. The output of the rounding procedure is an assignment to the vertices of C that satisfies a constant fraction of the constraints inside it.

4. **An iteration result.** By that, we mean a result saying that if we have a procedure as above that is able to find a $C \in \mathcal{C}$ and find an assignment to the vertices of C that satisfies a constant fraction of the constraints in it, then we can iterate it to satisfy a constant fraction of the constraints in the whole graph.

   This is achieved by randomizing the edges incident to C, which ensures that any subsequent C' we find can only have negligible overlap with the C's we have found thus far. The idea is that any C that we found has at least δ of the edges touching it staying inside it, out of which we manage to satisfy a constant fraction, hence overall we manage to satisfy a constant fraction of the edges touching C. In return for that, we randomize the edges touching C hence effectively giving up on the objective value coming from those edges. The point is that since the ratio between the number of edges we manage to satisfy, and the number of edges we give up on is $\Omega(1)$, after a few iterations we will manage to satisfy a constant fraction of the edges before we drop the objective value of our instance by too much.

Our of these 4 components, the second, third and fourth components work as is for any of the graphs that we care about. Indeed, the elimination of correlations and conditioning are completely generic, and the rounding procedure only hinges on the constraints being affine. As for the fourth component, it only relies on the properties of the family $\mathcal{C}$ guaranteed from our structure theorem and otherwise is also generic.

We next discuss the first component, which is a globally hypercontractive inequality, and for concreteness we consider the most basic version of it. This is a result saying that a set S which is not dense inside any $C \in \mathcal{C}$ has large expansion, and in all of our graphs of interest such results are known [KMMS18, KMS18, FKLM20, KLLM21, BHKL22b, GLL22]. For such a result to give us an algorithm, we also require the proof of this part to be in the SoS proof system of constant degree (which is the case in all of the above examples, but in principle may not be the case).

In the case of the Johnson graph the collection $\mathcal{C}$ is simply the collection of basic sets

$$\mathcal{C} = \{J|_a \mid |a| \le r\}, \qquad \text{where } J|_a = \left\{ u \in \binom{[n]}{\ell} \;\middle|\; u \supseteq a \right\},$$

the basic form of the global hypercontractive inequality is from [KMMS18] (in the appendix we give an SoS version of that proof), and we have shown in Section 4 how to conclude from it the edge covering theorem in a black-box manner.

A Proof of the Iteration Lemma

We give a proof of the iteration lemma. Let $\mathrm{val}_f(H)$ denote the fraction of edges inside H that are satisfied by the assignment f.

Lemma A.1. *Let $c, \gamma, \delta \in (0,1]$ and $r \in \mathbb{N}$. Let I be an affine UG instance on alphabet Σ on $J_{n,\ell,\alpha}$ with ℓ, n large enough and value at least c. Suppose we have a subroutine $\mathcal{A}$ which given as input any affine UG instance I' on $J_{n,\ell,\alpha}$ with $\mathrm{val}_\mu(I') \geq c - \gamma$, returns an $\leq r$-restricted subcube H on J and a partial assignment f such that, $val_f(H) \geq \delta$. Then if $\mathcal{A}$ runs in time $T(\mathcal{A})$, there is a $O(|V(J)|T(\mathcal{A}) + |V(J)|^3)$-time algorithm which finds a solution for I that satisfies an $\Omega(\delta\gamma(1-\alpha)^r)$-fraction of the edges of J.*

Proof. We will use the algorithm $\mathcal{A}$ as a subroutine. To get a full assignment, our algorithm below is a generalized version of the Algorithm 3.4, where we've replaced some of the steps in Algorithm 3.4 with an arbitrary subroutine $\mathcal{A}$ that finds a subcube and an assignment to it with high value. We include it here for completeness.

Algorithm A.2 (Partial to Full Assignment)**.**

1. Set $j = 1, I_0 = I, R_0 = \phi$.
2. While $|R_{j-1}| < \frac{\gamma}{2}|V(J)|$:
   (a) Run subroutine $\mathcal{A}$ on I_{j-1} to find an $\leq r$-restricted subcube H_j and partial assignment f_j.
   (b) Let S_j be the induced subgraph of H_j induced by the set $H_j \setminus R_{j-1}$ and assign them using f_j. Let $R_j = R_{j-1} \cup H_j$.
   (c) Randomize the constraints on edges incident on R_j and let the new instance by I_j.
   (d) Increment j.
3. Output any assignment to $V(G)$ that agrees with all partial assignments f_j (assigned to S_j's) considered above.

Let us analyse the jth-iteration of this procedure. We are given an instance I_{j-1}.

Claim A.3. $\mathrm{val}(I_{j-1}) \geq \mathrm{val}(I) - \frac{2|V(R_{j-1})|}{|V(G)|}$.

Proof. Suppose X is an assignment that achieves the value of I. We can check that X satisfies at least $\mathrm{val}(I) - \frac{2|V(R_{j-1})|}{|V(G)|}$-fraction of edges of I_{j-1}, hence the claim follows. To see this note that the fraction of edges incident on R_{j-1} and therefore the fraction that is randomized is at most $\frac{2|V(R_{j-1})|}{|V(G)|}$ and every edge not incident on R_{j-1} remains satisfied by X. □

We will show that the partial assignment f_j to $H_j \setminus R_{j-1}$ satisfies a large fraction of edges (as counted in the whole graph). Before that let us show that the randomization of edges ensures that the value of the edges randomized is small (even when restricted to some subcube). Let $E(S)$ denote the edges incident on S (including those with only one endpoint in S) and let $E(S,S)$ denote

the edges with both endpoints in S. For a set of edges E let $\mathrm{val}_f(E)$ denote the fraction of edges in E that are satisfied by f.

Claim A.4. *With probability $1-o(1)$ the instance I_{j-1} satisfies the following for all r-restricted subcubes C:*

$$\mathrm{val}(E(C \cap R_{j-1})) \leq \frac{2}{|\Sigma|},$$

where $\mathrm{val}(E(S))$ denotes the maximum fraction of $E(S)$ that can be satisfied by any UG assignment to the endpoints of $E(S)$.

Proof. Let k denote $|\Sigma|$. Fix an i-restricted subcube C for $i \leq r$. We know that R_{j-1} is a union of $\leq r$-restricted subcubes, therefore $C \cap R_{j-1}$ also forms a union of $\leq r$-restricted subcubes inside C. Let this collection of subcubes be denoted by $\mathcal{C}$. Consider the set of edges E inside C on which the constraints were randomized (in the previous iteration), and let A be the set of vertices in C that are endpoints of them. We claim that with probability $1-o(1)$, for all assignments F to A, F satisfies at most $2/k$ of the constraints. First, the number of assignments is $k^{|A|}$, so we fix one assignment F, analyze it, and then union bound over all assignments.

Fix F; for each $e \in E$, the probability that F satisfies e is $1/k$. Hence in expectation F satisfies $\mu = 1/k$ fraction of the constraints, and by Chernoff's bound

$$\Pr[\mathrm{val}_F(E) \geq 2\mu] \leq e^{-\frac{|E|}{3k}}.$$

Now we lower bound $|E|$. Note that if $v \in A$, then either v was in a subcube in $\mathcal{C}$ or v was adjacent to a vertex in a subcube from $\mathcal{C}$. Let the degree of v inside C be denoted by d which we know is $\binom{\ell-i}{\alpha(\ell-i)}\binom{n-\ell}{\alpha(\ell-i)}$.

1. If $v \in C'$ for some $C' \in \mathcal{C}$ then at least $(1-\alpha)^r$ of the edges incident to v remain in C' and therefore are in E.

2. If v is adjacent to $u \in R_{j-1} \cap C$ then $u \in C'$ for some $C' \in \mathcal{C}$. In particular there is a subset $I \subseteq u$ of size at most r such that C' is an I-restricted subcube. It follows that in fact v has at least $\binom{\ell-i}{\alpha(\ell-i)}\binom{n-\ell-r}{\alpha(\ell-i)-r}$ neighbours in C'. By a simple calculation this is at least $n^{-r}d$, and it follows that at least $n^{-r}d$ of the edges adjacent to v are in E.

In conclusion, in any case we get that at least $n^{-r}d$ of the edges adjacent to $v \in A$ are inside E, hence $|E| \geq n^{-r}d|A|/2 \gg 3n|\Sigma||A|$, so Chernoff gives a probability of $e^{-n|\Sigma||A|}$ which is good enough for a union bound over all ($k^{|A|}$ many) assignments to A. Additionally we union bound over all $\leq r$-restricted subcubes C which are only $n^{O(r)}$ many, to get the statement of the lemma. □

At iteration j of the while-loop, since the while condition is met, we know that I_{j-1} has value $\geq c-\gamma$ (Claim A.3) so inside the while-loop, $\mathcal{A}$ will always find an $\leq r$-subcube H_j and assignment f_j with $\mathrm{val}_{f_j}(H_j) \geq \delta$. Next, we find an assignment f_j to the set of vertices $V(S_j)$ that by definition don't intersect previously assigned vertices. Since f_j doesn't reassign any vertices, in the final step of the algorithm it is possible to output an assignment that is consistent with all previously considered partial assignments. We will now show that our final partial assignment satisfies a large fraction of the edges, where we say that an edge (u, v) is satisfied by a partial assignment f_j, if both vertices u, v have been assigned labels under f_j and the labels satisfy the edge.

Claim A.5. *The value of the partial assignment found at iteration j satisfies:*

$$\mathrm{val}(f_j) \geq \delta(1-\alpha)^r \frac{|V(H_j)|}{2|V(G)|},$$

where $\mathrm{val}(f_j)$ *denotes the fraction of edges (in* $E(G)$*) satisfied by the partial assignment* f_j*.*

Proof. Let $H_j = S_j \cup (H_j \cap R_{j-1})$ and let $E_1 = E(S_j, S_j)$ and $E_2 = E(H_j \cap R_{j-1})$ so that $E_1 \cup E_2 = E(H_j, H_j)$ denoted by E. By Claim A.4 we know that $\mathrm{val}_{f_j}(E_2) \leq 2/k$ where k denotes $|\Sigma|$. By an averaging argument:

$$\delta \leq \frac{\#\mathrm{sat}_{f_j}(E_1)}{|E|} + \frac{|E_2|}{|E|}\mathrm{val}_{f_j}(E_2) \leq \frac{\#\mathrm{sat}_{f_j}(E_1)}{|E|} + \frac{2}{k},$$

where $\#\mathrm{sat}_{f_j}(E_1)$ denotes the number of edges of E_1 satisfied by f_j. This implies that $\frac{\#\mathrm{sat}_{f_j}(E_1)}{|E|} \geq \delta/2$.

Since H_j is a $\leq r$-restricted subcube at least $(1-\alpha)^r$-fraction of edges stay inside H_j, hence $|E|/|E(H_j)|$ is at least $(1-\alpha)^r$. Using that $|E(H_j)|/|E(J)| \geq |V(H_j)|/|V(J)|$ we get that:

$$\mathrm{val}(f_j) = \frac{\#\mathrm{sat}_{f_j}(E_1)}{|E|} \cdot \frac{|E|}{|E(H_j)|} \cdot \frac{|E(H_j)|}{|E(J)|} \geq \frac{\delta}{2}(1-\alpha)^r \frac{|V(H_j)|}{|V(G)|},$$

where $\mathrm{val}(f_j)$ denotes the edges satisfied by f_j (assigned to S_j) as a fraction in the whole graph. □

Once we have these facts, the conclusion is immediate. Firstly there cannot be more than $V(G)$ iterations of the while-loop, since at each iteration we assign at least one new vertex. Each iteration takes time $T(\mathcal{A}) + |V(G)|^2$, hence the algorithm runs in time $|V(G)|(T(\mathcal{A}) + |V(G)|^2)$.

Suppose the algorithm exits the while loop at the $t+1$ iteration. Then we know that $R_t \geq \frac{\gamma}{2}|V(J)|$. Using claims A.5 above, we get that the value of the final assignment is proportional to the number of vertices assigned:

$$\sum_{j=1}^{t} \mathrm{val}(f_j) \geq \delta(1-\alpha)^r \sum_{j=1}^{t} \frac{|V(H_j)|}{2|V(J)|} \geq \delta(1-\alpha)^r \frac{|R_t|}{2|V(J)|} \geq \delta(1-\alpha)^r \frac{\gamma}{4},$$

thus proving the lemma. □

B Missing Proofs

The following claim bounds the Booleanity error term and shows that there is a SoS proof that they are negligible (provided that we take the parameter ν to be sufficiently small).

Claim B.1.

$$\mathcal{A}_I \quad \vdash_{\mathrm{poly}(\ell^r/\nu)} \quad \mathop{\mathbb{E}}_c[\sum_s B(H_s^c)] \leq |\Sigma|\ell^{O(r)}\sqrt{\nu}.$$

Proof. Recall that $B(F) = \frac{4}{3}\mathbb{E}_u[(F(u)^3 - F(u))(\Pi_{\geq\lambda_r}F)(u)] = \frac{4}{3}\langle F^3 - F, PF\rangle$, where Π is the projection operator $\Pi_{\geq\lambda_r}$. Using Cauchy-Schwarz we get that for all $\eta > 0$:

$$B(F) \leq \frac{4}{3}\left(\frac{\eta}{2}||F^3 - F||_2^2 + \frac{1}{2\eta}||\Pi F||_2^2\right) \leq \frac{2\eta}{3}\mathbb{E}_u[F(u) - F(u)^3] + \frac{2}{3\eta}\delta(F^2), \tag{40}$$

where the last inequality holds since $F \in [0,1]$ and Π is a projection operator. We will use this to bound $B(H_s^c)$ as follows. Fixing s, u, taking $F = H_s^c(u)$, the first term above is $\mathbb{E}_c[(H_s^c(u) - H_s^c(u)^3]$ and we upper bound it by $O(\ell^r\nu)$. Indeed, first as $0 \leq H_s^c(u) \leq 1$ we have

$$\mathbb{E}_c[H_s^c(u) - H_s^c(u)^3] = (1 + H_s^c(u))\,\mathbb{E}_c[H_s^c(u) - H_s^c(u)^2] \leq 2\,\mathbb{E}_c[H_s^c(u) - H_s^c(u)^2]. \tag{41}$$

Next, recall that $H_s^c(u)$ is a product of the terms $(1 - R_{s,a}^c)$ for $a \subset u, |a| \leq r$. Additionally each variable $R_{s,a}^c$ is a product of approximate indicators: namely there are $\beta_i \in (0,1)$ such that $R_{s,a}^c(u) = p_1^{c,s,a}(\delta_1^{c,s,a}) \cdot \ldots \cdot p_k^{c,s,a}(\delta_k^{c,s,a})$, where $p_i^{c,s,a} = p_{\beta_i + c\nu^2,\nu^2}$ or $p_{<\beta_i+c\nu^2,\nu^2}$ as c varies in $\{0,1,\ldots,\lfloor 1/\nu\rfloor\}$, and δ_i's are linear functions of the variables $G_s(u)$ and lie in $[0,1]$ under the axioms $\mathcal{A}_I$. By Claim B.2(applied twice) it follows that:

$$\mathbb{E}_c[H_s^c(u) - H_s^c(u)^2] \leq \sum_{a\subset u:|a|\leq r}\mathbb{E}_c[R_{s,a}^c - (R_{s,a}^c)^2] \leq \sum_{a\subset u:|a|\leq r}\sum_{i\in[k]}\mathbb{E}_c[p_i^{c,s,a}(\delta_i^{c,s,a}) - p_i^{c,s,a}(\delta_i^{c,s,a})^2],$$

so we can bound each of the terms above.

We will show that each term is smaller than $O(\nu)$. Let p be an approximate indicator polynomial $p = p_{\beta,\nu}$ and let p^c denote $p_{\beta+c\nu^2,\nu^2}$ for $c \in \{0,\ldots,\lfloor 1/\nu\rfloor\}$. For all $\delta \in [0,1]$, we argue that

$$\delta \in [0,1] \vdash_{\widetilde{O}(1/\nu^2)} \quad \mathbb{E}_c[p^c(\delta) - p^c(\delta)^2] \leq O(\nu). \tag{42}$$

Indeed, this follows by case analysis on $\delta \in [0,1]$ (that we show next), and then using Lúkacs theorem (Corollary 2.5) we get there is an SoS proof of degree $\deg(p^c)$.

If $\delta \in [0,\beta]\cup[\beta+\nu,1]$ then one can check that $p^c(\delta) \in [0,\nu^2]$ for all c. Therefore let $\delta \in [\beta,\beta+\nu]$, and specifically in one of the intervals - $[\beta + t\nu^2, \beta + (t+1)\nu^2]$ for $t \in \{0,\ldots,\lfloor 1/\nu\rfloor\}$. Then we get that $p^t(\delta)$ could be any number between $[0,1]$ and therefore $p^t(\delta) - p^t(\delta)^2 \leq 1/4$. But for every $t' \neq t$ we get that $p^{t'}(\delta)$ is either in the interval $[0,\nu^2]$ or $[1-\nu^2,1]$ implying that $p^{t'}(\delta) - p^{t'}(\delta)^2 \leq O(\nu^2)$. Averaging over a we therefore get:

$$\mathbb{E}[p^c(\delta) - p^c(\delta)^2] \leq \nu \cdot \frac{1}{4} + (1-\nu)O(\nu^2) \leq O(\nu).$$

In conclusion, we get that $\mathbb{E}_c[H_s^c(u) - H_s^c(u)^2] \leq O(k\ell^r\nu) = \ell^{O(r)}\nu$. Plugging this into (40) and taking $\eta = 1/\sqrt{\nu}$ we get that

$$\mathbb{E}_c[B(H_s^c)] \leq \sqrt{\nu}\ell^{O(r)} + O(\sqrt{\nu}\,\mathbb{E}_c[\delta(H_s^c)]) \leq \sqrt{\nu}\ell^{O(r)},$$

and summing over $s \in \Sigma$ finishes the proof. □

Claim B.2.

$$\{0 \leq x_i \leq 1\} \vdash_{2k} \prod_{i\in[k]} x_i - (\prod_{i\in[k]} x_i)^2 \leq x_1 - x_1^2 + \ldots + x_k - x_k^2.$$

Proof. We prove this by induction. For $k = 1$ this is clear, and for $k = 2$ we have

$$\begin{aligned} xy - (xy)^2 &= xy - xy^2 + xy^2 - x^2y^2 \\ &= x(y - y^2) + (x - x^2)y^2 \\ &\leq y - y^2 + x - x^2. \end{aligned}$$

For the inductive step, let $k \geq 3$ and denote $y = x_1 \cdots x_{k-1}$. Applying the base case and then the inductive hypothesis we get

$$\prod_{i\in[k]} x_i - \left(\prod_{i\in[k]} x_i\right)^2 = yx_k - (yx_k)^2 \leq y - y^2 + x_k - x_k^2 \leq x_1 - x_1^2 + \ldots + x_{k-1}^2 - x_{k-1}^2.$$

One can check that the SoS degree used is at most $2k$. □

C Sum-of-Squares Certificate of Expansion in the Johnson Graph

In this section we give an SoS proof of Theorem 3.17. Our proof follows the same lines as [KMMS18], except that we implement each step carefully by a low-degree SoS proof. For convenience, we restate Theorem 3.17 below.

Theorem C.1 (Expansion Theorem 3.17 for Johnson Graphs restated)**.** *For all $\alpha \in (0,1)$, all integers $\ell \geq 1/\alpha$ and $n \geq \ell$, the following holds: Let $J_{n,\ell,\alpha}$ be the α-noisy Johnson graph. For every constant $\gamma \in (0,1)$ and positive integer $r \leq O(\ell)$, every function $F : V(J) \to \mathbb{R}$ that is (r,γ)-pseudorandom has high expansion:*

1. $\{F(X) \in [0,1]\}_{X\in V(J)} \vdash_{O(1)}$
   $\langle F, LF\rangle \geq \delta(1-(1-\alpha)^{r+1})(1-\gamma^{1/3}\exp(r)) - \sum_{j=0}^{r} \frac{c_j \ell^j}{\gamma} \mathbb{E}_{a\sim\binom{[n]}{j}}[q_a(F)(\delta(F^2|_a) - \gamma)] + B(F)$
2. $\{F(X) \in [0,1]\} \vdash_2 0 \leq q_a(F) \leq 1,$

where for all $j \leq r$, c_j's are positive constants of size at most $\exp(r)$, for all $a \subseteq [n]$ of size j, $q_a(F)$ is a degree 2 polynomials, and $B(F) = \frac{4}{3}\mathbb{E}_X[(F^3 - F)\Pi_{\geq\lambda_r} F]$ for $\lambda_r = (1-\alpha)^r$. Here, $\Pi_{\geq\lambda}$ is the projection operator on the space of spanned by eigenvectors ofJ of eigenvalues at least λ.

It will be more convenient for us to move to a closely related Cayley graph, which we denote by $C_{n,\ell,\alpha}$. This graph is essentially the same as the Johnson graph, albeit viewed as a product domain. In Theorem C.18, we state an analogous structure theorem for "permutation-invariant" sets on $C_{n,\ell,\alpha}$ that we show in Theorem C.18, and it is easy to derive Theorem C.1 above from it.

Notation. We use $[n]$ to denote the set $\{0,\ldots,n-1\}$, and also the group $(\mathbb{Z}/n\mathbb{Z})$, the natural numbers modulo n. Generally, when we take a set S and raise it to a positive integer power ℓ, we mean the set of all ordered multisets of elements of S of size ℓ. We use χ_t for $t \in [n]$ to denote the characters of the group $\mathbb{Z}/n\mathbb{Z}$ (or the eigenvectors of the n-cycle), where $\chi_t : [n] \to \mathbb{C}$ is the function $\chi_t(x) = e^{\frac{2\pi i t x}{n}}$. We will use $\lambda_G(v)$ to denote the eigenvalue of v which is an eigenvector of the adjacency matrix of graph G. For a string $S \in \Sigma^m$, for some alphabet Σ, and a set $I \subseteq [m]$, we denote its restriction to the set of coordinates in I, by $S|_I$.

C.1 The Graph $C_{n,\ell,\alpha}$ and its Spectrum

Definition C.2 (Johnson-approximating Cayley Graph $C_{n,\ell,\alpha}$)**.** *Let α be a number in $(0,1)$ and ℓ be a positive integer. Let n be a positive integer such that $n > \ell$.*

1. *The vertex set of $C_{n,\ell,\alpha}$ is $[n]^\ell$. We will drop the subscript (n,ℓ,α) in $C_{n,\ell,\alpha}$ when these parameters are clear from context.*
2. *The edges are described by the following random process. For a vertex $X = (x_1, \ldots, x_\ell), x_i \in [n]$, the distribution over the neighbours of X is described by: choose $(y_1, \ldots, y_\ell)$ uniformly at random from $[n]^\ell$ and $b = (b_1, \ldots, b_\ell) \sim \{0,1\}^\ell$ such that the Hamming weight of b equals $\alpha\ell$, and output $Z = (x_1 + b_1 \cdot y_1, \ldots, x_l + b_\ell \cdot y_\ell)$ as the neighbour of X.*

It is easy to verify that the graph defined above is a weighted Cayley graph with vertex set being the elements of the group $[n]^\ell = (\mathbb{Z}/n\mathbb{Z})^\ell$. The natural group operation associated with this set is component-wise addition modulo n.

We next discuss the spectral properties of $C_{n,\ell\alpha}$. Overloading notations, we let $C = C_{n,\ell,\alpha}$ also denote the normalized adjacency matrix of the graph C. Note firstly that the eigenvectors of C are the characters of the group $[n]^\ell$, namely χ_T where $T = (T_1, \ldots, T_\ell) \in [n]^\ell$ defined as $\chi_T(x) = \chi_{T_1}(x_1) \cdot \ldots \cdot \chi_{T_\ell}(x_\ell)$ for $x \in [n]^\ell$ We next define the degree of an eigenvector.

Definition C.3 (Degree of χ_T)**.** *For all $T \in [n]^\ell$, where $T = (T_1, \ldots, T_\ell)$, define the degree of χ_T as:*

$$\mathsf{deg}(\chi_T) = |T| := |\{i \mid T_i \neq 0\}|.$$

The following lemma asserts that the eigenvalue of χ_T is roughly $(1-\alpha)^{|T|}$:

Lemma C.4. *Let $\lambda_C(\chi_T)$ denote the eigenvalue of C corresponding to the eigenvector χ_T for $T \in [n]^\ell$. We have that,*

$$\lambda_C(\chi_T) = \begin{cases} \frac{\binom{\ell-|T|}{(1-\alpha)\ell-|T|}}{\binom{\ell}{(1-\alpha)\ell}}, & |T| \leq (1-\alpha)\ell \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Let $T = (T_1, \ldots, T_\ell)$. For all $X \in [n]^\ell$, we have that,

$$\begin{aligned} C \cdot \chi_T(X) = \mathop{\mathbb{E}}_{y,b}[\chi_T(x_1 + b_1 y, \ldots, x_l + b_l y_l)] &= \chi_T(X) \mathop{\mathbb{E}}_{y,b}[\chi_{T_1,\ldots,T_l}(b_1 y_1, \ldots, b_l y_l)] \\ &= \chi_T(X) \mathop{\mathbb{E}}_{y,b}[\chi_{(b_1 T_1, \ldots, b_l T_l)}(y)]. \end{aligned}$$

For $y, S \in [n]$ and $S \neq 0$, we know that the eigenvector χ_S is orthogonal to the eigenvector χ_0, equivalently that $\mathbb{E}_y[\chi_S(y)] = 0$, whereas if $S = 0$ then $\mathbb{E}_y[\chi_S(y)] = 1$. So we get that,

$$\begin{aligned} \lambda_C(\chi_T) = \mathop{\mathbb{E}}_{y,b}[\chi_{(b_1 T_1, \ldots, b_\ell T_\ell)}(y)] &= \Pr_b[(b_1 T_1, \ldots, b_\ell T_\ell) = 0^\ell] \\ &= \begin{cases} \frac{\binom{l-|T|}{(1-\alpha)l-|T|}}{\binom{l}{(1-\alpha)l}}, & |T| \leq (1-\alpha)\ell \\ 0, & \text{otherwise.} \end{cases}. \end{aligned}$$

□

C.2 Analyzing non-expanding sets of the Johnson graph

Since we want to deal with sets in the Johnson graph $J_{n,\ell,\alpha}$ we will only consider "permutation-invariant" sets on $C_{n,\ell,\alpha}$. Notice that the vertices of the Johnson graph are subsets of $[n]$ of size ℓ, whereas the vertices of the Johnson-approximating graph are ordered ℓ-tuples of $[n]$. Therefore, given a set S in the Johnson graph, it has a natural mapping to the set S' which is a subset of the vertices of the Johnson-approximating graph C, $S' := \{(x_{\pi(1)}, \ldots, x_{\pi(\ell)}) \mid \pi : [\ell] \to [\ell], \{x_1, \ldots, x_\ell\} \in S\}$. This leads to the following definition:

Definition C.5 (Permutation-invariance)**.** *We say that a set $S \subseteq C_{n,\ell,\alpha}$ is permutation-invariant if for all permutations $\pi \in \mathcal{S}_\ell$ and all $X = (x_1, \ldots, x_\ell) \in S$ we have that $X_\pi = (x_{\pi(1)}, \ldots, x_{\pi(\ell)})$ belongs to S. Similarly, a function $F : V(C) \to \mathbb{R}$ is permutation invariant if for all inputs $X = (x_1, \ldots, x_\ell)$, we have that $F(x_1, \ldots, x_\ell) = F(x_{\pi(1)}, \ldots, x_{\pi(\ell)})$, for all $\pi \in S_\ell$. Further let $\mathcal{A}_{inv}$ denote the set of axioms that F is permutation-invariant, that is,*

$$\mathcal{A}_{inv} := \{F(x_1, \ldots, x_\ell) = F(x_{\pi(1)}, \ldots, x_{\pi(\ell)})\}_{\pi \in \mathcal{S}_\ell, X \in [n]^\ell}.$$

Since the set of vertices in C that correspond to some set of vertices in J are permutation invariant it will be enough to focus are attention on these special sets and from now on whenever we refer to a set in $V(C)$, the reader can assume that it is permutation-invariant.

To analyze non-expanding sets of C, we will consider permutation-invariant functions $F : V(C) \to [0, 1]$. Typically one would consider 0/1-valued functions F, where F is the indicator function of a set S, i.e. $F(X) = 1$ when $X \in S$. But since we need to analyze "approximate-sets" (the indicator function is approximated by a polynomial that takes values close to 0/1), $F(X)$ could take any value between $[0, 1]$.

Recall that the Fourier decomposition of F gives us that, $F(X) = \sum_T \hat{F}(T)\chi_T(X)$. We will now define the following for a function F:

Definition C.6. *Given a function $F \colon V(C) \to [0, 1]$, we have the following level decomposition:*

1. *We write $F = F_0 + \ldots + F_\ell$, where $F_i(X) = \sum_{T:|T|=i} \hat{F}(T)\chi_T(X)$. We will call F a level i function, if its Fourier decomposition has degree i characters only, i.e. $\hat{F}(T) = 0$, for all T such that $|T| \neq i$.*

2. *Let $f_{i,F} : [n]^i \to \mathbb{R}$ be a function defined as,*

$$f_{i,F}(x_1, \ldots, x_i) := \sum_{T_1, \ldots, T_i \in ([n] \setminus 0)^i} \hat{F}(T_1, \ldots, T_i, 0, \ldots)\chi_{T_1, \ldots, T_i}(x_1, \ldots, x_i),$$

Let $X = (x_1, \ldots, x_j) \in [n]^j$ and I be a subset of $\{1, \ldots, j\}$. Let $I = \{k_1, \ldots, k_{|I|}\}$ where $k_1 < k_2 < \ldots < k_{|I|}$. We will use $X|_I$ to denote the ordered tuple of elements $(x_{k_1}, \ldots, x_{k_{|I|}})$. We will now state some simple properties of F that are implied by permutation-invariance.

Lemma C.7. *For all functions $F : [n]^\ell \to \mathbb{R}$ that are permutation-invariant, we have that:*

1. *$\hat{F}(T_1, \ldots, T_\ell) = \hat{F}(T_{\pi(1)}, \ldots, T_{\pi(\ell)})$, for all $(T_1, \ldots, T_\ell) \in [n]^\ell$ and all permutations $\pi : [\ell] \to [\ell]$.*

2. *The functions F_i and $f_{i,F}$ are also permutation-invariant.*

3. $F_i(X) = \sum_{\substack{I \subseteq [\ell] \\ |I|=i}} f_{i,F}(X|_I)$.

Proof. A straightforward manipulation of the definitions. □

We note that the Fourier coefficients $\hat{F}(T), \forall T \in (\{0,1\}^k)^\ell$, the level-$i$ functions $F_i(X)$ and $f_i(X)$ are all linear functions of the indeterminates $\{F(X)\}$, hence we shall also think of them as indeterminates when arguing about SoS proofs.

C.3 Restrictions

In this section, we define restrictions of functions F and state several lemmas related to them.

Definition C.8 (r-restricted subcubes of C)**.** *Given an ordered tuple, $A = (a_1, \ldots, a_r)$ for $a_i \in [n]$ and $r \leq \ell - 1$, we let $C|_A$ denote the subset of vertices of C whose first r coordinates are restricted to be $(a_1, \ldots, a_r)$. We call such a subset an r-restricted subcube of C.*

Definition C.9 (Restrictions)**.** *Given a function $F : [n]^\ell \to \mathbb{R}$ and an ordered tuple, $A = (a_1, \ldots, a_r)$ for $a_i \in [n]$ and $1 \leq r \leq \ell - 1$, we define the restricted function $F|_A : [n]^{\ell-r} \to \mathbb{R}$ as,*

$$F|_A(x_1, \ldots, x_{\ell-r}) = F(a_1, \ldots, a_r, x_1, \ldots, x_{\ell-r}).$$

Further, let $\delta_A(F)$ denote the mass of the function restricted to A, that is,

$$\delta_A(F) := \delta(F|_A) = \mathop{\mathbb{E}}_{X \in [n]^{\ell-r}}[F|_A(X)].$$

For convenience, when $A = \phi$ ($r = 0$), define $F|_A(X) := F(X)$, and $\delta_A(F) := \delta(F) = \mathbb{E}_{X \in [n]^\ell}[F(X)]$.

The following lemma gives a relation between the level functions of a function F and the level function of its restrictions.

Lemma C.10. *Let F be a permutation-invariant function on $V(C)$. Then we have the following:*

1. *For all $a \in [n]$ and for all i such that $0 \leq i \leq \ell - 1$, and all $X \in [n]^i$, we have that,*

$$f_{i+1,F}(a, X) = f_{i,F|_{\{a\}}}(X) - f_{i,F}(X).$$

2. *For all integers i such that $0 \leq i \leq \ell$ and for all $X \in [n]^i$, we get an inclusion-exclusion formula for $f_i(X)$ in terms of restrictions of F:*

$$f_{i,F}(X) = \sum_{B \subseteq \{1,\ldots,i\}} (-1)^{i-|B|} \delta_{X|_B}(F),$$

where $X|_B$ is the ordered tuple of elements of X restricted to the indices in B.

Proof. We prove each item separately.

Proof of the first item. Using the definition, we can expand out $f_{i+1,F}$ to get that,

$$f_{i,F}(a,X) = \sum_{(T_1,\ldots,T_{i+1})\in([n]\setminus 0)^{i+1}} \hat{F}(T_1,\ldots,T_{i+1},0,\ldots,0)\chi_{T_1,\ldots,T_{i+1}}(a,X).$$

We can split this sum into two parts, one where T_1 can take *any* value (even 0) and the second where $T_1 = 0$. We get that,

$$f_{i,F}(a,X) = \sum_{\substack{T_1\in[n]\\ T\in([n]\setminus 0)^i}} \hat{F}(T_1,T,0,\ldots,0,\ldots,0)\chi_{T_1,T}(a,X) - \sum_{\substack{T_1=0\\ T\in([n]\setminus 0)^i}} \hat{F}(0,T,0,\ldots,0)\chi_T(X).$$

We will show that the first term equals $f_{i,F|_{\{a\}}}(X)$ and the second term equals $f_{i,F}(X)$. This implies the conclusion needed.

For the first term we have that,

$$\begin{aligned}
&\sum_{\substack{T_1\in[n]\\ T\in([n]\setminus 0)^i}} \hat{F}(T_1,T,0,\ldots,0)\chi_{T_1,T}(a,X)\\
&= \sum_{\substack{T_1\in[n]\\ T\in([n]\setminus 0)^i}} \mathop{\mathbb{E}}_{\substack{Y_1\in[n],\\ Y\in[n]^{\ell-1}}} \left[F(Y_1,Y)\overline{\chi_{T_1}}(Y_1)\overline{\chi_{(T,0,\ldots,0)}(Y)}\right]\chi_{T_1}(a)\chi_T(X)\\
&= \sum_{T\in([n]\setminus 0)^i} \mathop{\mathbb{E}}_{\substack{Y_1\in[n],\\ Y\in[n]^{\ell-1}}} \left[F(Y_1,Y)\overline{\chi_{(T,0,\ldots,0)}(Y)}\sum_{T_1\in[n]}\chi_{T_1}(a-Y_1)\right]\chi_T(X). \qquad (43)
\end{aligned}$$

We now have that $\sum_{T_1\in[n]}\chi_{T_1}(a-Y_1) = 0$ if $Y_1 \neq a$ and equals n otherwise. Using this fact we get that (43) is equal to

$$\sum_{T\in([n]\setminus 0)^i} \frac{1}{n}\cdot \mathop{\mathbb{E}}_{Y\in[n]^{\ell-1}}\left[F(a,Y)\overline{\chi_{(T,0,\ldots,0)}(Y)}\cdot n\right]\chi_T(X) = \sum_{T\in([n]\setminus 0)^i} \widehat{F|_{\{a\}}}(T,0,\ldots,0)\chi_T(X),$$

which is equal to $f_{i,F|_{\{a\}}}(X)$ by definition.

As for the second term, it is equal to

$$\begin{aligned}
&\sum_{\substack{T_1=0\\ (T_2,\ldots,T_{i+1})\in([n]\setminus 0)^i}} \widehat{F}(0,T_2,\ldots,T_{i+1},0,\ldots,0)\chi_{T_2,\ldots,T_{i+1}}(X)\\
&= \sum_{(T_2,\ldots,T_{i+1})\in([n]\setminus 0)^i} \hat{F}(T_2,\ldots,T_{i+1},0,\ldots,0)\chi_{T_2,\ldots,T_{i+1}}(X),
\end{aligned}$$

since by Lemma C.7 (1) we have that $\hat{F}(0,T_2,\ldots,T_{i+1},0,\ldots) = \hat{F}(0,T_2,\ldots,T_{i+1},0,\ldots)$. Since the last equality is the definition of $f_{i,F}(X)$, the conclusion follows.

Proof of the second item. We will prove this claim by induction on i. For the base case of $i = 0$, by definition, we have that,

$$f_{0,F}(\phi) = \hat{F}(0,\ldots,0) = \mathop{\mathbb{E}}_{X\in[n]^\ell}[F(X)] = \delta(F) = \delta_\phi(F) = \sum_{B\subseteq\phi} \delta_{\phi|_B}(F).$$

Now let us assume that for all permutation-invariant functions G the claim holds for $i-1$, i.e. for all $X \in [n^{i-1}]$, we have that $f_{i-1,G}(X) = \sum_{B\in\{1,\ldots,i-1\}} (-1)^{i-1-|B|}\delta_{X|_B}(G)$. Now we will prove the claim for $f_{i,F}$, thus completing the induction. Let $X = (x_1, X')$, where $X \in [n]^i, x_1 \in [n]$ and $X' \in [n]^{i-1}$. Then by property (1) of the same lemma, we have that,

$$f_{i,F}(X) = f_{i,F|_{\{x_1\}}}(X') - f_{i,F}(X').$$

Expanding the RHS using the induction hypothesis on the functions $F|_{\{x_1\}}$ and F, we get that,

$$\begin{aligned}
f_{i,F}(X) &= \sum_{B'\in\{1,\ldots,i-1\}} (-1)^{i-1-|B'|}\delta_{X'|_{B'}}(F|_{\{x_1\}}) - \sum_{B'\in\{1,\ldots,i-1\}} (-1)^{i-1-|B'|}\delta_{X'|_{B'}}(F) \\
&= \sum_{B'\in\{1,\ldots,i-1\}} (-1)^{i-(1+|B'|)}\delta_{(x_1,X'|_{B'})}(F) + \sum_{B'\in\{1,\ldots,i-1\}} (-1)^{i-|B'|}\delta_{X'|_{B'}}(F) \\
&= \sum_{B\in\{1,\ldots,i\}:1\in B} (-1)^{i-|B|}\delta_{X|_B}(F) + \sum_{B\in\{1,\ldots,i\}:1\notin B} (-1)^{i-|B|}\delta_{X|_B}(F) \\
&= \sum_{B\in\{1,\ldots,i\}} (-1)^{i-|B|}\delta_{X|_B}(F).
\end{aligned}$$

This completes the inductive step and the proof of the lemma. □

C.4 Calculating 2-nd Moments

Let $W_i(F)$ denote the weight of F on the ith level, that is, let $\eta_i = W_i[F] = \mathbb{E}_X[F_i(X)^2]$. Let $\delta(F)$ denote $\mathbb{E}_X[F(X)]$. Now we will derive some lemmas about the second moments. We get the following relation between the level-i Fourier weight η_i and the ℓ_2-norm of the f_i's.

Lemma C.11. *For all $i \in [\ell]$, given the variables $\{F(X)\}_{X\in V(C)}$:*

$$\mathcal{A}_{inv} \quad \vdash_2 \quad \mathop{\mathbb{E}}_{X\in[n]^i}[f_{i,F}(X)^2] = \frac{\eta_i}{\binom{\ell}{i}}.$$

Proof. Since $F_i(A) = \sum_{I\subseteq[\ell]} f_i(A|_I)$, we have that,

$$\begin{aligned}
\eta_i = \mathop{\mathbb{E}}_{A\sim[n]^\ell}[F_i(A)^2] = \mathop{\mathbb{E}}_{A}[(\sum_I f_i(A_I))^2] &= \sum_I \mathop{\mathbb{E}}_{A}[f_i(A_I)^2] + \sum_{I\neq I'} \mathop{\mathbb{E}}_{A}[f_i(A_I)f_i(A_{I'})] \\
&= \sum_I \mathop{\mathbb{E}}_{A}[f_i(A_I)^2] + 0 \\
&= \mathop{\mathbb{E}}_{(x_1,\ldots,x_i)}[f_i(X)^2] \cdot \binom{\ell}{i},
\end{aligned}$$

and rearranging implies the lemma. □

Lemma C.12. *For all $i \in [\ell]$, for all $a, b \in [i]$, for all $A \in (\{0,1\}^k)^a$, given the indeterminates $\{F(X)\}_{X\in V(C)}$:*

$$\mathcal{A}_{inv}(F) \quad \vdash_2 \quad \mathop{\mathbb{E}}_{X\in(\{0,1\}^k)^{i-a}}[f_i(A,X)^2] \leq \frac{\delta(F^2|_A)}{\binom{l-a}{i-a}}.$$

Proof. By Lemma C.11, we have that,

$$\mathop{\mathbb{E}}_{X\in(\{0,1\}^k)^{i-a}}[f_i(A,X)^2] = \mathop{\mathbb{E}}_{X\sim(\{0,1\}^k)^{i-a}}[f_{i-a,F|_A}(B)^2] = \frac{W^{i-a}(F|_A)}{\binom{l-a}{i-a}}. \tag{44}$$

We also have that,

$$W^{i-a}(F|_A) \leq \mathop{\mathbb{E}}_{X\in(\{0,1\}^k)^{i-a}}[F|_A(X)^2] = \delta(F^2|_A), \tag{45}$$

where the first inequality is a degree 2 SoS-inequality. The proof is concluded by combining (44), (45). □

C.5 Bounding 4-th moments

The next part of the proof is to prove a lower bound on the fourth moment of F_i (using the fact it is correlated with a Boolean valued function, hence is supposed to have a high 4th moment) as well as an upper bound on the fourth moment of F_i (using the fact F is pseudo-random). We begin with the following lemma which gives us the lower bound:

Lemma C.13 (Lower Bound)**.** *For all real $\epsilon > 0$ given the indeterminates $\{F(X)\}_{X\in V(C)}$ we have that,*

$$\mathop{\mathbb{E}}_{X}[F_i^4(X)] \geq 4\epsilon^3\eta_i - 3\epsilon^4\delta(F) + B(F),$$

where $B(F) = 4\varepsilon^3\,\mathbb{E}_X[(F^3-F)F_i] + 3\varepsilon^4\,\mathbb{E}_X[F-F^4]$.

Proof. We have that,

$$\eta_i = \langle F, F_i\rangle = \mathop{\mathbb{E}}_{X}[F(X)^3\cdot F_i(X)] + \mathop{\mathbb{E}}_{X}[(F(X)-F(X)^3)\cdot F_i(X)], \tag{46}$$

since $F = \sum F_i$ is an orthogonal decomposition of F. Using Fact 2.4 on the first term we get:

$$\begin{aligned}
\mathop{\mathbb{E}}_{X}[F(X)^3\cdot F_i(X)] &= \mathop{\mathbb{E}}_{X}[F(X)^3\cdot F_i(X)] \\
&\leq \frac{3\epsilon}{4}\mathop{\mathbb{E}}_{X}[F(X)^4] + \frac{1}{4\epsilon^3}\mathop{\mathbb{E}}_{X}[F_i(X)^4], \\
&= \frac{3\epsilon}{4}\mathop{\mathbb{E}}_{X}[F(X)] + \frac{3\epsilon}{4}\mathop{\mathbb{E}}_{X}[F(X)^4 - F(X)] + \frac{1}{4\epsilon^3}\mathop{\mathbb{E}}_{X}[F_i(X)^4],
\end{aligned}$$

for all real $\epsilon > 0$. Plugging this into (46) and rearranging gives the lemma. □

We will now prove an upper bound on $\mathbb{E}_X[F_i^4(X)]$. We have the following lemma which bounds the fourth moment of F_i in terms of the second moment of F_i:

Lemma C.14 (Upper Bound)**.** *For all integers $i \in [\ell]$, there exist positive constants $c_1,\ldots,c_i \leq \exp(i)$ such that for all F:*

$$\mathcal{A}_{inv} \vdash_{O(1)} \mathop{\mathbb{E}}_{X}[F_i(X)^4] \leq \exp(i)\gamma\eta_i + \sum_{j=0}^{i} c_j\ell^i \mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{\ell-j}}[f_i(a,b)^2]\cdot(\delta(F^2|_a)-\gamma)].$$

Proof. When the set X and the function F is clear from context, we use $f_i(I)$ in place of $f_i(X|_I)$. Thus,

$$\mathop{\mathbb{E}}_X[F_i^4(X)] = \sum_{\substack{I_1,\dots,I_4\subseteq[l]\\|I_j|=i,\forall j\in[4]}} \mathop{\mathbb{E}}_X[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)].$$

Denote,

$$D(i,d) = \{(I_1,\dots,I_4) \mid \forall j\in[4], I_j\subseteq[l], |I_j| = i, |I_1\cup I_2\cup I_3\cup I_4| = d\}.$$

Note that for all $d\in[i,4i]$,

$$|D(i,d)| \le \binom{\ell}{d}\cdot\binom{d}{i}^4 \le \left(\frac{\ell e}{d}\right)^d\left(\frac{de}{i}\right)^{4i} \le \ell^d\exp(i).$$

We will prove the following lemma about the four-wise products, for $(I_1,\dots,I_4)\in D(i,d)$. We defer the proof to Section C.6.

Lemma C.15. *Let i,d be integers such that $i\le d\le 4i$ and let $I_1,\dots,I_4\subseteq[l]$, such that $(I_1,I_2,I_3,I_4)\in D(i,d)$. Then there exist integers $t_1,t_2\in[i]$ such that for all γ:*

$$\begin{aligned}
&\mathop{\mathbb{E}}_X[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]\\
&\le \left(\frac{O(i)}{\ell}\right)^d\gamma\eta_i + \left(\frac{O(i)}{\ell}\right)^{d-i}\mathop{\mathbb{E}}_{a_1\sim[n]^{t_1}}[\mathop{\mathbb{E}}_{b_1\sim[n]^{i-t_1}}[f_i(a_1,b_1)^2]\cdot(\delta(F^2|_{a_1})-\gamma)]\\
&+\left(\frac{O(i)}{\ell}\right)^{d-i}\mathop{\mathbb{E}}_{a_2\sim[n]^{t_2}}[\mathop{\mathbb{E}}_{b_2\sim[n]^{i-t_2}}[f_i(a_2,b_2)^2](\delta(F^2|_{a_2})-\gamma)].
\end{aligned}$$

We now use Lemma C.15 to finish the proof of Lemma C.14. Indeed, we get that

$$\begin{aligned}
&\mathop{\mathbb{E}}_X[F_i^4(X)]\\
&= \sum_{\substack{I_1,\dots,I_4\subseteq[l]\\|I_j|=i,\forall j\in[4]}}\mathop{\mathbb{E}}_X[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]\\
&\le \sum_{d=i}^{4i}\sum_{\substack{I_1,\dots,I_4\subseteq[l]\\(I_1,\dots,I_4)\in D(i,d)}}\mathop{\mathbb{E}}_X[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]\\
&\le \sum_{d=i}^{4i}\sum_{\substack{I_1,\dots,I_4\subseteq[l]\\(I_1,\dots,I_4)\in D(i,d)}}\left(\frac{O(i)}{\ell}\right)^d\gamma\eta_i + \left(\frac{O(i)}{\ell}\right)^{d-i}\mathop{\mathbb{E}}_{a_1\sim[n]^{t_1}}[\mathop{\mathbb{E}}_{b_1\sim[n]^{i-t_1}}[f_i(a_1,b_1)^2]\cdot(\delta(F^2|_{a_1})-\gamma)]\\
&\qquad\qquad + \left(\frac{O(i)}{\ell}\right)^{d-i}\mathop{\mathbb{E}}_{a_2\sim[n]^{t_2}}[\mathop{\mathbb{E}}_{b_2\sim[n]^{i-t_2}}[f_i(a_2,b_2)^2](\delta(F^2|_{a_2})-\gamma)]\\
&\le \exp(i)\gamma\eta_i + \sum_{j=0}^{i}c_j\ell^i\mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2]\cdot(\delta(F^2|_a)-\gamma)],
\end{aligned}$$

for some positive constants $c_j\le\exp(i)$, where we have used that $|D(i,d)|\le\ell^d\exp(i)$. □

C.6 Proof of Lemma C.15

We begin by analyzing the simple case in which there is an index that appears in only one of the I_j's, and show that in this case the expectation is just equal to 0:

Proposition C.16. *If there is an index j that appears in only one of the I_ks, then,*

$$\mathop{\mathbb{E}}_{X}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] = 0.$$

Proof. Without loss of generality we can assume that 1 appears only in I_4. We have that,

$$\begin{aligned}
\mathop{\mathbb{E}}_{X}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] &= \mathop{\mathbb{E}}_{x_1,\ldots,x_l}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] \\
&= \mathop{\mathbb{E}}_{x_2,\ldots,x_l}[f_i(I_1)f_i(I_2)f_i(I_3)\mathop{\mathbb{E}}_{x_1}[f_i(I_4) \mid x_2 = x_2', \ldots, x_l = x_l']] \\
&= \mathop{\mathbb{E}}_{x_2,\ldots,x_l}[f_i(I_1)f_i(I_2)f_i(I_3)\delta_{(x_2',\ldots,x_l')\cap I_4}(f_i)] \\
&= 0,
\end{aligned}$$

where the last equality follows by Lemma C.7. □

For the rest of the proof we assume each index in $I_1 \cup I_2 \cup I_3 \cup I_4$ appears in at least two sets. Without loss of generality we can assume that $\cup_{j=1}^4 I_j = [d]$. Let $H_2, H_3, H_4 \subseteq [d]$ be the set of elements that appear in $2, 3, 4$ of the sets $(I_1, \ldots, I_4)$ respectively. Furthermore let $H_2 = X_{12} \cup X_{13} \ldots X_{34}$, where X_{jk} is the set of elements that occur in I_j and in I_k. We will abuse notation and use H_j to also denote the set of random variables $(x_{j_1}, \ldots, x_{j_m}), m \in H_j$ and $x_m \sim \{0,1\}^k$. Firstly we get that,

$$\begin{aligned}
&\mathop{\mathbb{E}}_{H_2}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] \\
=&\mathop{\mathbb{E}}_{X_{12},\ldots,X_{34}}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] \\
=&\mathop{\mathbb{E}}_{\substack{X_{13},X_{14},\\X_{23},X_{24}}}[\mathop{\mathbb{E}}_{X_{12}}[f_i(I_1)f_i(I_2)]\mathop{\mathbb{E}}_{X_{34}}[f_i(I_3)f_i(I_4)]] \\
\leq&\frac{\varepsilon}{2}\mathop{\mathbb{E}}_{\substack{X_{13},X_{14},\\X_{23},X_{24}}}[\mathop{\mathbb{E}}_{X_{12}}[f_i(I_1)f_i(I_2)]^2] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{\substack{X_{13},X_{14},\\X_{23},X_{24}}}[\mathop{\mathbb{E}}_{X_{34}}[f_i(I_3)f_i(I_4)]^2] &(47)\\
\leq&\frac{\varepsilon}{2}\mathop{\mathbb{E}}_{\substack{X_{13},X_{14},\\X_{23},X_{24}}}[\mathop{\mathbb{E}}_{X_{12}}[f_i(I_1)^2]\mathop{\mathbb{E}}_{X_{12}}[f_i(I_2)^2]] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{\substack{X_{13},X_{14},\\X_{23},X_{24}}}[\mathop{\mathbb{E}}_{X_{34}}[f_i(I_3)^2]\mathop{\mathbb{E}}_{X_{34}}[f_i(I_4)^2]] &(48)\\
\leq&\frac{\varepsilon}{2}\mathop{\mathbb{E}}_{X_{12},X_{13},X_{14}}[f_i(I_1)^2]\mathop{\mathbb{E}}_{X_{12},X_{23},X_{24}}[f_i(I_2)^2] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{X_{34},X_{13},X_{23}}[f_i(I_3)^2]\mathop{\mathbb{E}}_{X_{34},X_{14},X_{24}}[f_i(I_4)^2] \\
=&\frac{\varepsilon}{2}\mathop{\mathbb{E}}_{H_2}[f_i(I_1)^2]\mathop{\mathbb{E}}_{H_2}[f_i(I_2)^2] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{H_2}[f_i(I_3)^2]\mathop{\mathbb{E}}_{H_2}[f_i(I_4)^2], &(49)
\end{aligned}$$

where inequalities (47), (48) follow by Cauchy Schwarz (Lemma 2.2 and Lemma 2.3).

We will now bound the expression $\mathbb{E}_X[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]$. We have that,

$$\mathop{\mathbb{E}}_{X}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]$$

$$
\begin{aligned}
&= \mathop{\mathbb{E}}_{H_4,H_3}[\mathop{\mathbb{E}}_{H_2}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)]] \\
&\leq \frac{\varepsilon}{2}\mathop{\mathbb{E}}_{H_4,H_3}[\mathop{\mathbb{E}}_{H_2}[f_i(I_1)^2]\mathop{\mathbb{E}}_{H_2}[f_i(I_2)^2]] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{H_4,H_3}[\mathop{\mathbb{E}}_{H_2}[f_i(I_3)^2]\mathop{\mathbb{E}}_{H_2}[f_i(I_4)^2]] \\
&= \frac{\varepsilon}{2}\mathop{\mathbb{E}}_{A}[\mathop{\mathbb{E}}_{B_1}[f_i(I_1)^2]\mathop{\mathbb{E}}_{B_2}[f_i(I_2)^2]] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{C}[\mathop{\mathbb{E}}_{D_1}[f_i(I_3)^2]\mathop{\mathbb{E}}_{D_2}[f_i(I_4)^2]], \\
&= \frac{\varepsilon}{2}\mathop{\mathbb{E}}_{A}[\mathop{\mathbb{E}}_{B_1}[f_i(a,b_1)^2]\mathop{\mathbb{E}}_{B_2}[f_i(a,b_2)^2]] + \frac{1}{2\varepsilon}\mathop{\mathbb{E}}_{C}[\mathop{\mathbb{E}}_{D_1}[f_i(c,d_1)^2]\mathop{\mathbb{E}}_{D_2}[f_i(c,d_2)^2]], \qquad (50)
\end{aligned}
$$

with $A = H_4 \cup (H_3 \cap I_1 \cap I_2)$, $B_1 = (H_2 \cap I_1) \cup ((H_3 \cap I_1) \setminus I_2)$, $B_2 = (H_2 \cap I_2) \cup ((H_3 \cap I_2) \setminus I_1)$, $C = H_4 \cup (H_3 \cap I_3 \cap I_4)$, $D_1 = (H_2 \cap I_3) \cup ((H_3 \cap I_3) \setminus I_4)$, $D_2 = (H_2 \cap I_4) \cup ((H_3 \cap I_4) \setminus I_3)$. The sets a, b_1, b_2, c, d_1, d_2 are instantiations of the corresponding random variables, $A, \ldots, D_2$. We have that,

$$
\begin{aligned}
\mathop{\mathbb{E}}_{A}[\mathop{\mathbb{E}}_{B_1}[f_i(a,b_1)^2]\mathop{\mathbb{E}}_{B_2}[f_i(a,b_2)^2]] &\leq \mathop{\mathbb{E}}_{A}[\frac{\delta(F^2|_a)}{\binom{\ell-|A|}{i-|A|}}\mathop{\mathbb{E}}_{B_2}[f_i(a,b_2)^2]] \\
&= \frac{1}{\binom{\ell-|A|}{i-|A|}}\mathop{\mathbb{E}}_{A}[\gamma\mathop{\mathbb{E}}_{B_2}[f_i(a,b_2)^2]] + \mathop{\mathbb{E}}_{A}[(\delta(F^2|_A)-\gamma)\left(\frac{\mathbb{E}_{B_2}[f_i(a,b_2)^2]}{\binom{\ell-|A|}{i-|A|}}\right)], \\
&\leq \frac{\gamma}{\binom{\ell-|A|}{i-|A|}}\cdot\frac{\eta_i}{\binom{\ell}{i}} + \mathop{\mathbb{E}}_{A}[(\delta(F^2|_A)-\gamma)\left(\frac{\mathbb{E}_{B_2}[f_i(a,b_2)^2]}{\binom{\ell-|A|}{i-|A|}}\right)],
\end{aligned}
$$

where in the first inequality we have used Lemma C.12 to bound $\mathbb{E}_{B_1}[f_i(a,b_1)^2]$ for each $a \sim A$ and the last inequality uses Lemma C.11. Similarly we can bound the second term in equation (50):

$$
\mathop{\mathbb{E}}_{C}[\mathop{\mathbb{E}}_{D_1}[f_i(c,d_1)^2]\mathop{\mathbb{E}}_{D_2}[f_i(c,d_2)^2]] \leq \frac{\gamma}{\binom{\ell-|C|}{i-|C|}}\cdot\frac{\eta_i}{\binom{\ell}{i}} + \mathop{\mathbb{E}}_{C}[(\delta(F^2|_C)-\gamma)\left(\frac{\mathbb{E}_{D_2}[f_i(c,d_2)^2]}{\binom{\ell-|C|}{i-|C|}}\right)].
$$

Plugging in these bounds into equation (50) and setting $\varepsilon = \left(\frac{O(i)}{\ell}\right)^{\frac{|A|-|C|}{2}}$ we get:

$$
\begin{aligned}
&\mathop{\mathbb{E}}_{X}[f_i(I_1)f_i(I_2)f_i(I_3)f_i(I_4)] \\
&\leq \left(\frac{O(i)}{\ell}\right)^{\frac{4i-|A|-|C|}{2}}\gamma\eta_i + \mathop{\mathbb{E}}_{A}[\left(\frac{O(i)}{\ell}\right)^{\frac{2i-|A|-|C|}{2}}\mathop{\mathbb{E}}_{B_2}[f_i(a,b_2)^2]\cdot(\delta(F^2|_a)-\gamma)] \\
&+ \mathop{\mathbb{E}}_{C}[\left(\frac{O(i)}{\ell}\right)^{\frac{2i-|A|-|C|}{2}}\mathop{\mathbb{E}}_{D_2}[f_i(c,d_2)^2](\delta(F^2|_c)-\gamma)].
\end{aligned}
$$

We know that $4i - |A| - |C| = 2d$ and substituting this gives the lemma statement.

C.7 Combining the Upper and Lower Bounds on $\|F_i\|_4$: the Level i Inequality

Combining the upper and lower bounds on $\mathbb{E}[F_i(X)^4]$ we can now prove an upper bound on η_i.

Lemma C.17 (Level i inequality)**.** *There exist positive constants $c_0, \dots, c_i \le \exp(i)$ such that given the indeterminates $\{F(X)\}_{X \in V(C)}$,*

$$\mathcal{A}_{inv} \vdash_{O(1)} \eta_i \le \exp(i)\gamma^{1/3}\delta + B(F) + \sum_{j=0}^{i} \frac{c_j \ell^i}{\gamma} \mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2] \cdot (\delta(F^2|_a) - \gamma)],$$

where $B(F) = \frac{4}{3}\mathbb{E}_X[(F-F^3)F_i] + \exp(i)\gamma^{1/3}\mathbb{E}_X[F^4 - F]$.

Proof. Let $\delta = \mathbb{E}_X[F(X)]$ and $\eta_i = \mathbb{E}_X[F_i^2(X)]$. Under the Booleanity axioms on F, we have proved an upper and lower bound on $\mathbb{E}_X[F_i^4(X)]$,

- Upper Bound in Lemma C.14:

$$\mathop{\mathbb{E}}_X[F_i(X)^4] \le \exp(i)\gamma\eta_i + \sum_{j=0}^{i} c'_j \ell^i \mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2] \cdot (\delta(F^2|_a) - \gamma)], \tag{51}$$

  for some positive constants $c'_j \le \exp(i)$.

- Lower Bound in Lemma C.13: For all $\varepsilon \in \mathbb{R}$ we get,

$$\mathop{\mathbb{E}}_X[F_i^4(X)] \ge 4\varepsilon^3\eta_i - 3\varepsilon^4\delta + B_1(F), \tag{52}$$

  where $B_1(F) = 4\varepsilon^3\mathbb{E}_X[(F^3-F)F_i] + 3\varepsilon^4\mathbb{E}_X[F - F^4]$.

Combining (51) and (52), setting $\varepsilon = (\exp(i)\gamma)^{1/3}$ and dividing by $3\varepsilon^3$ yields

$$\eta_i \le \exp(i)\gamma^{1/3}\delta + \sum_{j=0}^{i} \frac{c'_j}{3\gamma\exp(i)} \ell^i \mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2] \cdot (\delta(F^2|_a) - \gamma)] - \frac{B_1(F)}{3\gamma\exp(i)}.$$

The proof is thus completed by setting $c_j = \frac{c'_j}{3\exp(i)}$ and $B(F) = \frac{4}{3}\mathbb{E}_X[(F-F^3)F_i] + \exp(i)\gamma^{1/3}\mathbb{E}_X[F^4 - F]$. □

C.8 Expansion of Pseudorandom sets

Lemma C.17 directly implies an expansion theorem for $C_{n,\ell,\alpha}$, as follows:

Theorem C.18 (Expansion Theorem for $C_{n,\ell,\alpha}$)**.** *For all $\alpha \in (0,1)$, all integers $\ell \ge 1/\alpha$ and $n \ge \ell$, the following holds: Let $C_{n,\ell,\alpha}$ be the Johnson-approximating Cayley graph. For every constant $\gamma \in (0,1)$ and positive integer $r \le O(\ell)$, every permutation-invariant function $F : V(C) \to \mathbb{R}$ that is (r,γ)-pseudorandom:*

1. $\mathcal{A}_{inv}(F) \cup \{F(X) \in [0,1]\}_{X \in V(C)} \vdash_{O(1)}$
   $\langle F, LF\rangle \ge \mathbb{E}[F](1-(1-\alpha)^{r+1})(1-\gamma^{1/3}\exp(r)) - \sum_{j=0}^{r} \frac{c_j\ell^j}{\gamma} \mathbb{E}_{a\sim[n]^j}[q_a(F)(\delta(F^2|_a) - \gamma)] + B(F)$.
2. $\{F(X) \in [0,1]\} \vdash_2 0 \le q_a(F) \le 1$,

where for all $j \le r$, c_j's are positive constants $\le \exp(r)$, for all $a \in [n]^j, j \le r$, $q_a(F)$ are degree 2 polynomials and $B(F) = \frac{4}{3}\mathbb{E}_X[(F^3-F)\sum_{i=0}^{r} F_i] + c\,\mathbb{E}_X[F - F^4]$, for a positive constant $c \le \exp(r)\gamma^{1/3}$.

Proof. We first have that: $\langle F, LF\rangle = \mathbb{E}[F^2] - \langle F, CF\rangle$, where L and C denote the normalized Laplacian and adjacency matrix of the graph $C_{k,\ell,\alpha}$ respectively. Let $1 = \lambda_1 \geq \ldots \lambda_\ell \geq -1$ denote the eigenvalues of C. We will henceforth bound $\langle F, CF\rangle$:

$$\begin{aligned}\langle F, CF\rangle = \sum_{i\in[l]} \lambda_i\eta_i &\leq \sum_{i=0}^{r} \lambda_i\eta_i + (\mathbb{E}[F^2] - \sum_{i=1}^{r}\eta_i)\lambda_{r+1}\\ &\leq (1-\lambda_{r+1})\sum_{i=0}^{r}\eta_i + \mathbb{E}[F]\lambda_{r+1},\end{aligned}$$

where in the last step we use that $F(X) \in [0,1]$. Let δ denote $\mathbb{E}[F]$. Using Lemma C.17 we get:

$$\begin{aligned}&\langle F, CF\rangle\\ &\leq (1-\lambda_{r+1})\sum_{i=0}^{r}\left(\exp(i)\gamma^{1/3}\delta + B_i(F) + \sum_{j=0}^{i}\frac{c'_{ij}\ell^i}{\gamma}\mathop{\mathbb{E}}_{a\sim[n]^j}[\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2]\cdot(\delta(F^2|_a)-\gamma)]\right) + \delta\lambda_{r+1}\\ &\leq (1-\lambda_{r+1})(\exp(r)\gamma^{1/3}\delta) + (1-\lambda_{r+1})\sum_i B_i(F)\\ &+\sum_{j=0}^{r}\mathop{\mathbb{E}}_{a\sim[n]^j}[(\delta(F^2|_a)-\gamma)(\sum_{i=j}^{r}(1-\lambda_{r+1})\frac{c'_{ij}\ell^i}{\gamma}\mathop{\mathbb{E}}_{b\sim[n]^{i-j}}[f_i(a,b)^2])] + \delta\lambda_{r+1},\end{aligned}$$

for some positive constants $c'_{ij} \leq \exp(i)$ and $B_i(F) = \frac{4}{3}\mathbb{E}_X[(F-F^3)F_i] + \exp(i)\gamma^{1/3}\mathbb{E}_X[F^4 - F]$.

Let us now look at the multipliers of $\delta(F^2|_a) - \gamma$, for each $a \in [n]^j$ consider the polynomial: $q_a(F) = \frac{1}{c_j\ell^j}\sum_{i=j}^{r}(1-\lambda_{r+1})c'_{ij}\ell^i\,\mathbb{E}_{b\sim[n]^{i-j}}[f_i(a,b)^2]$, for fixed constants $c_j \leq \exp(r)$ to be determined later. Using Lemma C.12 to bound $\mathbb{E}_{b\sim[n]^{i-j}}[f_i(a,b)^2]$ we get that:

$$\{F(X)\in[0,1]\} \vdash_2 \qquad 0 \leq q_a(F) \leq \frac{1}{c_j}\sum_{i=j}^{r}(1-\lambda_{r+1})c'_{ij}\ell^{i-j}\frac{\delta(F^2|_a)}{\binom{\ell-j}{i-j}} \leq 1,$$

for $c_j = \sum_{i=j}^{r}(1-\lambda_{r+1})c'_{ij}\frac{\ell^{i-j}}{\binom{\ell-j}{i-j}} \leq \exp(r)$. Plugging in the polynomials $q_a(F)$ and rearranging yields

$$\langle F, LF\rangle \geq \delta(1-\lambda_{r+1})(1-\gamma^{1/3}\exp(r)) - \sum_{j=0}^{r}\frac{c_j\ell^j}{\gamma}\mathop{\mathbb{E}}_{a\sim[n]^j}[q_a(F)(\delta(F^2|_a)-\gamma)] + B(F),$$

for some positive constants $c_j \leq \exp(r)$ and $B(F) = \frac{4}{3}\mathbb{E}_X[(F^3-F)\sum_{i=0}^{r}F_i] + c\,\mathbb{E}_X[F-F^4]$, for a positive constant $c \leq \exp(r)\gamma^{1/3}$.

We will now plug in upper bounds for λ_i. From Lemma C.4, we have that $\lambda_i = \frac{\binom{\ell-|T|}{(1-\alpha)\ell-|T|}}{\binom{\ell}{(1-\alpha)\ell}}$ for $i \leq (1-\alpha)\ell$ and 0 otherwise. One can check that $\lambda_i \leq (1-\alpha)\lambda_{i-1}$ for all i between 1 and ℓ. Since $\lambda_0 = 1$, we get that $\lambda_i \leq (1-\alpha)^i$, thus completing the proof of the lemma.

□

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